Analytics-Driven Decision Support Framework for Enterprise Food Operations and Employee Health Management

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Project Type: **Academic Project (Type 3)**

Project Area:

Business Analytics, Employee Health Management, Food Operations Analytics, Supply Chain Analytics

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Analytics-Driven Decision Support Framework for Enterprise Food Operations and Employee Health Management

Executive Summary of Mid-Semester Progress

This section documents the research framing, dataset preparation, implementation evidence, preliminary findings, risks, and next steps completed at the mid-semester stage. It is written as an academic progress report for an MBA in Business Analytics project. The approved Project Report structure that follows is retained without alteration; this summary and the chapters after each corresponding approved section add the evidence gathered since the proposal stage.

The project develops a food intelligence decision-support prototype for enterprise food operations and employee wellness. Its current contribution is a traceable catalogue that connects nutrition facts with recipe evidence, ingredient information, declared allergens, semantic representations, and dashboard-based analysis. The work is exploratory and decision-support oriented; it does not provide clinical advice, diagnostic conclusions, or production deployment claims.

Graph RAG remains an optional advanced extension in the approved outline. It is deliberately excluded from the current achievement claims because its retrieval quality and operational complexity require further research validation. This decision narrows the mid-semester deliverable to reliable data preparation, semantic retrieval, graph modelling, and a transparent dashboard prototype.

Enterprise cafeterias and institutional food services must balance nutritional transparency, recipe evidence, food safety awareness, and practical menu planning. Yet food information is commonly fragmented across nutrition tables, recipe sources, ingredient records, and manually maintained menu documents. This project addresses that fragmentation by developing an analytics-driven food intelligence prototype for decision support.

The mid-semester implementation has produced a processed catalogue containing 1,014 uniquely identified food records and 97 analytical fields. The catalogue contains macronutrient and micronutrient facts, engineered ratios, recipe-match evidence, ingredient details were available, source-declared allergens, semantic text, embedding readiness, and nutrition-cluster labels. A total of 449 records has

confirmed exact or alias recipe matches, while 187 records are retained in a review queue and 378 records remain unlinked rather than being assigned unsupported evidence.

Semantic representations have been generated for all 1,014 catalogue records using the all-MiniLM-L6-v2 sentence-transformer model. The resulting embedding matrix has dimensions 1,014 x 256 and supports natural-language discovery of comparable foods. The project also includes a Neo4j graph ingestion design and a Streamlit dashboard prototype. The dashboard presents menu-level nutrition patterns, evidence coverage, allergen visibility, semantic exploration, and constraint-based menu planning.

The preliminary result is not a claim that the system optimizes live cafeteria operations. Rather, it demonstrates that disparate food datasets can be integrated into an auditable analytical foundation. The next work focuses on evaluation: confirming fuzzy recipe matches, validating graph ingestion, constructing a labelled semantic-retrieval test set, and gathering user feedback on dashboard usefulness.

1. Background

Modern organizations are increasingly investing in employee wellness, workplace experience, and data-driven operational management. Food services such as enterprise cafeterias, institutional kitchens, and corporate catering facilities are no longer limited to meal preparation activities; they influence employee satisfaction, workplace health, operating efficiency, and organizational productivity.

Food operations face practical business challenges. Cafeteria and kitchen managers must plan menus, estimate preparation time, manage ingredient availability, control operational bottlenecks, minimize wastage, and ensure safe food choices for employees with dietary restrictions or allergen concerns. These decisions are often made using manual judgment, static menu records, and fragmented data sources.

At the same time, modern food-service environments require stronger nutritional transparency and explainable recommendations. Recipes, ingredients, nutrients, allergens, dietary categories, regions, and courses are naturally connected entities. A graph-based analytical approach can therefore support more structured reasoning than flat menu records.

This project proposes an analytics-driven decision support framework for enterprise food operations and employee health management. The framework is positioned as a Business Analytics solution that combines predictive analytics, semantic menu intelligence, knowledge graph modeling, and explainable natural language decision support.

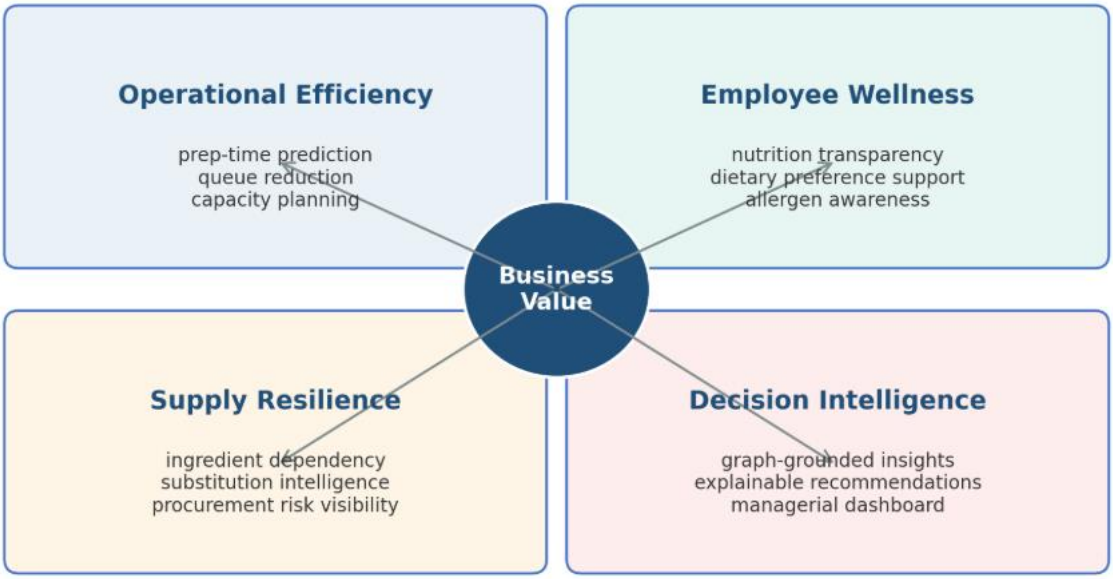


Figure 1: Business Analytics value framework for the proposed abstract.

2. Problem Statement

Enterprise cafeterias and institutional food operations typically operate under multiple constraints related to time, cost, nutrition, safety, and supply continuity. However, many such environments lack an integrated analytics framework that converts food, ingredient, nutritional, and operational data into actionable business insights.

1. **Operational planning problem:** Limited ability to predict preparation time, kitchen workload, and peak-hour resource requirements.
2. **Menu intelligence problem:** Menu records usually do not capture semantic similarity, ingredient substitutes, regional variations, and overlapping raw material dependencies.
3. **Nutrition and wellness problem:** Employees increasingly require dietary transparency, but traditional cafeteria systems may not provide explainable nutritional or allergen guidance.
4. **Supply chain dependency problem:** Ingredient shortages or price fluctuations can affect multiple dishes, but dependency visibility is often weak.
5. **AI reliability problem:** Generic AI responses can be unsupported. In food and health contexts, decision support must be grounded in validated data relationships.

Therefore, there is a requirement for an integrated Business Analytics framework that supports cafeteria managers, employee wellness teams, and business stakeholders through predictive analytics, explainable recommendations, and dashboard-based decision support.

Chapter 1: Expanded Requirement Statement and Problem Context

1.1 Enterprise Food Operations Landscape

Enterprise cafeterias, institutional kitchens, and corporate food-service environments operate as recurring service systems. Their menu decisions influence employee experience, perceived wellness support, food availability, and kitchen workload. Unlike a simple retail catalogue, a cafeteria menu must be interpreted in relation to nutrition, ingredient composition, recipe complexity, allergy declarations, substitutions, and operational feasibility.

Food operations data is usually distributed across separate artifacts. Nutrition sources provide nutrient composition but rarely contain recipe ingredients or preparation methods. Recipe sources provide textual instructions and ingredients but may differ in naming conventions, serving assumptions, and cultural terminology. Allergen information can be incomplete or inconsistent. These conditions make it difficult to present reliable, comprehensible, and evidence-aware information to managers and users.

1.2 Operational, Wellness, and Analytics Challenges

Table 1. Operational, wellness, and analytics challenges mapped to the project's response.

Challenge area	Observed decision problem	Analytics response in this project
Menu information	Nutrition and recipe details exist in separate sources.	Create a canonical food catalogue with source-aware evidence fields.
Wellness transparency	Users need understandable information about nutrients and declared allergens.	Provide filtering and readable nutrition summaries without clinical claims.
Evidence quality	Dish names can be ambiguous across datasets.	Separate confirmed matches, review candidates, and unmatched records.
Menu discovery	Keyword search does not	Generate sentence embeddings

	identify semantically comparable dishes.	for similarity-based exploration.
Relationship visibility	Ingredients, allergens, and nutrients are connected but hard to query together.	Model these entities and relations through a knowledge-graph design.

1.3 Refined Problem Statement

The central research problem is the absence of a transparent and integrated analytics layer that can connect food nutrition, recipe evidence, ingredients, declared allergens, and semantic similarity for enterprise food decision support. Existing raw datasets provide useful individual facts but do not directly support auditable menu exploration or managerial comparison.

Accordingly, the project investigates whether public food and recipe data can be transformed into a structured, evidence-aware catalogue and made accessible through analytics, semantic retrieval, graph modelling, and dashboard interaction. The project is evaluated as an applied prototype, not as a deployed operational system or medical recommendation engine.

3. Motivation for this project

The motivation for this project comes from the intersection of Business Analytics, AI-enabled decision support, employee wellness, and food operations optimization.

From a business perspective, enterprise cafeterias are high-volume service systems where small inefficiencies in preparation planning, food availability, or menu design can affect employee satisfaction, waiting time, food wastage, and operating cost. A data-driven framework can help managers move from reactive decision-making to proactive planning.

From an employee wellness perspective, workplace food environments can influence dietary choices and health awareness. An analytics framework that improves transparency around ingredients, nutrients, dietary categories, and allergens can support broader employee well-being initiatives.

From an MBA Business Analytics perspective, this project applies predictive modeling, data preparation, machine learning, dashboard development, decision support, supply chain analytics, and stakeholder communication. It converts technical capability into managerial value and demonstrates how analytics can solve practical business problems.

4. Project Objectives

The primary objective is to design and develop an analytics-driven decision support framework that improves enterprise food operations and supports employee health management through predictive analytics, knowledge graph intelligence, and explainable AI.

- **Objective 1 - Operational Analytics:** Develop machine learning models that predict food preparation time and operational complexity.
- **Objective 2 - Menu Intelligence:** Implement semantic similarity capabilities to identify similar dishes, possible substitutes, and menu redundancies.
- **Objective 3 - Knowledge Graph Modeling:** Create a structured food knowledge graph representing relationships among dishes, ingredients, nutrients, allergens, dietary categories, regions, courses, and operational dependencies.
- **Objective 4 – Exploration of Explainable Decision Support:** Use a Graph RAG-based query layer to enable graph-grounded, evidence-based, and auditable responses.

- **Objective 5 - Supply Chain and Dependency Analytics:** Identify high-dependency ingredients, ingredient overlap across dishes, and potential single points of failure in procurement and menu planning.
- **Objective 6 - Employee Wellness Analytics:** Support nutrition-aware food recommendations and dietary transparency.
- **Objective 7 - Dashboard and Business Communication:** Deliver an interactive dashboard that communicates operational, nutritional, and managerial insights to business users.

The success of the project will be evaluated using:

- Prediction Accuracy (MAE / RMSE)
- Model Explainability
- Reduction in manual menu analysis effort
- Ability to identify ingredient dependencies
- Quality of generated business insights
- Dashboard usability and decision-support effectiveness

Chapter 2: Project Objectives and Success Measures

2.1 Objective Interpretation at Mid-Semester

Table 2. Approved objectives interpreted against mid-semester evidence.

Approved objective	Mid-semester interpretation	Current evidence
Operational analytics	Prepare data and derived features that can later support operational models.	Recipe time and ingredient fields are available in source data; no final prediction model is claimed.
Menu intelligence	Enable comparable-food discovery beyond literal keywords.	1,014 semantic documents and embeddings have been prepared.
Knowledge graph modelling	Define a relationship model and reproducible ingestion path.	Neo4j schema constraints and ingestion script implemented; validation remains pending.
Explainable decision support	Make evidence status and applied filters visible.	Dashboard presents match status, nutrition fields, declared allergens, and source-aware warnings.
Supply-chain dependency analytics	Establish ingredient-level evidence for later dependency analysis.	Ingredients are available for matched records; network analysis is future work.
Employee wellness analytics	Support non-clinical nutrition-aware food exploration.	Rule-based wellness filters are implemented in the dashboard.
Dashboard communication	Present the catalogue to non-technical decision makers.	Functional Streamlit prototype with manager, governance, planner, and explorer views.

2.2 SMART Objective Mapping

Table 3. SMART mapping of core project objectives.

Objective	Specific	Measure	Time-bound mid-semester evidence
Build canonical catalogue	Integrate nutrition and recipe evidence.	Unique food codes, evidence fields, quality checks.	Completed prototype catalogue.
Support semantic discovery	Embed food records for similarity retrieval.	Embedding coverage and retrieval test cases.	Embedding coverage completed; formal relevance evaluation pending.
Create graph foundation	Represent food relationships in Neo4j.	Constraints, node/edge ingestion, validation queries.	Schema and script completed; execution validation pending.
Deliver dashboard	Expose useful filters and evidence to users.	Working views, KPI outputs, user feedback.	Prototype completed; usability study pending.

The success measures have been deliberately separated into implementation measures and outcome measures. Implementation measures, such as record count, match coverage, embedding coverage, and dashboard availability, can be verified now. Outcome measures, such as reduced manual analysis effort, improved planning quality, or user satisfaction, require a future user study or a live organizational setting and are not claimed at this stage.

5. Project Scope

In-Scope Items

- Collection and integration of Indian food and nutrient datasets.

- Data cleaning, missing value treatment, standardization, and feature engineering.
- Predictive modeling for preparation-time and kitchen-complexity analytics.
- Semantic recipe representation for similarity search and menu intelligence.
- Knowledge graph modeling for dishes, ingredients, nutrients, allergens, regions, courses, and diets.
- Executive dashboard and business insight presentation.
- Evaluation of model outputs using prediction accuracy, recommendation relevance, explainability, and business usefulness.

Exploratory Scope Items

- Graph RAG-based natural language query interface for explainable decision support.

Out-of-Scope Items

- Real-time IoT kitchen sensor integration.
- Direct integration with live Point of Sale payment systems.
- Clinical diagnosis or medical treatment recommendation.
- Replacement of certified dietitians, doctors, or nutrition professionals.
- Full-scale enterprise deployment across multiple physical cafeteria locations.
- Real-time employee health record integration.

This project is positioned as a decision support and analytics framework, not as a clinical diagnosis tool.

Chapter 3: Scope, Assumptions, and Boundaries

3.1 In-Scope Work

- Integration of a nutrition-oriented food catalogue with publicly available recipe and allergen-supporting datasets.
- Data profiling, duplicate control, text normalization, evidence-status labelling, and feature engineering.
- Recipe title matching using exact, alias, and review-oriented fuzzy matching stages.
- Nutrition clustering for descriptive portfolio analysis.
- Sentence-transformer embeddings and semantic similar-food retrieval.
- Knowledge graph schema and repeatable Neo4j ingestion design for dishes, nutrients, ingredients, and allergens.
- A Streamlit dashboard with manager, data-quality, menu-planning, catalogue, and semantic-exploration views.

3.2 Out-of-Scope Work

- Live point-of-sale, procurement, inventory, kitchen sensor, employee-health, or personal medical data integration.
- Clinical nutrition assessment, diagnosis, treatment, or allergy guarantees.
- Claims of business savings, waste reduction, revenue improvement, or operational throughput without organizational data.
- Production rollout across multiple cafeterias and real-time automated menu decisions.
- Graph RAG as a completed project deliverable; it is retained only as a future optional extension.

3.3 Research Assumptions and Constraints

The work assumes that the available datasets are suitable for exploratory analytics, while recognizing that public datasets are heterogeneous. Nutrient values are treated as dataset-provided estimates. Recipe matches are treated as evidence only when the match status is confirmed; review and unmatched records must not be presented as verified recipe links.

The main constraints are limited project time, incomplete recipe linkage, possible differences in recipe serving conventions, restricted access to live operational data, and local-compute limits for model and database experiments. These constraints shape the project toward a decision-support prototype rather than a production system.

Chapter 4: Literature Review and Research Gap

4.1 Food Analytics and Nutrition-Aware Decision Support

Food informatics combines structured nutrient data, food descriptions, recipes, ingredients, and consumer-facing decision support. The challenge is not only to store food facts, but to preserve their source, units, context, and uncertainty. In an enterprise setting, this creates a need for interfaces that make food information understandable without oversimplifying safety or nutrition evidence.

Recommendation systems can help users navigate large catalogues, but nutrition-aware settings require transparent constraints. A ranked food should be accompanied by the facts and filters that produced the ranking. This principle motivates the present project's evidence-status fields and visible nutrition filters rather than opaque recommendation scores alone.

Min et al. (2019) survey food computing broadly, covering recognition, retrieval, recommendation, and nutrition monitoring, and note that most systems specialize in a single stage rather than connecting nutrient facts, recipe evidence, and user-facing explanation end to end. Freyne and Berkovsky (2010) show that ingredient-level recommendation is more acceptable to users than opaque recipe-level ranking, which supports this project's decision to expose ingredients and nutrients rather than a single unexplained score. Trattner and Elsweiler (2017) further argue that food recommenders must be evaluated against health-aware and diversity-aware criteria, not accuracy alone, reinforcing the project's calibrated, non-clinical framing.

4.2 Workplace Wellness and Food Transparency

Workplace wellness research indicates that food environments may influence dietary behavior, but intervention effects depend on context, design, and employee participation. The project therefore avoids causal claims. Instead, it treats nutrition-aware menu discovery and transparent information display as enabling capabilities that can be evaluated with users in a later stage.

The dashboard language is intentionally non-clinical. For example, a high-protein option is defined as a food satisfying a configured threshold in the dataset; it is not represented as an individually suitable health recommendation. This distinction is central to responsible analytics in a food-service setting.

Peñalvo et al. (2021) find mixed but generally positive effects of workplace wellness programmes on

dietary habits, contingent on programme design and voluntary participation, which supports treating this project's nutrition filters as an enabling tool rather than a guaranteed behavior-change intervention. Ge, Ricci, and Massimo (2015) similarly propose health-aware recommendation as a decision-support layer that should be combined with, not substituted for, professional dietary guidance, a position consistent with the out-of-scope boundary defined in Chapter 3.

4.3 Semantic Search and Sentence Embeddings

Traditional keyword search depends on literal token overlap. Food naming is particularly variable because recipes may use regional names, preparation methods, ingredient references, or synonymous dish terms. Transformer architectures introduced contextual language representation, and Sentence-BERT subsequently enabled efficient sentence-level similarity comparison. These developments make semantic retrieval appropriate for food catalogue exploration.

In this project, semantic text includes dish identity, recipe evidence when available, allergens, ingredients, instructions, and selected nutrient facts. The embedding does not establish nutritional equivalence or ingredient safety. It acts as a discovery mechanism that surfaces comparable records for human review.

Devlin et al. (2019) introduced bidirectional transformer pretraining (BERT), which underlies most modern sentence-embedding models, including the all-MiniLM-L6-v2 model used in this project. Teng, Lin, and Adamic (2012) show that ingredient-co-occurrence networks can reveal substitution patterns in recipes, a complementary signal to text-based similarity that could extend this project's semantic layer in later phases.

4.4 Knowledge Graphs and Explainability

Knowledge graphs represent entities and relationships explicitly. A graph model is useful in food analytics because a dish may have many nutrients, ingredients, and allergens, while the same ingredient may occur in many dishes. Graph modelling supports relationship-oriented questions that are difficult to express through a single flat table.

The project adopts a property-graph approach using Neo4j. It places nutrient amounts on relationships between Dish and Nutrient nodes, retains declared allergen evidence separately, and uses unique constraints to make ingestion repeatable. The immediate value is structural transparency, even before

advanced graph analytics or natural-language generation is attempted.

Ehrlinger and Wöß (2016) note that knowledge graphs are most useful when entity and relationship semantics are explicit and queryable, rather than merely stored as connected records; the constraint-based schema in this project follows that recommendation. Robinson, Webber, and Eifrem (2015) similarly argue that graph databases suit domains with many-to-many relationships, such as dishes sharing ingredients or nutrients, which are awkward to express in normalized relational tables.

4.5 Explainable and Responsible AI

Explainable AI emphasizes that an analytical output should be interpretable by the people who must act on it. In the present prototype, explainability is delivered through visible evidence status, source-aware fields, declared filters, and inspectable details. This approach is preferable at the current stage to presenting a black-box recommendation without provenance.

Responsible AI also requires an appropriate boundary for automated outputs. The project identifies uncertainty rather than concealing it: unmatched records stay unmatched, review records are marked for validation, declared allergens are not promoted to guarantees, and no medical or clinical advice is generated.

Ribeiro, Singh, and Guestrin (2016) argue that trust in a model's output depends on the user being able to see which evidence drove a specific prediction; this project applies the same principle at the data layer by keeping match method, match score, and evidence status visible alongside every record. Doshi-Velez and Kim (2017) caution that interpretability requirements differ by application stakes, which supports treating food-safety-adjacent outputs, such as allergen exclusions, with stricter evidence requirements than descriptive nutrition clustering.

4.6 Research Gap

Table 4. Research gap identified between existing literature streams and this project.

Research stream	Typical contribution	Gap addressed by this project
Nutrition databases	Provide nutrient composition.	Often lack linked recipe evidence and business-facing

		exploration.
Recipe recommendation	Suggest recipes or similar dishes.	May not preserve nutrition provenance and match confidence.
Knowledge graphs	Represent connected entities.	Often not combined with a practical menu-planning dashboard.
Workplace wellness	Studies food environment interventions.	Needs transparent tools for nutrition and menu information exploration.
Semantic retrieval	Discovers contextually similar text.	Requires domain-specific safeguards against unsupported equivalence claims.

The identified gap is a lightweight, traceable prototype that integrates nutrition data, recipe evidence, semantic discovery, graph structure, and managerial dashboarding while explicitly exposing uncertainty. The contribution is integration and auditability rather than a claim of novel machine-learning architecture.

6. Proposed methodology

The project will follow a structured Business Analytics methodology consisting of six phases. The focus of this abstract is to define the analytical approach and expected business value, not to provide detailed implementation code or system deployment steps.

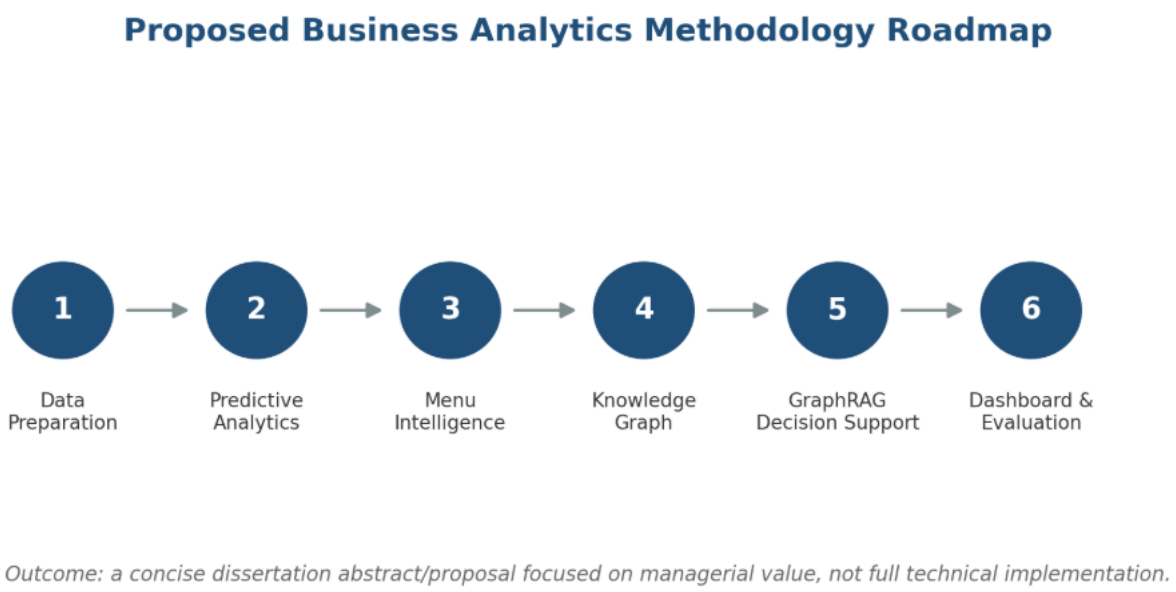


Figure 2: Proposed Business Analytics methodology roadmap.

Phase 1: Data Collection and Preparation

Collect and integrate food, ingredient, and nutrient datasets. Clean, standardize, transform, and engineer variables such as total preparation time, ingredient count, complexity score, course category, regional category, and dietary classification.

Phase 2: Predictive Analytics and Machine Learning

Build predictive models for operational planning. Regression models such as Random Forest, Gradient Boosting, or XGBoost may be evaluated using metrics such as Mean Absolute Error, Root Mean Squared Error, and R-squared.

Phase 3: Semantic Search and Menu Intelligence

Create semantic representations of dishes using dish names, ingredients, diet type, preparation method, course, and region. Use these representations for similarity search, recipe clustering, and substitution discovery.

Phase 4: Knowledge Graph Construction

Develop a graph-based representation of food operations where entities such as Dish, Ingredient, Nutrient, Allergen, Diet Type, Region, and Course are connected through meaningful relationships.

Phase 5: Graph RAG-Based Explainable Decision Support (Optional Advanced Extension)

Integrate graph retrieval with a natural language interface so that user responses are grounded in structured facts rather than unsupported generation.

Phase 6: Dashboard Development and Business Evaluation (Prototype Dashboard)

Build a dashboard covering preparation-time prediction, menu similarity, ingredient dependency, nutrition and allergen insights, and executive summaries.

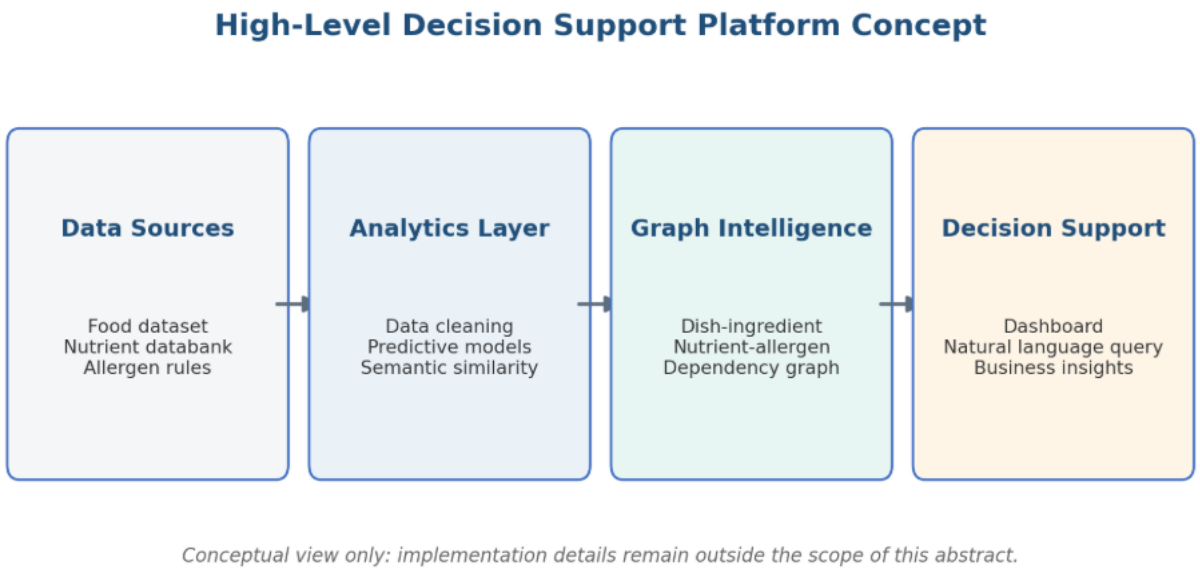


Figure 3: Conceptual framework *flow for the abstract/proposal.*

Chapter 5: Research Methodology

5.1 Research Design

The project follows an applied analytics and design-science orientation. It addresses a practical decision-support problem by constructing and evaluating an artefact: an evidence-aware food intelligence prototype. Quantitative methods are used for profiling, matching coverage, descriptive clustering, and semantic-retrieval preparation. Design-science elements are reflected in iterative creation of data structures, retrieval components, graph design, and dashboard views.

The unit of analysis is a canonical food record. A record is identified by food_code and may be linked to zero or one supporting recipe record. The design intentionally separates data facts from derived features and confidence labels, enabling later audit and revision.

5.2 Data Processing Workflow

Table 5. Seven-step data processing workflow from source profiling to dashboard delivery.

Step	Method	Output
1. Source profiling	Inspect dimensions, columns, identifiers, and missingness.	Documented source roles and quality checks.
2. Canonicalization	Standardize identifiers, names, numeric fields, and text.	Canonical food catalogue.
3. Recipe matching	Exact/alias matching followed by fuzzy review candidates.	Matched, review, and unmatched evidence labels.
4. Feature engineering	Create ratios, ingredient counts, semantic text, and cluster labels.	Analysis-ready fields.
5. Semantic representation	Encode semantic text using all-MiniLM-L6-v2.	1,014 x 256 embedding matrix.
6. Graph modelling	Define nodes, relationships,	Neo4j-ready ingestion pipeline.

	constraints, and batch ingestion.	
7. Dashboard prototype	Expose KPIs, filters, details, and semantic search.	Streamlit decision-support interface.

5.3 Research Questions

- RQ1: Can heterogeneous nutrition and recipe data be consolidated into an evidence-aware food catalogue suitable for analytics?
- RQ2: Can sentence embeddings support discovery of semantically comparable food records beyond keyword search?
- RQ3: Can a knowledge-graph model make dish, nutrient, ingredient, and declared-allergen relationships more visible and queryable?
- RQ4: Can a dashboard present food evidence in a form useful for menu exploration without making unsupported health or operational claims?

5.4 Evaluation Plan

Mid-semester evaluation focuses on artefact completeness and traceability: unique catalogue identifiers, non-negative constrained nutrient fields, match-status counts, recipe evidence coverage, embedding coverage, graph schema existence, and dashboard functionality. Subsequent evaluation will use manual review of fuzzy recipe matches, labelled retrieval queries for Precision@K, graph node/edge parity checks, and a structured usability review with potential users.

5.5 Hypotheses

The following hypotheses operationalize the research questions above. Each is stated as testable and is intended to be confirmed or rejected using the evaluation methods in Section 5.4 during the remaining project phases; none is treated as already proven at the mid-semester stage.

- H1 (linked to RQ1): A canonical catalogue built from heterogeneous nutrition and recipe sources can achieve unique-key integrity and traceable evidence status without manual record-by-record reconciliation.
- H2 (linked to RQ2): Sentence-embedding retrieval returns food records that domain-plausible

reviewers judge as topically relevant more often than a literal keyword match on food names alone.

- H3 (linked to RQ3): A property-graph representation of dishes, nutrients, ingredients, and allergens allows relationship queries (for example, shared-ingredient or allergen-exclusion queries) that are not directly expressible as a single filter on the flat catalogue.
- H4 (linked to RQ4): A dashboard that exposes evidence status and applied filters is preferred by reviewers over an equivalent view that only shows a ranked recommendation without supporting evidence.

7. Plan of Work

Chapter 6: Plan Versus Progress

Table 6. Plan versus progress across the six project phases.

Phase	Planned deliverable	Actual mid-semester status	Progress
1. Data collection and preparation	Integrated, cleaned, feature-ready catalogue.	Completed core catalogue, matching outputs, ratios, clusters, semantic fields, and quality checks.	90%
2. Predictive analytics	Preparation-time and complexity models.	Data fields and features identified; final benchmark models and metrics not yet exported.	35%
3. Semantic search	Embedding generation and similarity exploration.	All catalogue records embedded; local and Qdrant-oriented retrieval paths	80%

		implemented.	
4. Knowledge graph	Neo4j graph and validation.	Schema constraints and ingestion script implemented; database execution and count validation pending.	45%
5. Graph RAG (optional)	Graph-grounded language interface.	Deferred from achievement scope pending quality evaluation and simplification.	0%
6. Dashboard	Managerial decision-support prototype.	Functional Streamlit dashboard and reproducible KPI experiment completed; formal usability evaluation pending.	75%

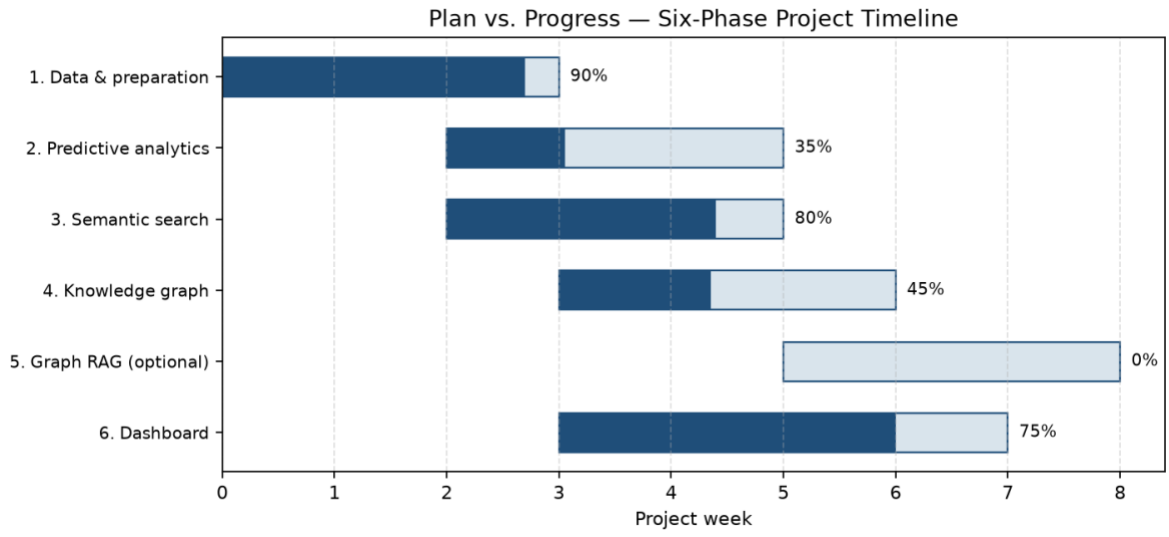


Figure 4. Plan vs. progress Gantt chart across the six project phases (weeks on the x-axis).

6.1 Milestone Interpretation

The project has progressed fastest in the data and interface layers because these provide the reliable foundation required for later modelling. The deliberate deferral of Graph RAG is a scope-control decision. It reduces the risk of presenting unvalidated generated recommendations and directs the remaining effort toward data quality, semantic evaluation, graph validation, and user-facing usefulness.

The main schedule deviation is that predictive modelling and graph validation are behind the initial conceptual plan. The corrective action is to use the established canonical catalogue as the single input for those tasks, avoiding further data restructuring. This ensures that future experiments use the same identifiers, match-status policy, and feature definitions as the dashboard.

Chapter 7: Detailed Work Completed till Mid-Semester

This chapter is organized by project phase, consistent with the six-phase methodology defined in the approved outline and the phase-wise implementation plan maintained for this project. Phase 4 (Graph RAG) is intentionally excluded from achievement claims in this chapter; it is retained only as an optional, deferred extension because its retrieval quality has not yet been validated.

Table 7. Mapping of report subsections to the project's six-phase methodology.

Project phase	Phase goal	Report subsection	Mid-semester status
Phase 0 - Foundation and governance	Freeze schema, nutrient dictionary, and validation rules.	7.2 Data Cleaning and Evidence Management	Core validation checks implemented; formal contract documents pending.
Phase 1 - Data and classical ML	Clean data, engineer features, and benchmark baseline models.	7.1-7.4	Cleaning, matching, and features completed; baseline model metrics pending.
Phase 2 - Embeddings	Generate dish	7.5 Semantic Search	Embedding generation

and semantic retrieval	embeddings and enable similarity search.	Design and Implementation	completed for the full catalogue.
Phase 3 - Knowledge graph construction	Model dishes, nutrients, ingredients, and allergens in Neo4j.	7.6 Knowledge Graph Design	Schema and ingestion script implemented; ingestion validation pending.
Phase 4 - Graph RAG (optional, deferred)	Graph-grounded natural-language decision support.	Not claimed in this chapter	Excluded from mid-semester achievements pending quality evaluation.
Phase 5 - Dashboard and delivery	Deliver a decision-support interface for business users.	7.7 Dashboard Design and Implemented Views	Functional Streamlit prototype completed; usability evaluation pending.

7.1 Dataset Identification and Roles

Table 8. Datasets used in the prototype and their primary roles.

Dataset	Records	Primary role	Use in prototype
Processed canonical catalogue	1,014 food records; 97 fields	Central analytical dataset	Dashboard, embeddings, graph ingestion.
Indian Food Dataset	5,938 recipes	Recipe composition and operational context	Ingredients, instructions, cuisine, time, ingredient count.
Allergen support dataset	399 food-product records	Allergen seed information	Supporting allergen identification and review.

Recipe matching source	2,231,142 raw rows examined	Recipe title evidence	Exact/alias and fuzzy review linkage.

The dataset architecture assigns one primary responsibility to each source. The canonical nutrition catalogue is the authority for nutrient facts. Recipe sources enrich records only when a match is supported. This source-role policy reduces the risk of blending incompatible facts or treating a similarly named recipe as a confirmed nutritional equivalent.

7.2 Data Cleaning and Evidence Management

Data cleaning included identifier checks, text handling, numeric-field inspection, and duplicate treatment. The processed catalogue contains 1,014 unique food_code values. Core nutrition fields checked for this report, including energy, protein, carbohydrate, fat, fiber, and sodium, have no negative recorded values. This is a basic but important validity condition before visualization or filtering.

Recipe-linking evidence is stored explicitly. Of the 1,014 foods, 449 are confirmed through exact or alias matching. Another 187 are plausible fuzzy candidates retained for review, and 378 have no accepted match. This policy protects the integrity of recipe and allergen details: a food record is not given recipe-derived facts merely because a name appears superficially similar.

7.3 Feature Engineering

Table 9. Engineered features and their analytical purpose.

Feature	Definition	Analytical purpose
ingredient_count	Count derived from a non-empty matched ingredient list.	Supports recipe-complexity and filter analysis.
protein_to_carb_ratio	protein_g divided by carb_g with numerical protection.	Compares macronutrient balance.

fat_to_carb_ratio	fat_g divided by carb_g with numerical protection.	Supports descriptive nutrition analysis.
calorie_to_protein_ratio	energy_kcal divided by protein_g with numerical protection.	Supports protein-density comparison.
semantic_text	Structured textual representation of food facts and recipe evidence.	Input for sentence embedding.
nutrition_cluster	Unsupervised label based on nutrition-profile features.	Portfolio segmentation, not clinical classification.
match_status	matched, review, or unmatched.	Controls appropriate use of recipe evidence.

7.4 Exploratory Analysis and Nutrition Clusters

Exploratory visualization was used to understand energy distributions, source composition, correlations among major nutrition features, and the quality of cluster separation. Clustering produces two descriptive menu segments in the current processed catalogue. Cluster 0 contains 709 records and has lower median energy and protein values than Cluster 1, which contains 305 records. These labels are used for portfolio comparison only and should not be interpreted as good-versus-bad food categories.

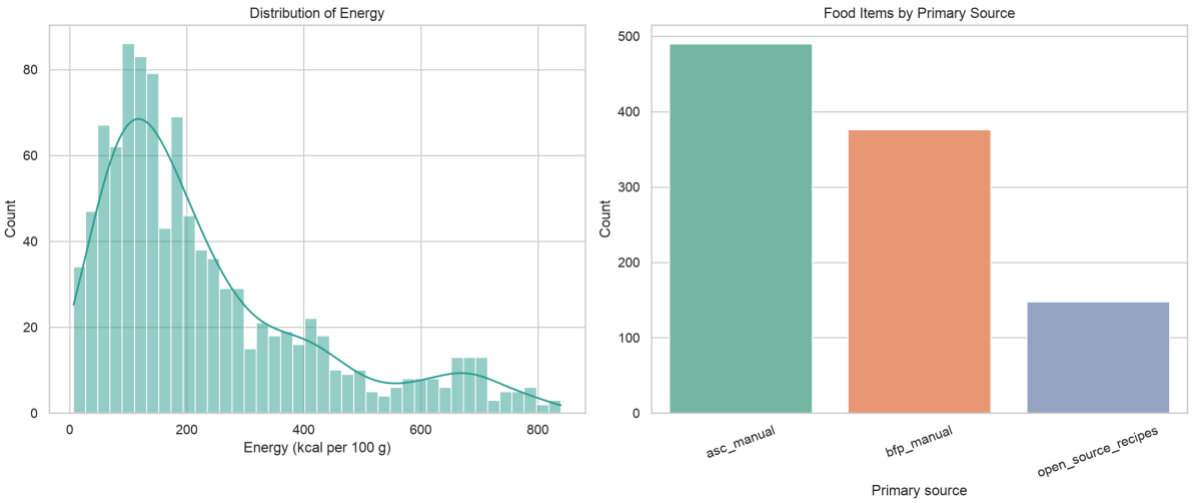


Figure 5. Energy and source distribution from the project exploratory analysis.

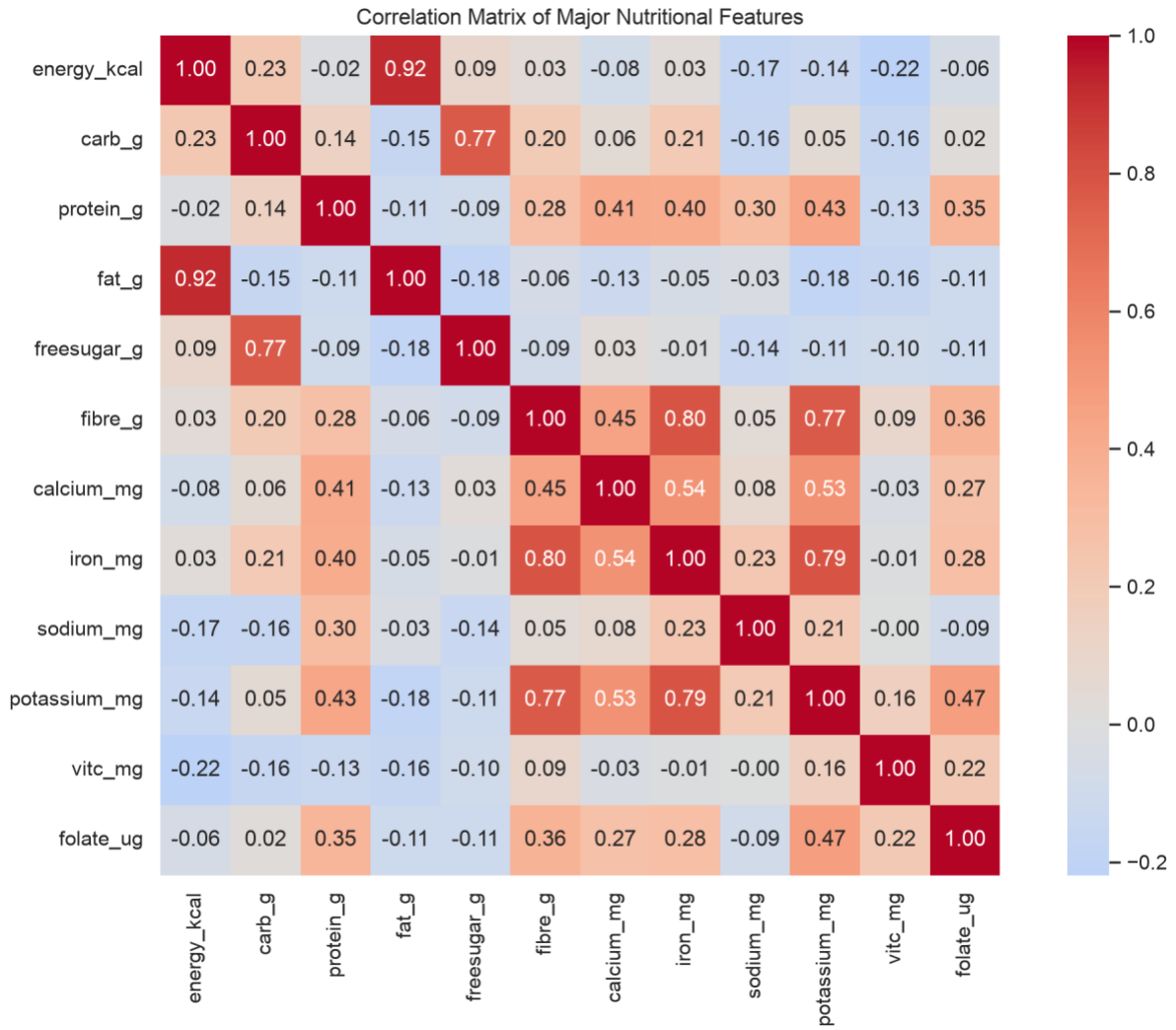


Figure 6. Major nutrition-feature correlations used for exploratory analysis.

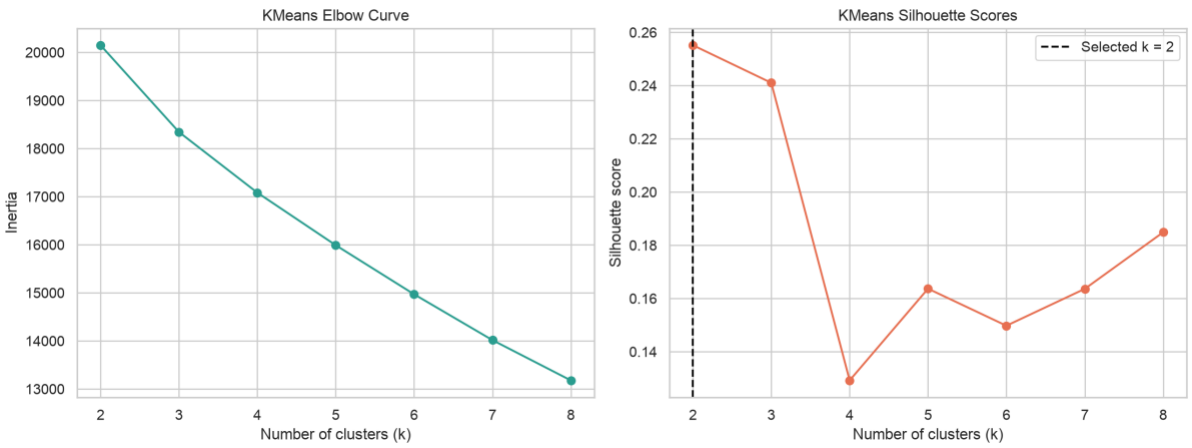


Figure 7. Clustering quality exploration from the project analysis.

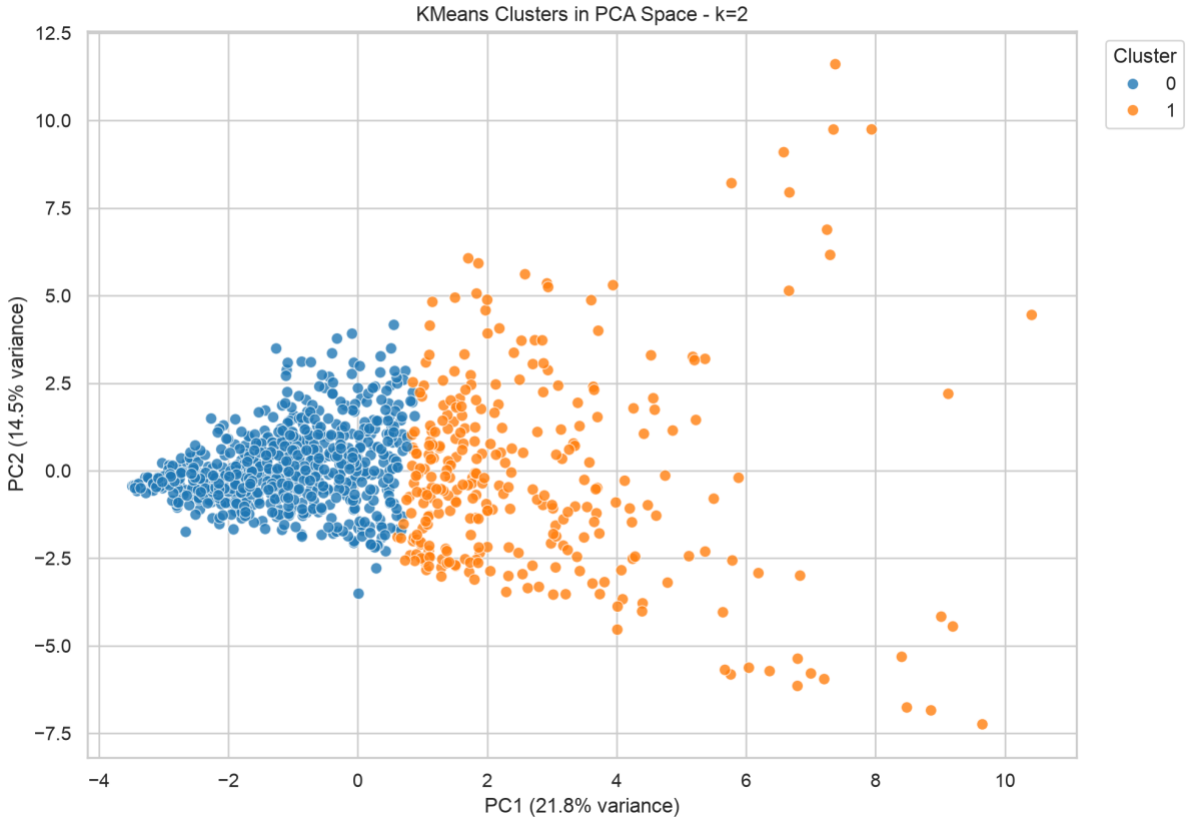


Figure 8. K-means cluster visualization using principal-component projection.

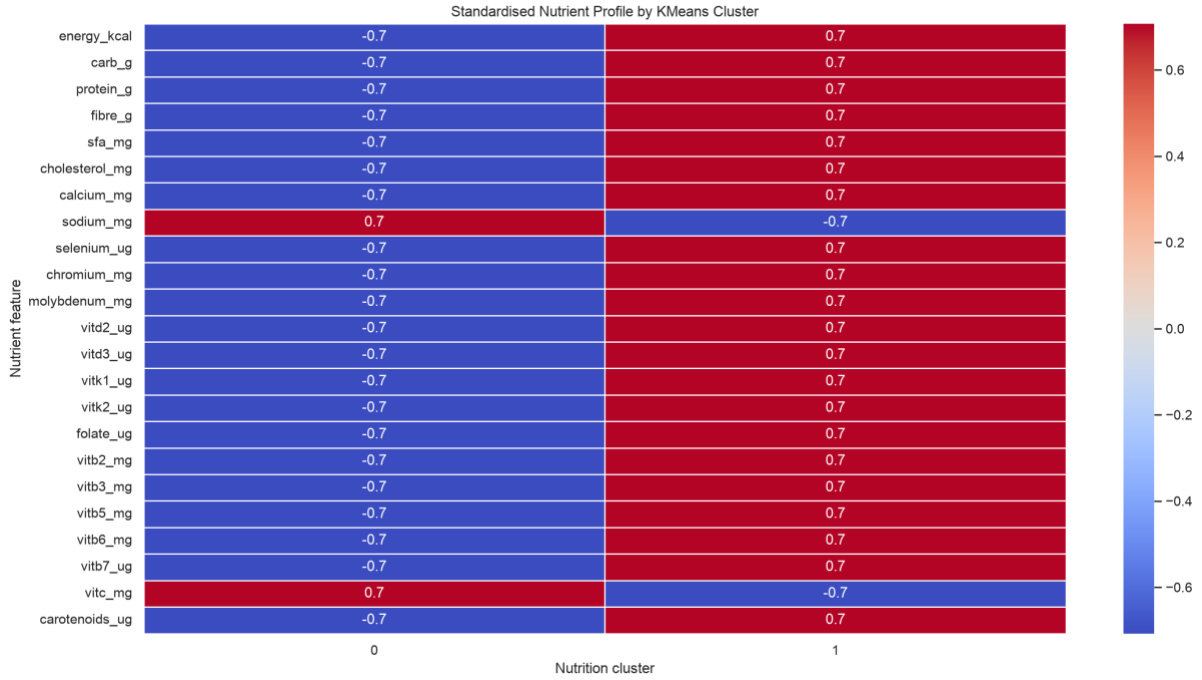


Figure 9. Nutrition-profile comparison across descriptive clusters.

7.5 Semantic Search Design and Implementation

For semantic search, each catalogue record is transformed into semantic text. The text combines food name, food code, linked recipe name, ingredients and instructions when verified, declared allergens, and selected nutrition facts. This design permits a user query to express an intention, such as finding a higher-protein food or discovering comparable dishes, without requiring exact dish-name wording.

The all-MiniLM-L6-v2 model creates a 256-dimensional embedding for each record. The saved matrix has one row per catalogue record, resulting in a 1,014 x 256 artifact. The system supports a Qdrant-oriented vector-database path and a local matrix fallback. The fallback is important for reproducible demonstrations because it allows semantic ranking without requiring a running external vector service.

Retrieval results are treated as candidates for exploration. A high semantic similarity score does not prove that two dishes are interchangeable, share ingredients, have identical cost, or are equally suitable for allergies. The dashboard therefore presents similarity as a discovery signal alongside evidence and nutrition fields.

7.6 Knowledge Graph Design

The implemented Neo4j ingestion design defines Dish, Nutrient, Ingredient, and Allergen nodes. A Dish node is keyed by food_code. Nutrient amounts and units are stored as properties on HAS_NUTRIENT relationships, which preserves a natural representation of the statement that a given dish has a particular nutrient quantity. Ingredient and declared-allergen relationships are added only from available catalogue evidence.

The graph design uses uniqueness constraints for Dish.food_code, Nutrient.name, Ingredient.name, and Allergen.name. This makes the ingest process re-runnable and reduces accidental duplication. Graph ingestion counts and relationship integrity checks are planned for the remaining semester work; they are not represented as completed results in this report.

Table 10. Knowledge-graph node and relationship types and their evidence policy.

Node / relation	Meaning	Evidence policy
Dish	Canonical food record.	Created for each unique food_code.
Nutrient	Named nutrition attribute.	Linked where numeric catalogue value is present.
Ingredient	Recipe ingredient token.	Linked only where recipe evidence is available.
Allergen	Declared allergen term.	Source-declared information; not inferred clinical safety.
HAS_NUTRIENT	Dish-to-nutrient relationship with amount and unit.	Derived from processed nutrition fields.
HAS_INGREDIENT	Dish-to-ingredient relationship.	Derived from linked recipe

		evidence.
HAS_DECLARED_ALLERGEN	Dish-to-allergen relationship.	Derived from catalogue allergen field.

7.7 Dashboard Design and Implemented Views

The dashboard is designed for four audiences: general users seeking a meal, cafeteria planners selecting candidates, managers reviewing overall coverage, and governance teams examining data quality. The primary interaction pattern is progressive filtering: users begin with a practical goal and add nutrition, evidence, allergen, or ingredient constraints only when required.

The Manager Overview reports catalogue size, confirmed recipe evidence, allergen information coverage, data-source count, nutrition clusters, and summary nutrition values. The Business, Operations and Strategy Scorecard intentionally distinguishes currently available menu evidence from unavailable commercial measures such as sales, inventory, waste, supplier performance, and kitchen timings. This avoids presenting missing business data as analytics results.

The Wellness Menu Planner applies transparent AND-based rules. For example, a high-protein profile filters for protein values at or above 10 g, while a heart-conscious profile applies sodium and cholesterol ceilings. The Food Intelligence Workspace permits keyword, nutrient, recipe-evidence, declared-allergen, and ingredient-count filters. A Semantic Similar-Food Explorer presents embedding-based comparable foods for review.

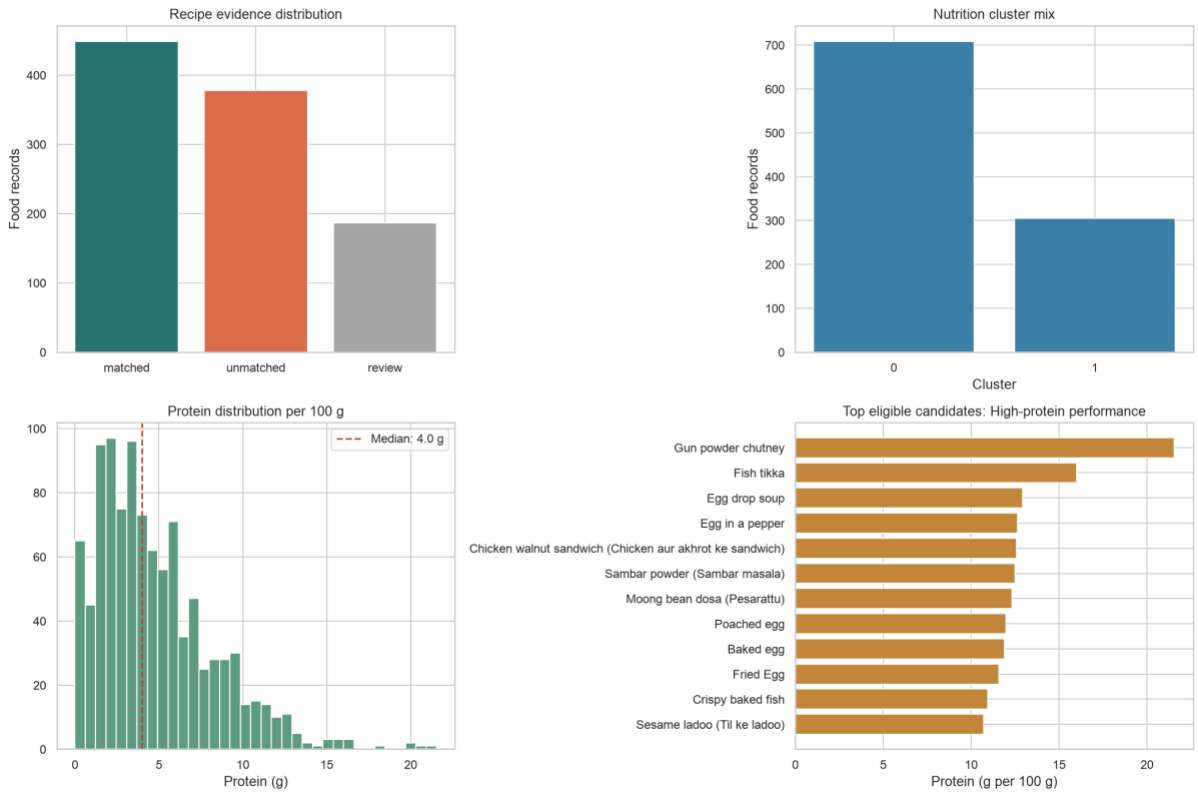


Figure 10. Reproducible dashboard KPI experiment and eligible-candidate analysis.

Phase	Activity	Key Deliverable	Business Relevance
Phase 1	Data collection, cleaning, and standardization	Cleaned master analytical dataset	Reliable foundation for analysis
Phase 2	Predictive model development	Preparation-time prediction model	Better kitchen planning
Phase 3	Semantic similarity and recipe clustering	Menu intelligence module	Improved menu planning and substitutes
Phase 4	Knowledge graph schema and ingestion	Food operations knowledge graph	Relationship-based decision support

Phase 5	Exploratory Graph RAG Integration	Conceptual / Prototype Explainable Query Framework	Auditable natural language insights
Phase 6	Dashboard creation and evaluation	Interactive analytics dashboard	Stakeholder-ready business communication
Final	Documentation and validation	Dissertation report and presentation	Academic submission package

8. Resource Requirements and Plan for their availability

Human Resources

- *Student researcher*
- *Academic supervisor*
- *Optional inputs from food operations or nutrition domain experts*

Data Resources

- *Indian Food Dataset*
- *Indian Nutrient Databank*
- *Food ingredient and nutrition reference data*
- *Curated dietary and allergen information*

Software Resources

- *Python, Pandas, NumPy, Scikit-learn*

- *Machine learning models such as XGBoost or Random Forest*
- *Sentence embedding and similarity search tools*
- *Graph database such as Neo4j*
- *Lang Chain or Graph RAG framework*
- *Streamlit or equivalent dashboard framework*
- *Visualization library such as Plotly*

Hardware Resources

- *Standard laptop or workstation*
- *Cloud-hosted database environment, if required*

Most software resources are open-source or available through free academic/developer tiers. The datasets are expected to be publicly available or academically usable. The project can be executed using a standard development environment without requiring specialized hardware.

Chapter 8: Resources and Availability

Table 11. Resource categories, availability, and handling approach.

Resource category	Required resource	Availability at mid-semester	Handling approach
Human	Student researcher and academic supervisor.	Available.	Iterative development and academic review.
Data	Nutrition, recipe, ingredient, and allergen-supporting datasets.	Available as local project data.	Use source-role and evidence-status policy.
Software	Python, Pandas, scikit-learn, Sentence Transformers, Streamlit.	Available through project environment.	Versioned dependency specification in pyproject.toml.
Vector retrieval	Qdrant or local embedding matrix.	Local artifact available; service optional.	Use local fallback during demonstrations.
Graph database	Neo4j local or managed instance.	Ingestion code ready; validated instance pending.	Complete validation in remaining work.
Hardware	Standard laptop/workstation.	Available.	Batch embedding and local artifacts limit compute requirements.

The resource strategy emphasizes reproducibility and low-cost tooling. Open-source Python libraries and locally stored artifacts allow the majority of the prototype to run without paid cloud services. Where infrastructure is unavailable, the design exposes clear fallback behavior rather than silently returning incomplete results.

9. Risks and Mitigation Plan

Risk	Impact	Likelihood	Mitigation Plan
Data quality issues such as missing values or inconsistent dish names	High	High	Apply data cleaning, imputation, standardization, and validation rules.
Inaccurate preparation-time prediction	Medium	Medium	Compare multiple models, tune parameters, validate results, and document limitations.
Ingredient name ambiguity across Indian regional foods	High	Medium	Use standardized naming, aliases, and entity resolution.
AI hallucination in nutrition or allergen responses	High	Medium	Restrict responses to graph-validated outputs and highlight source-grounded reasoning.
Knowledge graph complexity	Medium	Medium	Start with a manageable schema and expand iteratively.
Project timeline and integration complexity	Medium	Medium	Use phased development and prioritized a working minimum viable product.
Health-related interpretation risk	High	Low	Avoid medical diagnosis and position the output as decision support only.

Chapter 9: Risks, Mitigations, and Ethical Controls

Table 12. Identified risks, impact, mitigation, and contingency plans.

Risk	Potential impact	Mitigation	Contingency
Ambiguous recipe names	Incorrect ingredients or allergens could be linked.	Use confirmed, review, and unmatched evidence states.	Exclude review records from safety-sensitive use.
Missing allergen information	False confidence in food safety.	Display Not reported and source-declared wording.	Require manual recipe verification before approval.
Unit or serving inconsistencies	Misleading nutrition comparison.	Keep per-100 g and serving fields separate.	Document basis in all dashboard labels.
Semantic false similarity	Users may assume two foods are interchangeable.	Describe semantic search as discovery only.	Inspect recipe evidence and nutrition facts before decision.
Graph ingestion failure	Graph evidence may be incomplete.	Use constraints and count-validation checks.	Continue dashboard using canonical catalogue.
Scope expansion	Optional advanced features delay core deliverables.	Defer Graph RAG and prioritize evaluation.	Limit final scope to validated prototype capabilities.
Medical misuse	Users may treat dashboard output as health advice.	Use non-clinical language and visible disclaimers.	Refer users to qualified professionals for clinical concerns.

Ethically, the prototype follows the principle of calibrated claims. It reports what the dataset and implementation can support, labels what is uncertain, and avoids health, operational, or commercial conclusions that would require additional evidence. This approach is especially important because food

information affects wellbeing and can be interpreted as safety guidance.

Chapter 10: Issues Faced and Resolutions

Table 13. Issues encountered during data preparation and their resolutions.

Issue	Resolution applied	Status
Large and duplicated recipe source	Removed duplicate titles and used a controlled matching pipeline.	Completed for current catalogue.
Dish-name variation	Applied exact/alias confirmation plus separate fuzzy review candidates.	Review queue remains for manual validation.
Partial recipe coverage	Preserved unmatched nutrition records without manufactured recipe facts.	Handled through evidence-status display.
Allergen uncertainty	Used declared data and not reported labels rather than guarantees.	Implemented in data and dashboard.
Vector-service dependency	Implemented local saved-embedding fallback.	Available for local demonstration.
Graph RAG complexity and unreliable early output	Marked as optional and removed from current achievement scope.	Deferred for later evaluation.

The key technical lesson is that evidence management is more important than forcing full coverage. Preserving unmatched and review states creates a more honest system and provides a clear backlog for future data validation. A second lesson is that advanced generated interfaces should not be prioritized ahead of trustworthy data, transparent filters, and reproducible retrieval behavior.

10. Expected Benefits

Chapter 11: Preliminary Conclusions and Recommendations

11.1 Preliminary Conclusions

The mid-semester work demonstrates the feasibility of constructing an evidence-aware food intelligence catalogue from heterogeneous public sources. The prototype integrates 1,014 nutrition records with recipe evidence, ingredients where supported, allergen declarations, derived analytical features, semantic embeddings, and dashboard-ready structures.

The recipe matching results show that linkage quality must be treated as a first-class analytical variable. A confirmed match rate of 44.28% is useful but not sufficient to justify attaching recipe evidence to every food record. The additional 18.44% review queue increases potential coverage, but it should remain visibly unconfirmed until manual validation is completed.

Semantic retrieval is technically feasible for the complete catalogue, with 100% embedding coverage. Its value is in expanding food discovery beyond keyword search. However, formal relevance testing is needed before claiming that it improves menu decisions. The graph design provides a transparent structure for connected food facts, but successful Neo4j ingestion and integrity validation remain required.

The Streamlit dashboard is a meaningful prototype outcome because it turns the dataset into an inspectable decision-support surface. It makes uncertainty visible and explicitly distinguishes currently available menu analytics from unavailable sales, cost, inventory, waste, and operational data.

11.2 Recommendations for Remaining Work

- Manually validate a sample of fuzzy recipe-match candidates and publish precision estimates for the matching process.
- Train and evaluate a preparation-time or recipe-complexity model only after confirming target completeness and preventing feature leakage.
- Build a labelled semantic-query set and report Precision@5 or a comparable retrieval metric.
- Execute the Neo4j ingestion pipeline, record node and relationship counts, and run integrity queries.
- Conduct a small usability evaluation with representative dashboard users and document task completion, clarity, and perceived usefulness.

- Keep Graph RAG optional until it can be evaluated against deterministic dashboard and graph-query baselines.

Annexure A: Dataset Dictionary

This annexure documents the processed canonical catalogue used by the dashboard, embedding pipeline, and graph ingestion design. Field definitions are concise descriptions for report interpretation; the source dataset remains the authoritative technical record.

Table 14. Dataset dictionary, fields 1-24.

Field	Description
food_code	Unique identifier for the canonical food record.
food_name	Canonical dish or food name from the nutrition source.
match_status	Evidence status: matched, review, or unmatched.
match_method	Method used to link a food to a recipe record.
match_score	Similarity or confidence score for the recipe match.
allergens	Source-declared allergen information; not a clinical guarantee.
document_id	Processed catalogue field retained for analysis and traceability.
semantic_text	Text payload used to generate the semantic embedding.
embedding_status	Readiness state for semantic retrieval.
recipe_name	Linked recipe title when supporting evidence is available.
ingredients	Recipe ingredient list when a recipe link exists.
ingredient_count	Derived count of available recipe ingredients.
instructions	Recipe preparation instructions when available.
primarysource	Origin or source category of the nutrition record.
energy_kj	Energy measurement in kilojoules, generally per 100 g unless prefixed by unit_serving.

energy_kcal	Energy measurement in kilocalories, generally per 100 g unless prefixed by unit_serving.
carb_g	Nutrient amount in grams.
protein_g	Nutrient amount in grams.
fat_g	Nutrient amount in grams.
freesugar_g	Nutrient amount in grams.
fibre_g	Nutrient amount in grams.
sfa_mg	Nutrient amount in milligrams.
mufa_mg	Nutrient amount in milligrams.
pufa_mg	Nutrient amount in milligrams.

Annexure A: Dataset Dictionary (continued)

Table 15. Dataset dictionary, fields 25-48.

Field	Description
cholesterol_mg	Nutrient amount in milligrams.
calcium_mg	Nutrient amount in milligrams.
phosphorus_mg	Nutrient amount in milligrams.
magnesium_mg	Nutrient amount in milligrams.
sodium_mg	Nutrient amount in milligrams.
potassium_mg	Nutrient amount in milligrams.
iron_mg	Nutrient amount in milligrams.
copper_mg	Nutrient amount in milligrams.
selenium_ug	Nutrient amount in micrograms.
chromium_mg	Nutrient amount in milligrams.
manganese_mg	Nutrient amount in milligrams.
molybdenum_mg	Nutrient amount in milligrams.
zinc_mg	Nutrient amount in milligrams.
vita_ug	Nutrient amount in micrograms.
vite_mg	Nutrient amount in milligrams.
vitd2_ug	Nutrient amount in micrograms.
vitd3_ug	Nutrient amount in micrograms.
vitk1_ug	Nutrient amount in micrograms.
vitk2_ug	Nutrient amount in micrograms.

folate_ug	Nutrient amount in micrograms.
vitb1_mg	Nutrient amount in milligrams.
vitb2_mg	Nutrient amount in milligrams.
vitb3_mg	Nutrient amount in milligrams.
vitb5_mg	Nutrient amount in milligrams.

Annexure A: Dataset Dictionary (continued)

Table 16. Dataset dictionary, fields 49-72.

Field	Description
vitb6_mg	Nutrient amount in milligrams.
vitb7_ug	Nutrient amount in micrograms.
vitb9_ug	Nutrient amount in micrograms.
vitc_mg	Nutrient amount in milligrams.
carotenoids_ug	Nutrient amount in micrograms.
servings_unit	Processed catalogue field retained for analysis and traceability.
unit_serving_energy_kj	Nutrient value expressed for the reported serving unit.
unit_serving_energy_kcal	Nutrient value expressed for the reported serving unit.
unit_serving_carb_g	Nutrient value expressed for the reported serving unit.
unit_serving_protein_g	Nutrient value expressed for the reported serving unit.
unit_serving_fat_g	Nutrient value expressed for the reported serving unit.
unit_serving_freesugar_g	Nutrient value expressed for the reported serving unit.
unit_serving_fibre_g	Nutrient value expressed for the reported serving unit.
unit_serving_sfa_mg	Nutrient value expressed for the reported serving unit.
unit_serving_mufa_mg	Nutrient value expressed for the reported serving unit.
unit_serving_pufa_mg	Nutrient value expressed for the reported serving unit.
unit_serving_cholesterol_mg	Nutrient value expressed for the reported serving unit.
unit_serving_calcium_mg	Nutrient value expressed for the reported serving unit.
unit_serving_phosphorus_mg	Nutrient value expressed for the reported serving unit.

unit_serving_magnesium_mg	Nutrient value expressed for the reported serving unit.
unit_serving_sodium_mg	Nutrient value expressed for the reported serving unit.
unit_serving_potassium_mg	Nutrient value expressed for the reported serving unit.
unit_serving_iron_mg	Nutrient value expressed for the reported serving unit.
unit_serving_copper_mg	Nutrient value expressed for the reported serving unit.

Annexure A: Dataset Dictionary (continued)

Table 17. Dataset dictionary, fields 73-96.

Field	Description
unit_serving_selenium_ug	Nutrient value expressed for the reported serving unit.
unit_serving_chromium_mg	Nutrient value expressed for the reported serving unit.
unit_serving_manganese_mg	Nutrient value expressed for the reported serving unit.
unit_serving_molybdenum_mg	Nutrient value expressed for the reported serving unit.
unit_serving_zinc_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vita_ug	Nutrient value expressed for the reported serving unit.
unit_serving_vite_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vitd2_ug	Nutrient value expressed for the reported serving unit.
unit_serving_vitd3_ug	Nutrient value expressed for the reported serving unit.
unit_serving_vitk1_ug	Nutrient value expressed for the reported serving unit.
unit_serving_vitk2_ug	Nutrient value expressed for the reported serving unit.
unit_serving_folate_ug	Nutrient value expressed for the reported serving unit.
unit_serving_vitb1_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vitb2_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vitb3_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vitb5_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vitb6_mg	Nutrient value expressed for the reported serving unit.
unit_serving_vitb7_ug	Nutrient value expressed for the reported serving unit.
unit_serving_vitb9_ug	Nutrient value expressed for the reported serving unit.

unit_serving_vitc_mg	Nutrient value expressed for the reported serving unit.
unit_serving_carotenoids_ug	Nutrient value expressed for the reported serving unit.
protein_to_carb_ratio	Derived protein-to-carbohydrate ratio.
fat_to_carb_ratio	Derived fat-to-carbohydrate ratio.
calorie_to_protein_ratio	Derived energy-to-protein ratio.

Annexure A: Dataset Dictionary (continued)

Table 18. Dataset dictionary, fields 97-97.

Field	Description
nutrition_cluster	Unsupervised nutrition-profile segment for descriptive analysis.

Annexure B: Reproducible KPI Snapshot

Table 19. Reproducible KPI snapshot computed directly from project artifacts.

Metric	Value
Catalogue dishes	1014
Unique food codes	1014
Confirmed recipe matches	449
Confirmed match rate	44.28%
Review candidates	187
Review or confirmed coverage	62.72%
Unmatched records	378
Allergen declaration coverage	59.07%
Semantic document coverage	100.00%
Records with ingredient counts	636
Data-source categories	3
Raw recipe rows examined	2,231,142

These values are derived from local project artifacts at report-generation time. They support reproducibility because the report can be regenerated after any validated change to the catalogue or matching pipeline.

Annexure C: Selected Implementation Excerpts

The following excerpts are included to document the implemented analytical logic. Full source code is retained in the project repository. Code excerpts are provided as technical evidence and should be read together with the data-quality and scope limitations described in the main report.

C.1 Evidence-aware menu filtering

```
def apply_menu_filters(catalog, profile, sources=None, statuses=None, allergens_to_avoid=None):
    candidates = catalog.copy()
    if sources:
        candidates = candidates[candidates["primarysource"].isin(sources)]
    if statuses:
        candidates = candidates[candidates["match_status"].isin(statuses)]
    for allergen in allergens_to_avoid or []:
        candidates = candidates[~candidates["allergens"].str.contains(
            allergen, case=False, na=False, regex=False
        )]
    for field, (minimum, maximum) in WELLNESS_GOALS[profile].items():
        if minimum is not None:
            candidates = candidates[candidates[field].ge(minimum)]
        if maximum is not None:
            candidates = candidates[candidates[field].le(maximum)]
    return candidates
```

C.2 Neo4j schema constraints

```
statements = [
    "CREATE CONSTRAINT dish_food_code IF NOT EXISTS "
    "FOR (dish:Dish) REQUIRE dish.food_code IS UNIQUE",
    "CREATE CONSTRAINT ingredient_name IF NOT EXISTS "
    "FOR (ingredient:Ingredient) REQUIRE ingredient.name IS UNIQUE",
    "CREATE CONSTRAINT nutrient_name IF NOT EXISTS "
    "FOR (nutrient:Nutrient) REQUIRE nutrient.name IS UNIQUE",
    "CREATE CONSTRAINT allergen_name IF NOT EXISTS "
    "FOR (allergen:Allergen) REQUIRE allergen.name IS UNIQUE",
]
```

C.3 Graph nutrient relationship ingestion

```
UNWIND $rows AS row
MATCH (dish:Dish {food_code: row.food_code})
MERGE (nutrient:Nutrient {name: row.name})
MERGE (dish)-[measurement:HAS_NUTRIENT]->(nutrient)
SET measurement.amount = row.amount, measurement.unit = row.unit
```

C.4 Local embedding retrieval fallback

```
catalog = pd.read_csv(DATA_PATH).dropna(subset=["semantic_text"]).reset_index(drop=True)
embeddings = np.load(EMBEDDINGS_PATH)
if len(catalog) != len(embeddings):
    raise ValueError("Catalog rows and saved embedding rows do not match.")
scores = embeddings @ query_vector
indices = np.argsort(scores)[-top_k:][::-1]
return [{"food_code": catalog.iloc[index]["food_code"],
         "score": round(float(scores[index]), 4)} for index in indices]
```

Annexure D: References Added for the Expanded Report

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Annexure E: Summary of How Project Outline Feedback Was Addressed

Table 20. Mapping of project outline feedback to actions taken in this report.

Outline feedback / requirement	Action taken in the mid-semester report
Keep the work aligned to objectives	Each implemented component is mapped to an approved objective and its current evidence level.
Use an academic-project methodology	Added research design, research questions, evaluation plan, literature review, and research gap.
Show actual progress	Added reproducible counts for catalogue size, matching outcomes, embedding coverage, and dashboard scope.
Explain scope and limitations	Added explicit in-scope, out-of-scope, assumptions, constraints, risks, and ethical controls.
Provide detailed work completed	Documented data preparation, evidence policy, feature engineering, semantic retrieval, graph design, and dashboard implementation.
Avoid unsupported claims	Deferred Graph RAG, operational savings, clinical guidance, and unvalidated model-performance assertions.

The report has been expanded using project-specific evidence and original explanatory writing. It should be reviewed by the student and faculty guide before submission, particularly for institutional formatting, certificate wording, final page numbering, and confirmation that all claims match the latest validated artifacts.

- **Operational Benefits:** Improved preparation-time forecasting, workload planning, service scheduling, and reduction of peak-hour bottlenecks**.**

- **Cost and Supply Chain Benefits:** Better visibility into ingredient dependencies, high-frequency raw materials, procurement risk, and menu redundancy.
- **Employee Wellness Benefits:** Improved nutritional transparency, allergen awareness, dietary preference support, and healthier menu decision-making.
- **Decision Support Benefits:** Conversion of fragmented food and nutrition data into structured, explainable, and manager-friendly insights.
- **Business Analytics Benefits:** Practical application of predictive modeling, semantic analytics, knowledge graph modeling, dashboarding, and stakeholder communication.
- **Strategic Benefits:** Prototype for analytics-led transformation in enterprise food services and employee-centric workplace systems.

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Glossary

Term	***Meaning***
Business Analytics	*Use of data, statistical methods, and analytical models to support business decision-making.*
Decision Support System	*A system that helps managers make informed decisions using data, models, and insights.*
Machine Learning	*Algorithms that learn patterns from historical data and make predictions.*
Predictive Analytics	*Analytics technique used to forecast likely future outcomes.*
Knowledge Graph	*Structured representation of entities and relationships.*
Graph RAG	*Retrieval-Augmented Generation using graph-based retrieval and reasoning.*
Semantic Search	*Search method that identifies meaning and context rather than only matching keywords.*

Vector Embedding	*Numerical representation of text or entities used for similarity comparison.*
Allergen	*A substance that may trigger allergic reactions in sensitive individuals.*
Supply Chain Dependency	*Reliance of multiple products or operations on common raw materials.*
Employee Wellness Analytics	*Use of data and analytics to support workplace health and well-being programs.*
Dashboard	*Visual interface that presents key metrics and insights for business users.*

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

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Programme Name	MBA in Business Analytics
Semester	Final semester
Project Title	**Analytics-Driven Decision Support Framework for Enterprise Food Operations and Employee Health Management**

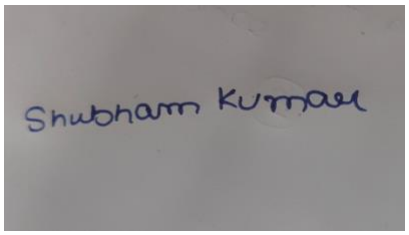
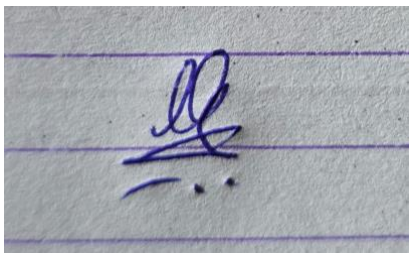
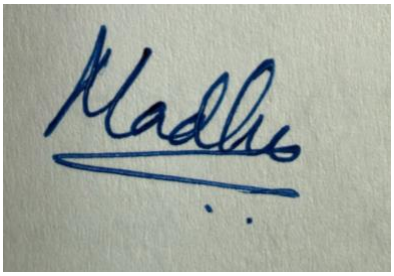
Particulars of the Supervisor and Additional Examiner

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Remarks of the Supervisor on Project Report

The proposed project addresses a relevant business analytics problem and demonstrates appropriate application of data analytics, machine learning, and decision-support concepts. The scope, methodology, and expected outcomes are suitable for an MBA in Business Analytics dissertation and provide potential value in the areas of food operations, employee wellness, and business intelligence.

		
Signature of Student	**Signature of Supervisor**	**Signature of Additional Examiner**
Name: Shubham Kumar	**Name: Mrigank Sharma**	**Name: Madhu Khiani**

PART II

FINAL REPORT

Analytics-Driven Decision Support Framework for Enterprise Food Operations and Employee Health Management

From a mid-semester prototype to an evidenced, governed decision-support service

Chapters 12 to 24 and Annexures F to K | November 2026

Part II block	Question answered	Main evidence
Chapters 12-13	What was achieved after mid-semester, and how can the claims be checked?	Verified run of 8 November 2026
Chapters 14-16	Why does the service matter, how is it built, and how is it operated?	Source code, API contract, executed UI walk-through
Chapters 17-18	What does the analytics do, and does it stand up to testing?	172 executed checks, catalogue profile, forecasting study
Chapters 19-21	How are behaviour, ethics, value and KPIs governed?	Management frameworks applied to measured facts
Chapters 22-24	What should management do next, and what are the limits?	Pilot design, risk register, conclusions
Annexures F-K	Evidence register, API catalogue, logic excerpts, demonstration script, references, glossary	Traceable to artefacts

How Part II relates to Part I

Part I (Chapters 1-11 and Annexures A-E) is reproduced exactly as submitted at mid-semester. Part II continues its numbering (chapters, tables, figures) and adds the work completed afterwards: integration into the Enterprise Cafeteria Operating System (ECOS), a verification run, quantitative profiling of the food catalogue, a forecasting study, a financial and governance appraisal, and a pilot roadmap.

Chapter 12: Final-Stage Executive Summary

12.1 Purpose of Part II

Part I recorded the project as it stood at mid-semester: a processed catalogue of 1,014 foods, semantic embeddings and a first dashboard, with the knowledge graph and Graph RAG still ahead. Part II records what the project became. The prototype was integrated into a role-based service, the **Enterprise Cafeteria Operating System (ECOS)**, and was then examined as a manager would examine any proposed operating change: does it work, what does it rely on, what can it not yet claim, what would it cost, and what should be done next?

The report keeps one discipline throughout. Every statement is tied to an evidence class (Chapter 13): it was **executed** on 8 November 2026, **code-inspected**, **documented**, **user-reported**, **proposed**, or is an **illustrative assumption**. The sections below summarise the answers; the later chapters give the evidence.

12.2 What changed since mid-semester

Table 21 compares the six phases of the Part I plan with their final status, and Figure 11 shows the same comparison graphically. Four phases were delivered or completed, one was re-scoped from preparation-time modelling to demand forecasting, and two ambitions in the approved outline, a property-graph knowledge base and Graph RAG, were not delivered. Declaring that openly is part of the contribution: the assessed artefact is a working decision-support service, not a research graph platform.

Table 21. Part I plan phases and their final status

Plan phase	At mid-semester	Final status (8 Nov 2026)	Where
1. Data and preparation	Catalogue of 1,014 records, 97 analytical fields	Completed and now consumed by the application; profiled in Chapter 15	Chapter 15
2. Predictive analytics	Planned as preparation-time and complexity models	Re-scoped to demand forecasting: Random Forest against a same-weekday baseline with a serving gate	Chapter 17
3. Semantic search	Embeddings ready	Delivered: 256-dimension cosine index in Qdrant; similarity contributes up to +0.30 to the recommendation score	Chapter 17
4. Knowledge graph	Schema designed	Design only: relationships are document references in MongoDB and Qdrant payloads, not a graph database	Chapter 15, Chapter 24
5. Graph RAG (optional)	Excluded from claims	Not delivered; the copilot answers from eight rule-based intents with optional LLM wording	Chapter 17
6. Dashboard	First dashboard and KPI experiment	Delivered as ECOS: 5 employee and 7 manager views over 42 API operations	Chapter 16

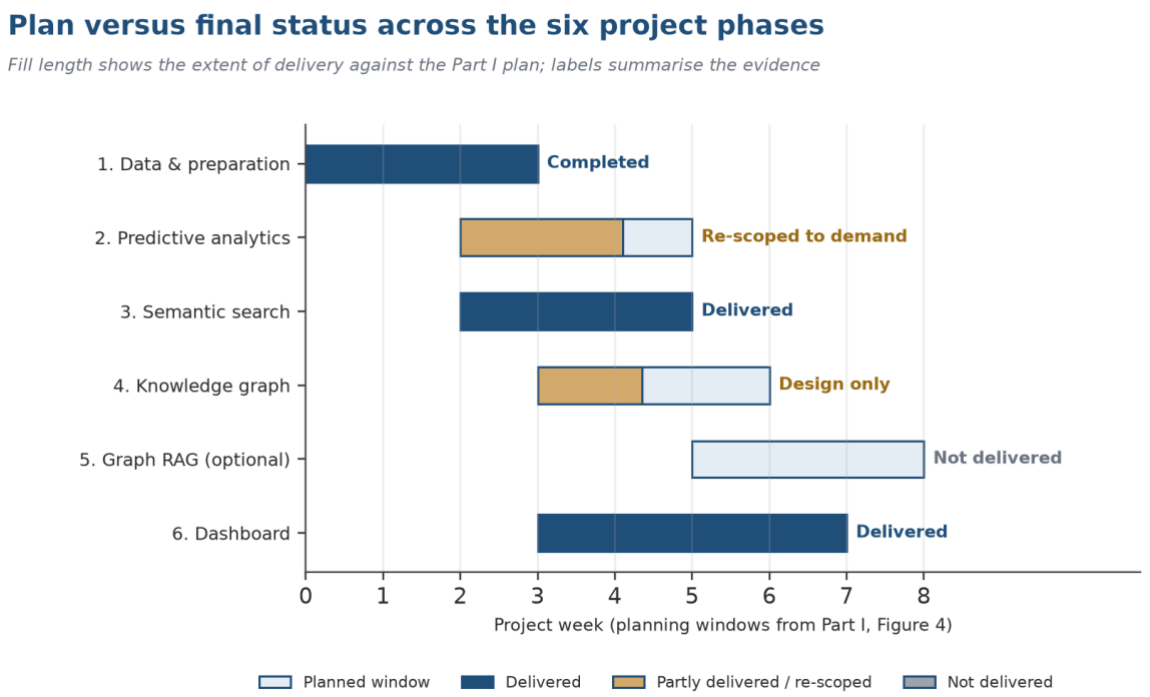


Figure 11. Plan versus final status across the six project phases (planning windows from Part I, Figure 4).

12.3 ECOS at a glance

Table 22. ECOS at a glance, with the evidence class of each fact

Dimension	Fact	Evidence class
Application surface	Two role workspaces: five employee pages and seven manager pages (listed in Chapter 16)	Executed
Service interface	14 routers, 42 operations on 39 paths under `/api/v1`	Code-inspected
Data stores	MongoDB, 9 collections, synchronous PyMongo; Qdrant `food_nutrients` with 1,014 vectors (256 dimensions, cosine)	Executed
Ordering controls	Idempotent checkout, conditional stock decrement with rollback, at most 10 per dish and 15 lines, lifecycle confirmed, preparing, ready, collected	Executed
Recommendation	Hard filters (availability, diet, allergen, intent, budget, energy) followed by an additive score; match percentage capped at 99	Executed
Forecasting	Random Forest (200 trees, depth 8) against a same-weekday-4 baseline, chronological 80/20 split, served only if MAE is not above baseline	Executed
Verification	138 backend and 34 user-interface checks, all passed	Executed
Forecast study	Customers: MAE 2.231 against baseline 1.641 (gate fails). Dishes: 0.792 against 0.793 (gate passes by 0.001)	Executed, synthetic data
Finance	Base case NPV +7.89 lakh, IRR 44.6%; downside NPV -4.79 lakh	Illustrative assumption

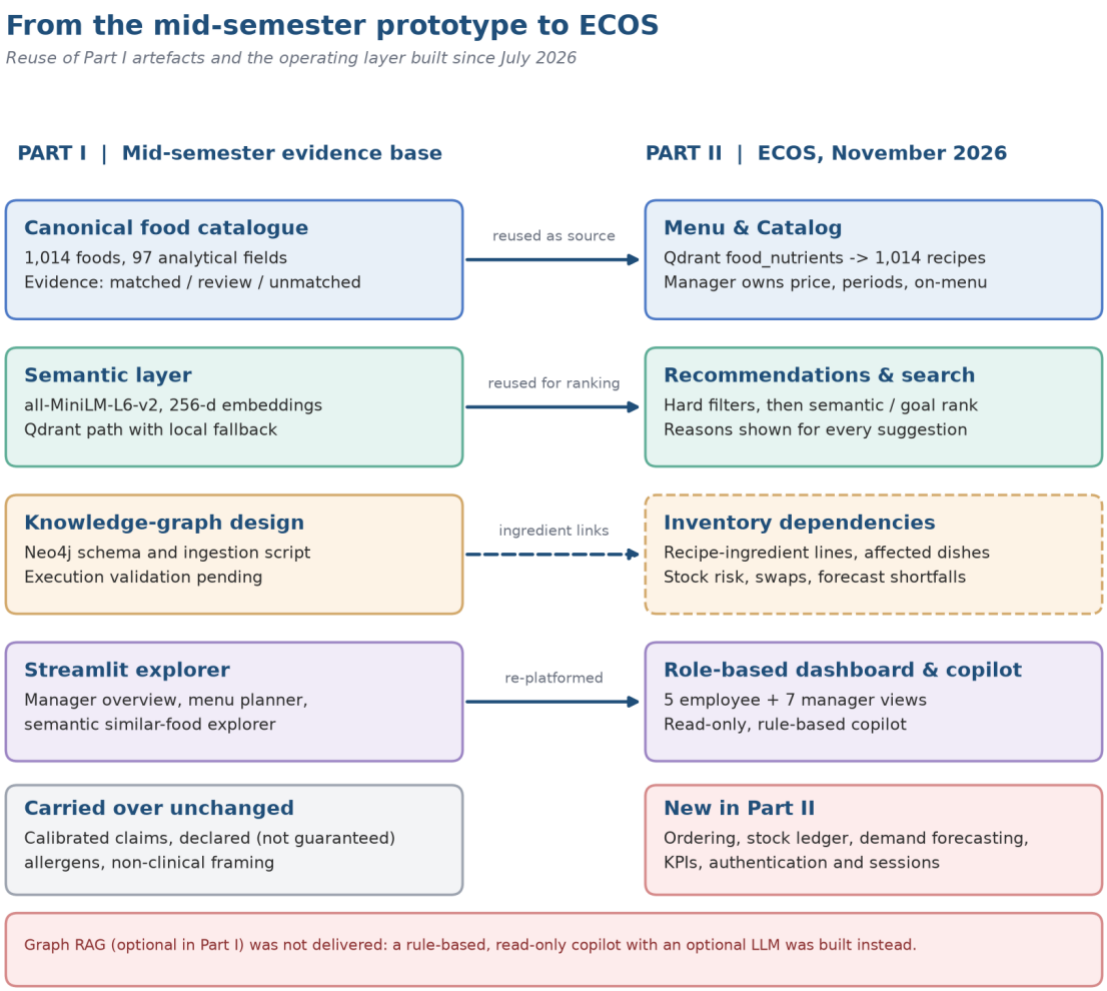


Figure 12. From the mid-semester catalogue to the ECOS operating service.

12.4 Management decision brief

Two questions need an answer. Is the prototype fit to serve as a structured demonstration and learning aid? And is further evidence-gathering worth sponsoring? To the first, the evidence says yes: stakeholders can inspect an integrated operating concept through the interface, and Chapter 18 shows it works as designed on synthetic data. A firm business case, a service-level promise, or confidence that its suggestions outperform current practice would go beyond what has been shown.

A staged approval is sensible. Begin by keeping the demonstration on synthetic or approved data, labelled as such. Before going live, settle the site, the evaluation window, the people responsible, a data-quality checklist, privacy limits and the conditions for stopping. Collect before-and-after measures with identical

definitions, and treat any change seen during a pilot as an association unless a sound comparison exists. Finish with a go/no-go meeting that includes operations, finance, purchasing, IT, privacy and staff representatives.

Three principles should bind any pilot. For **safety**, a named manager confirms stock, menu changes, reorders and any action based on a forecast; the software never places supplier orders on its own. For **information**, every menu, allergen and nutrition statement has an accountable source and a review date, and wellness indicators make no medical claim. For **money**, demonstration order values are not audited revenue, the mock payment collects nothing, and savings remain scenarios until procurement and finance confirm them.

> **Recommendation**
>
> **Advance to a governed single-site pilot; do not scale.** The prototype is sound enough to learn from and not yet sound enough to rely on. Five conditions should be met first (findings B1 to B4 and B7 in Chapter 18): correct the wellness score and its caption (B1), carry the Part I evidence status into the application (B2), harden the forecast serving gate (B3), define a stricter healthy-dish rule (B4), and limit managers to aggregate wellness data (B7).
>
> The strongest positive finding is the integrity of the ordering and stock path; the clearest warning is that a forecast gate that passes on a coin-flip is not yet a control.

Chapter 13: Research Method and Evidence Base

13.1 Research design

The project follows a design-science logic (Hevner, March, Park and Ram, 2004; Peffers et al., 2007): an artefact is built to address a managerial problem and then evaluated against explicit criteria. Because the artefact is a prototype running on synthetic operational data without live users, the evaluation is a **demonstration with a critical audit**, not a field experiment. The MBA lens enters through the questions put to the artefact: strategic fit (Porter, 1996), service design and operations (Shostack, 1984; Parasuraman, Zeithaml and Berry, 1985), analytics governance (Davenport and Harris, 2007; Sculley et al., 2015), financial appraisal (net present value, internal rate of return, payback) and behaviour and ethics (Thaler and Sunstein, 2008; Davis, 1989).

The four research questions and hypotheses of Part I (Section 5.3 and Section 5.5) are carried forward unchanged and answered in Chapter 24.

13.2 Evidence classes

Part I used a three-state match status for catalogue records. Part II adds a classification of the **claims themselves**, so that a reader can see what each statement rests on (Table 23).

Table 23. Evidence classes used in Part II and the inferences they permit

Class	Basis	May be stated	May not be inferred
Executed	A test, run or measurement performed on 8 November 2026 on the environment in Table 24	The result of that run	Behaviour in production, with real users or on other data
Code-inspected	A named source file or function was read	A rule, threshold or workflow exists in code	That it ran correctly unless it is also executed
Documented	README, API documentation, Part I	Design intent and published contract	That the running system still matches
User-reported	Statement by the project owner	The statement, attributed	Independent verification
Proposed	Management design in this report	A recommended practice or gate	Current capability
Illustrative assumption	Hypothetical figures	How a formula behaves	A forecast, saving or return

13.3 Verification environment and data

All executed evidence was produced on one laptop in isolated, throwaway databases created for this report. The owner's own databases, containers and credentials were not read or modified. The food catalogue is **real** (the 1,014-record Qdrant collection built in Part I). Users, orders, stock and history are **synthetic demonstration data**: 30 users, eight weeks of history and 920 orders, all flagged `source = demo_seed`. Figures derived from them describe the behaviour of the software, not of any cafeteria.

Table 24. Verification environment

Component	Version or setting
Machine	Apple M3 Pro, macOS 26.7
Language and libraries	Python 3.12.10; FastAPI 0.162.0; Uvicorn 0.52.4; Streamlit 1.62.0; Pydantic 2.13.4; pandas 3.0.5; NumPy 2.5.2; scikit-learn 1.9.0
Data services	MongoDB 8.3.4 (private instance, three throwaway databases); PyMongo 4.16.0; Qdrant client 1.19.0 against the local `food_nutrients` collection
Test harnesses	Backend smoke test (`backend/smoke_test.py`); user-interface test with Streamlit AppTest (`dashboard/ui_smoke_test.py`)
Data	Catalogue: 1,014 real records. Operations: synthetic, 30 users, 920 orders, 14 September to 6 November 2026
Screenshots	Captured from the running application with a headless browser (Part II, Chapter 16)

13.4 Limits of the evidence

- **No users were studied.** Usability, adoption and behaviour change remain hypotheses; no survey or interview data were collected.
- **Operational data are synthetic.** Forecast errors, order counts and revenue-like figures demonstrate mechanics, not performance.
- **One run, one machine.** The checks establish that the software behaves as designed in this environment, not that it is robust under load or failure.
- **Financial figures are hypothetical.** They illustrate method; none is derived from cafeteria records.

Chapter 14: Business Context, Stakeholders and Strategic Fit

14.1 The operating problem and value proposition

Think of a cafeteria as a compact production system whose parts depend on one another. Changing the menu changes what must be bought; what is in stock limits what can be substituted; expected attendance and habits set how much to cook; those decisions decide queues and leftovers; and the accuracy of nutrition and allergen labels decides whether diners trust the counter. If the relevant records sit in separate sheets, sales files, recipe documents and chat threads, someone must stitch them together by hand, and the cost shows up as repeated effort, exceptions found too late, mismatched menu descriptions and decisions that are hard to justify afterwards.

This report therefore treats the issue as one of coordination and decision quality rather than assuming a service has failed. Nothing has been measured at a real site for waste, stockouts, diner dissatisfaction, lost revenue or preparation delay. The working proposition is that one shared, role-appropriate view of the facts could make problems visible sooner and decisions easier to retrace. Evidence from a real site is still needed before any benefit is asserted.

Value arises on both sides of the counter. Diners get quick access to the menu, their own order history and declared dietary details, with suggestions they may ignore. Managers get orders, stock, menu operations, past behaviour and planning signals in one place for their reviews. The aim is not to showcase AI but to cut avoidable handoffs, expose exceptions earlier and tighten follow-up, on condition that the underlying data and controls are dependable.

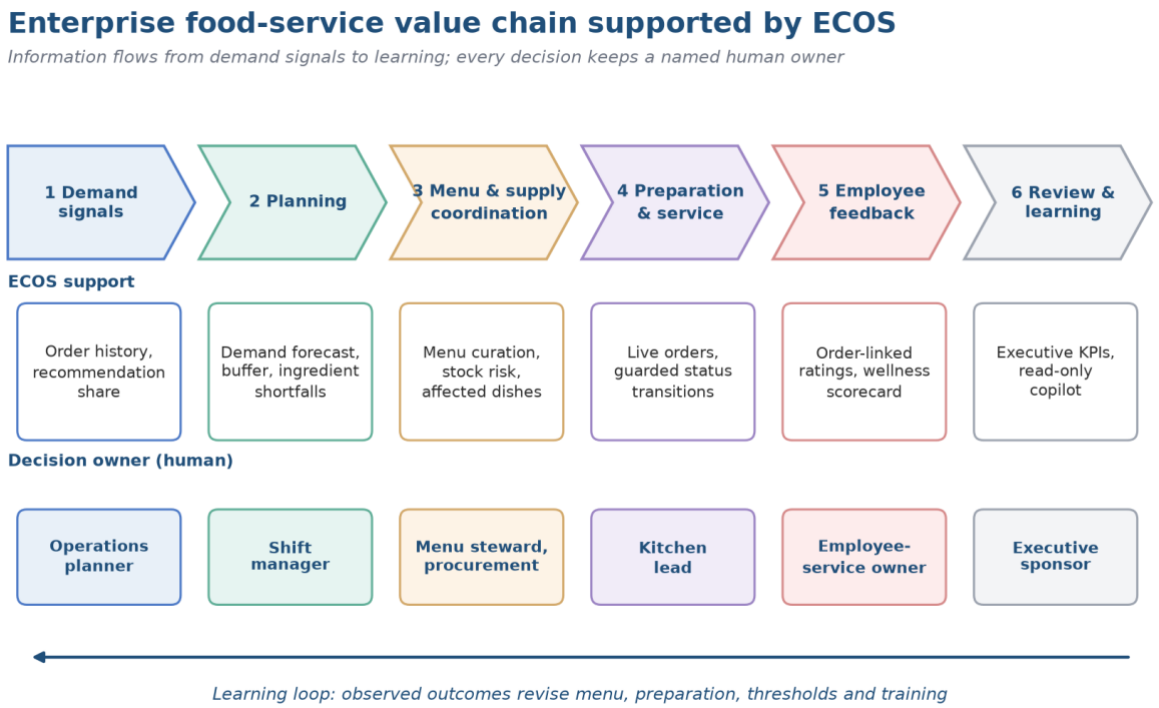


Figure 13. Enterprise food-service value chain and the ECOS layers that support it.

14.2 Stakeholders and their influence

Different groups want different things from the platform. Diners ask for easy choice, dependable availability, accurate descriptions, clear prices and respectful treatment of dietary data. Supervisors want an up-to-date picture and the power to fix wrong data. Menu planners trade off variety, nutrition, cost, ingredients and kitchen capacity. Buyers and suppliers need stable specifications and lead times, finance needs costs that reconcile, and IT and governance staff need access control, supportability and privacy protection. Staff representatives want a guarantee that wellness indicators will not be turned into surveillance or a basis for discrimination.

Engagement works best when tied to decisions. For each group, name the question being asked, the evidence offered and the decision the group may influence. Diners can test menu clarity; operators can check that stock and order views suit a real shift; purchasing can say how a restock hint would turn into a request; finance can settle cost definitions; privacy and HR can draw the boundary around wellness data; IT can state what service and recovery it needs.

Table 25. Stakeholders, needs and decision rights

Stakeholder	Engagement evidence	Decision right to clarify	Primary need
Employee	Usability session (none held yet)	Consent, preferences, how feedback is given	Reliable choice and trustworthy menu facts
Shift manager	Observed walk-through of a shift task	Whether to hold, amend or reject a suggestion	Quick handling of exceptions
Menu owner	Menu change log	Publishing and recipe versions	A feasible menu with safe substitutes
Procurement	Requisition and approval record	Releasing any supplier-facing order	Specifications and timing it can act on
Finance	Reconciliation and sign-off	Accounting definitions	Cost and benefit logic that can be audited
IT and privacy	Readiness review	Go-live and data access	A secure service that can be supported
Employee representative	Governance meeting notes	Raising objections to wellness practices	Fair and non-coercive use

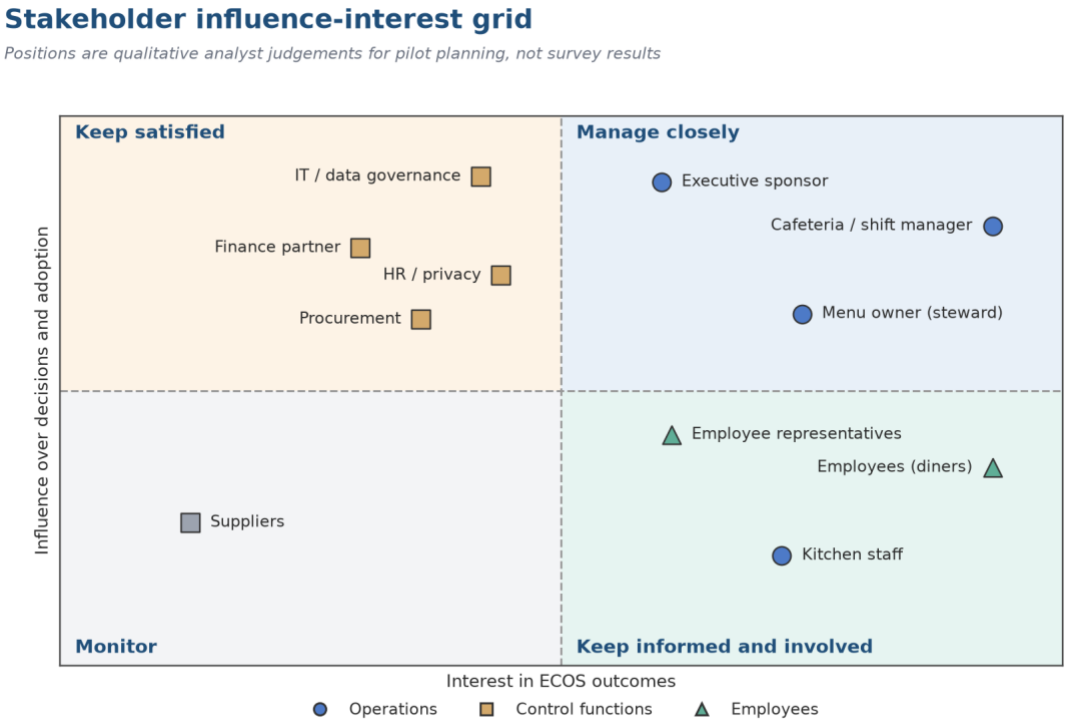


Figure 14. Stakeholder map: influence on and interest in the ECOS service.

14.3 Strategic fit and positioning

A strategy is a set of deliberate choices with costs attached, not a feature list. The sensible place for ECOS is as a decision aid that links what diners see with what managers run. Presenting it as a substitute for

purchasing systems, as health care, or as an automatic production planner would overreach. Keeping to the narrower role leaves one answerable question: does shared, role-specific information improve coordination and learning at an acceptable level of privacy, operational and financial risk?

Porter separates operational effectiveness from strategy, and the distinction matters here. A new interface may speed up what already happens, yet durable advantage needs a distinctive, mutually reinforcing set of activities that suit the organisation. For a cafeteria that set includes curated menu data, accessible choice, timely order visibility, regular stock review, human planning informed by forecasts, and a standing performance review. A gap in any one, such as stale stock records or an unowned menu attribute, leaves the dashboard exposing a weak process rather than repairing it.

Put as a hypothesis: when managers get timely, reasonably complete information along with clear decision rights, their calls on demand, stock and menu become more consistent; when diners see accurate information and keep control of their choice, trust and convenience may improve. Each link has to be measured. Having the technology does not mean it is used, and being used does not mean outcomes improve.

Table 26. Strategic choices and their costs

Choice	Trade-off to manage	Benefit sought
Support decisions rather than automate them	Needs manager time and training	Managers stay answerable
Keep recommendations optional	Click-through may stay modest, which is acceptable	Employee autonomy
Release in phases	Some handoffs stay manual	Controls stay manageable
Publish food attributes openly	More stewardship effort; uncertainty must be disclosed	Informed choice
Scale only on evidence	Visible growth is slower	Limits early commitment

14.4 Objectives and scope

Objectives should distinguish deliverables, operational outcomes and long-term impacts (Table 27). A screen or an endpoint is a deliverable; a shorter reconciliation delay is an outcome; better wellbeing is an impact that needs a stronger design and a longer horizon. Part II claims deliverables only, and states the evidence that would be needed for the other two levels.

Table 27. Levels of objective for the next evaluation stage

Level	Evidence needed	Example objective
Deliverable	Versioned build with acceptance criteria	Role-specific menu, order, stock and planning views exist
Process outcome	Timestamped queue and disposition records	Exceptions reach the responsible manager
Service outcome	Defined denominator and baseline	More orders completed; items available
Resource outcome	Physical counts and measured waste	Better stock planning, fewer avoidable leftovers
Employee outcome	Privacy-safe, non-coercive measurement	Informed, voluntary choice

The final scope is narrower than the approved vision and broader than the mid-semester prototype (Table 28). Two items from the approved outline remain outside it: a property-graph knowledge base and Graph RAG.

Table 28. Final scope boundary

Inside the delivered and verified scope	Outside (not assumed delivered)
Role-based employee and manager views with 42 API operations	Ten independently deployable production services
Ordering with idempotency, stock control and a four-state lifecycle	Payment settlement; revenue or cash collection
Inventory edit with validation before commit, movement history and risk view	Supplier master, purchase orders, supplier transmission
Forecast pipeline, evaluation and serving gate on synthetic history	A validated, continuously monitored production forecast
Rule-based, read-only copilot with optional LLM wording	Autonomous actions; general-purpose generative assistant
Semantic ranking over the 1,014-record catalogue	Property-graph knowledge base; Graph RAG

Chapter 15: System Architecture and Data Foundation

15.1 Architecture in one view

ECOS has three tiers. A Streamlit dashboard renders the employee and manager workspaces. A FastAPI service exposes 42 operations under `/api/v1`, authenticated with a session token in the `X-Session-Token` header. State lives in MongoDB (nine collections, synchronous PyMongo). The food catalogue is read from a Qdrant collection and synchronised into MongoDB, and a local sentence-embedding model (`all-MiniLM-L6-v2`, truncated to 256 dimensions) ranks dishes semantically when it is ready. An optional language model may rephrase copilot answers; nothing depends on it. The root `app.py` starts the API on the loopback interface, waits for its health check and then launches the dashboard. Figure 15 shows the components and their boundaries.

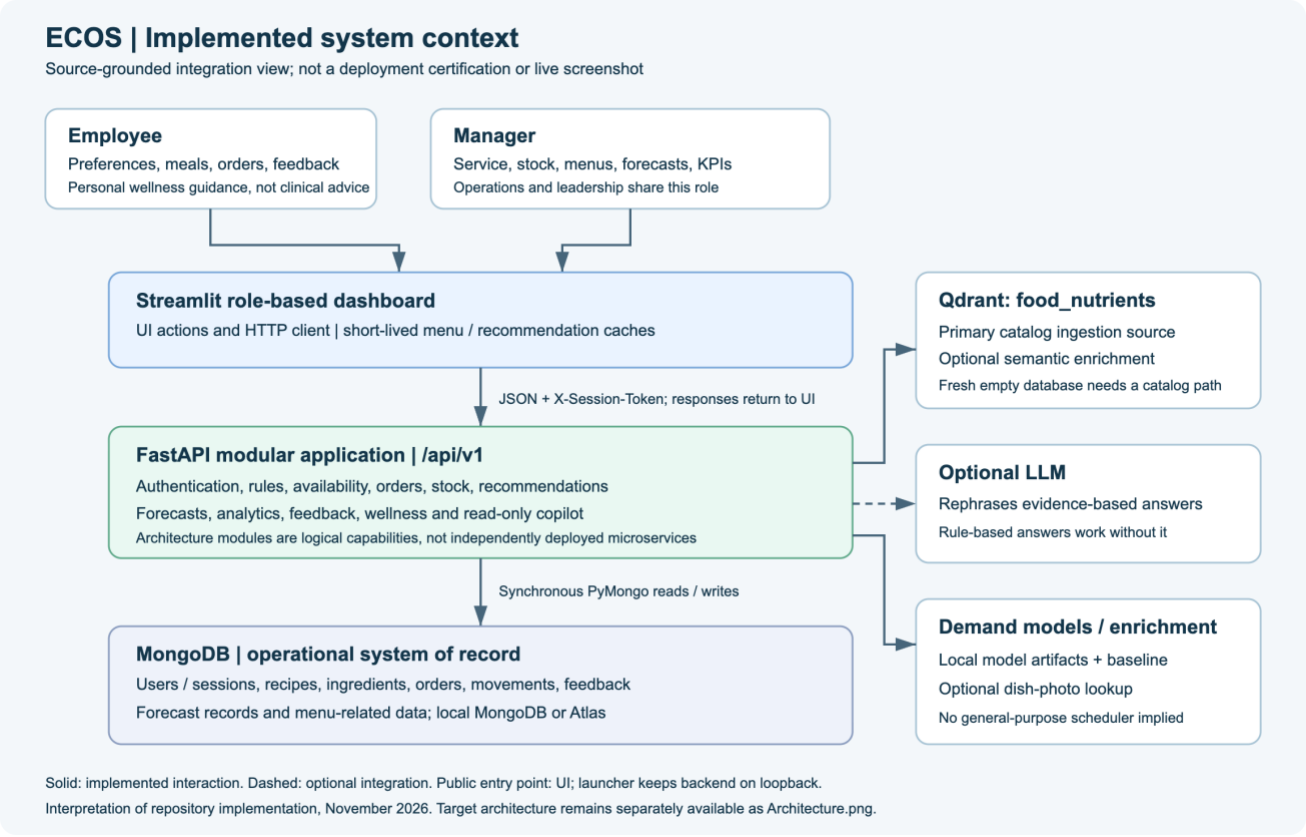


Figure 15. System context of ECOS as implemented (derived from the project asset docs/assets/system-context.svg).

15.2 Target architecture and what was built

The architecture poster of the project (Figure 16) is the target picture: ten backend modules, four user groups, an asynchronous MongoDB Atlas layer, scheduled background jobs and three business outcomes. It remains a valid communication artefact, but several of its elements are targets rather than features. Table 29 reconciles the poster with the verified implementation so that the picture is not read as a statement of capability.

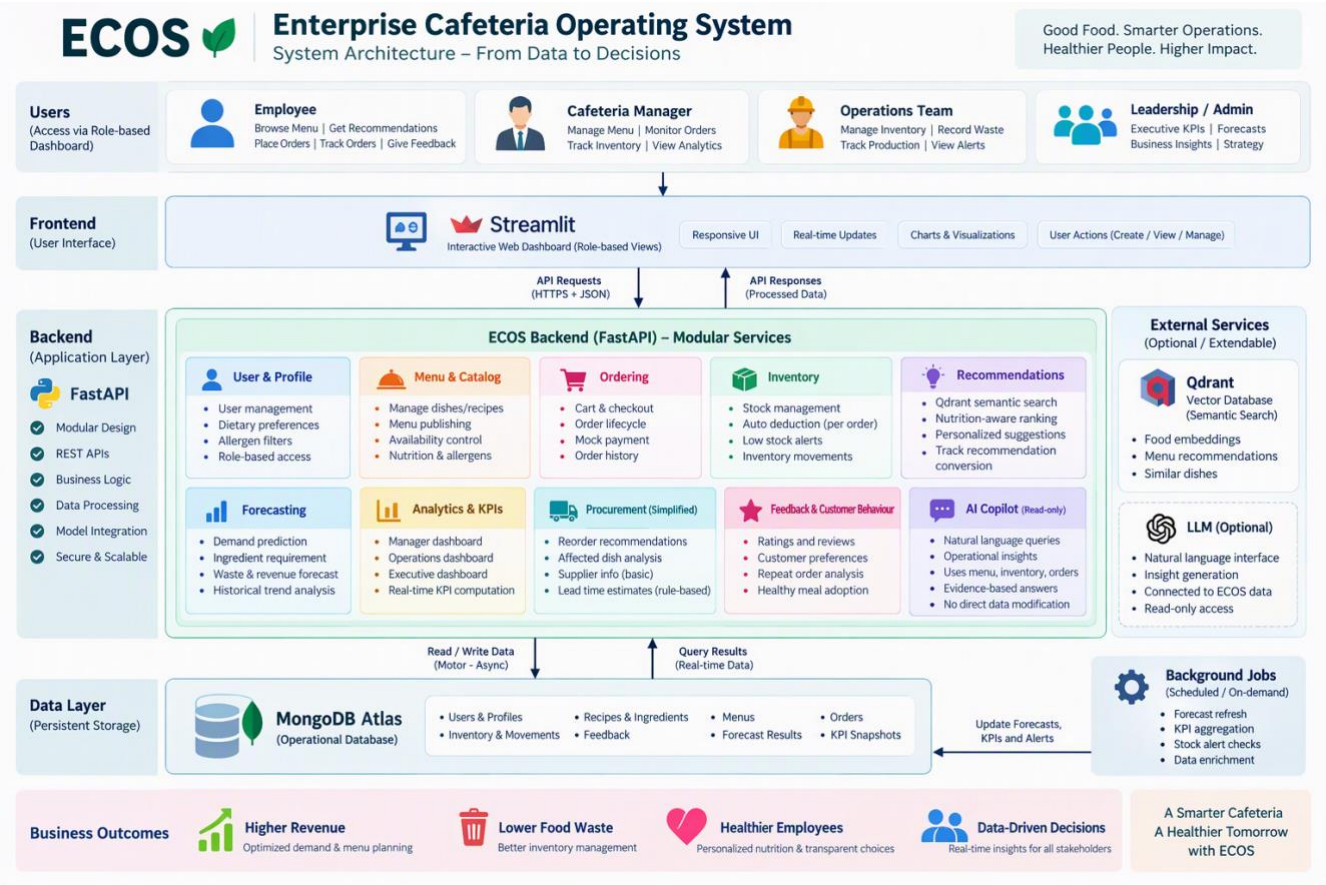


Figure 16. Target architecture poster for ECOS (existing project artefact; reconciled with the implementation in the table below).

Table 29. Target architecture compared with the verified implementation

Poster element	As built and verified	Management consequence
Four user groups (employee, manager, operations team, leadership)	Two roles, employee and manager; operations and executive views are manager pages	Do not imply four role types; finer role separation is future work

Poster element	As built and verified	Management consequence
Streamlit with real-time updates	Streamlit re-runs on interaction; periodic refresh only for the catalogue panel (10 s) and a reconnect banner (5 s)	No push channel; managers see state as of their last action or refresh
Ten backend modules	14 routers; procurement exists only as replenishment suggestions and affected-dish analysis inside inventory	No supplier master or purchase orders; keep human approval
MongoDB Atlas with Motor (async)	MongoDB with synchronous PyMongo; no Atlas dependency in the verified run	Review concurrency and connection pooling before load
Menus and KPI snapshots as stored data	No such collections: the menu is a flag on recipes; KPIs are computed on demand	No historical KPI series unless snapshots are added
Qdrant as external semantic search	Qdrant holds the 1,014 catalogue vectors; ranking proceeds without semantics until the model is ready	Semantic ranking is an enhancement, not a dependency
Scheduled and on-demand background jobs	On-demand only (catalogue sync, forecast run, model retraining); no scheduler	Assign a trigger, an owner and an alert for every job
Outcomes: higher revenue, lower waste, healthier employees	Targets; no waste logging, no revenue (mock payment), no health measurement	Measure before claiming (Chapters 20 and 21)

15.3 Technology stack

Table 30. Technology stack and evidence class

Layer	Technology	Role in ECOS	Evidence
Presentation	Streamlit 1.62.0	Role-based workspaces, forms and charts; session restore; reconnect banner	Executed
Service	FastAPI 0.162.0, Uvicorn 0.52.4, Pydantic 2.13.4	14 routers, 42 operations; request models enforce limits (10 per dish, 15 lines)	Executed
Persistence	MongoDB 8.3.4, PyMongo 4.16.0	Nine collections; unique and partial unique indexes; TTL index on sessions	Executed
Catalogue and semantics	Qdrant client 1.19.0; all-MiniLM-L6-v2 (256 dimensions)	1,014 vectors with cosine distance; catalogue sync; cached query embeddings	Executed
Analytics	scikit-learn 1.9.0, pandas 3.0.5, NumPy 2.5.2	Random Forest demand model, same-weekday baseline, evaluation	Executed
Optional language model	Configured through `utilities.py` and `ECOS_LLM_MODEL`	Rephrases rule-based copilot answers; absent in the verified run	Code-inspected
Packaging	Dockerfile, docker-compose.dev.yml, start.sh	UI on port 7860 in the container, mapped to 8501 on the host	Documented

15.4 Module design and ownership

Dividing a system into modules clarifies who owns what, how far a change spreads and where a fault can be contained. The interface documentation groups functions into user profiles, menu and catalogue, ordering, inventory, recommendations, forecasting, analytics and KPIs, a simplified purchasing view, feedback and behaviour, and the copilot, alongside platform services. Management's question is whether the result is one coherent product with clean handoffs or a set of adjacent features that define things differently.

It helps to follow the business information rather than the network calls. Suggestions depend on reliable menu attributes and availability. Ordering depends on item identity, price and order state. Stock risk depends on units, consumption figures and links from recipes to ingredients. Forecasts depend on dated history and a clear baseline. Analytics depend on consistent order events, and the copilot depends on whatever rules and records it may read. Where an upstream definition is shaky, a polished screen downstream can make the doubt look like certainty.

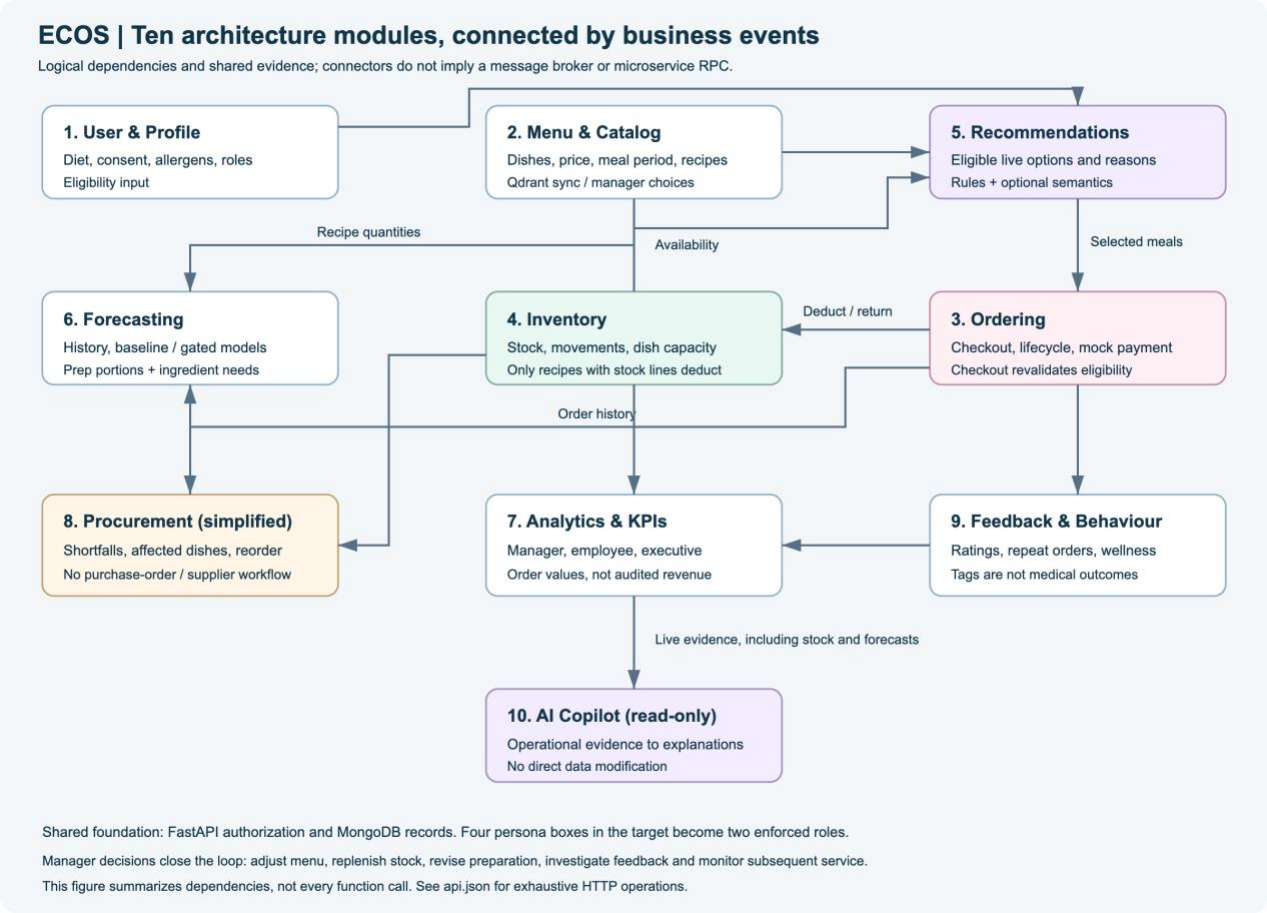


Figure 17. Module connections of ECOS (derived from the project asset docs/assets/module-connections.svg).

Table 31. Suggested owners by capability

Capability	Key contract	Technology or data owner	Business owner
Menu and catalogue	Versioned dish with declared attributes	Product or data team	Menu steward
Ordering	Lifecycle state; idempotent event	Application owner	Service operations
Inventory	Item, unit, balance, movement	Data or API owner	Stock controller
Forecasting	Horizon, unit, model or baseline status	Analytics owner	Operations planner

Capability	Key contract	Technology or data owner	Business owner
Recommendations	Explainable rule; opt-out	Product or analytics owner	Employee-service owner
Executive KPIs	Approved metric dictionary	Finance or data owner	Leadership sponsor
Copilot	Read-only scope; cited basis; fallback	Product owner	Operations owner

15.5 Data model

Table 32. MongoDB collections and their integrity rules

Collection	Purpose	Key fields and indexes
users	Accounts and preferences	`user_id` unique; role; scrypt password hash; diet, goals, declared allergens, favourites; failed-login counter and lock time
ingredients	Stock master	`ingredient_id` unique; name, unit, quantity, reorder point
recipes	Dishes synchronised from the catalogue plus menu settings	`dish_id` unique; `catalog_code`; index on (`on_menu`, `meal_periods`); nutrition, declared allergens, tags, ingredient lines, price, component and portion flags
orders	Order header and lines	`order_id` unique; (`user_id`, `date`); (`status`, `date`); partial unique (`user_id`, `request_id`) for idempotency; `created_at` and one timestamp per status
inventory_movements	Stock ledger	(`ingredient_id`, `created_at`); `order_id`; movement types such as `order_consumption` and `order_cancelled`
feedback	One rating per order	`order_id` unique; `user_id`; rating
forecasts	Stored forecast runs	(`date`, `created_at`)
sessions	Server-side sessions	`token_hash` unique (SHA-256 of the token); `expires_at` with a TTL index
meta	Catalogue-sync status	Single document with `_id` = `catalog_sync`

15.6 Data foundation: the 1,014-dish catalogue

The catalogue built in Part I is the only real dataset in the project. Figure 18 shows how it reaches the application: records are read from Qdrant, normalised to per-portion values where the payload allows, given an inferred diet label where the source has none, tagged by transparent nutrition rules, assigned an estimated price and meal periods, and written to the `recipes` collection without overwriting menu choices made by managers. Table 33 and Figure 19 profile the result.

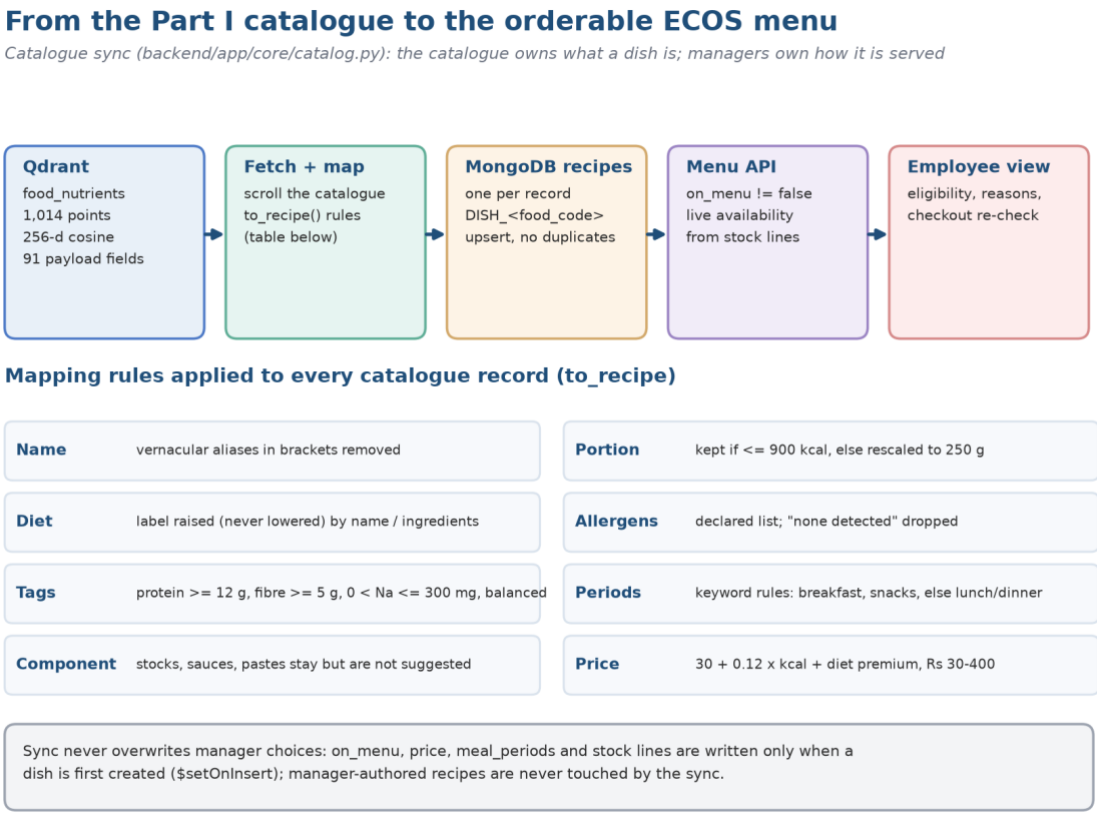


Figure 18. From Qdrant payload to menu-ready recipe: the catalogue synchronisation pipeline.

Table 33. Profile of the synchronised catalogue (1,014 dishes, run of 8 November 2026)

Attribute	Measure	Reading
Diet	Vegetarian 695 (68.5%), egg 160 (15.8%), non-vegetarian 159 (15.7%)	Diet label inferred by rule for 382 dishes (37.7%); 632 carry a source label
Declared allergens	599 dishes (59.1%) declare at least one: milk 422, egg 196, wheat 98, tree nuts 63, mustard 59, celery 36, peanut 28	No declaration is not the same as allergen-free: 415 dishes declare nothing
Nutrition tags	Lower sodium 729, high protein 224, high fibre 205, balanced 171	Rules: protein at least 12 g; fibre at least 5 g; sodium above 0 and at most 300 mg; balanced = protein 7 g, fibre 3 g, at most 550 kcal
Tag coverage	At least one tag on 941 dishes (92.8%); none on 73; tags per dish 0: 73, 1: 669, 2: 169, 3: 90, 4: 13	The healthy-dish test is permissive (Finding B4)
Meal periods	Lunch and dinner 539, snacks 357, breakfast 62, breakfast and snacks 56	Assigned by rule, not by kitchen practice
Data quality flags	Components 106; portion-normalised 201; ingredient text 636 (62.7%)	Components (sauces, chutneys, doughs) are not suggested unless searched for
Estimated price	Median 70 rupees, interquartile range 50 to 100, mean 86.8, range 30 to 290	Price = 30 + 0.12 x kcal + diet premium (0, 10, 40), rounded to 5, clipped to 30 to 400
Energy and nutrients	Energy median 256 kcal (mean 408); protein median 5.7 g; fibre 1.9 g; sodium 163 mg	Values are catalogue estimates, not laboratory measurements

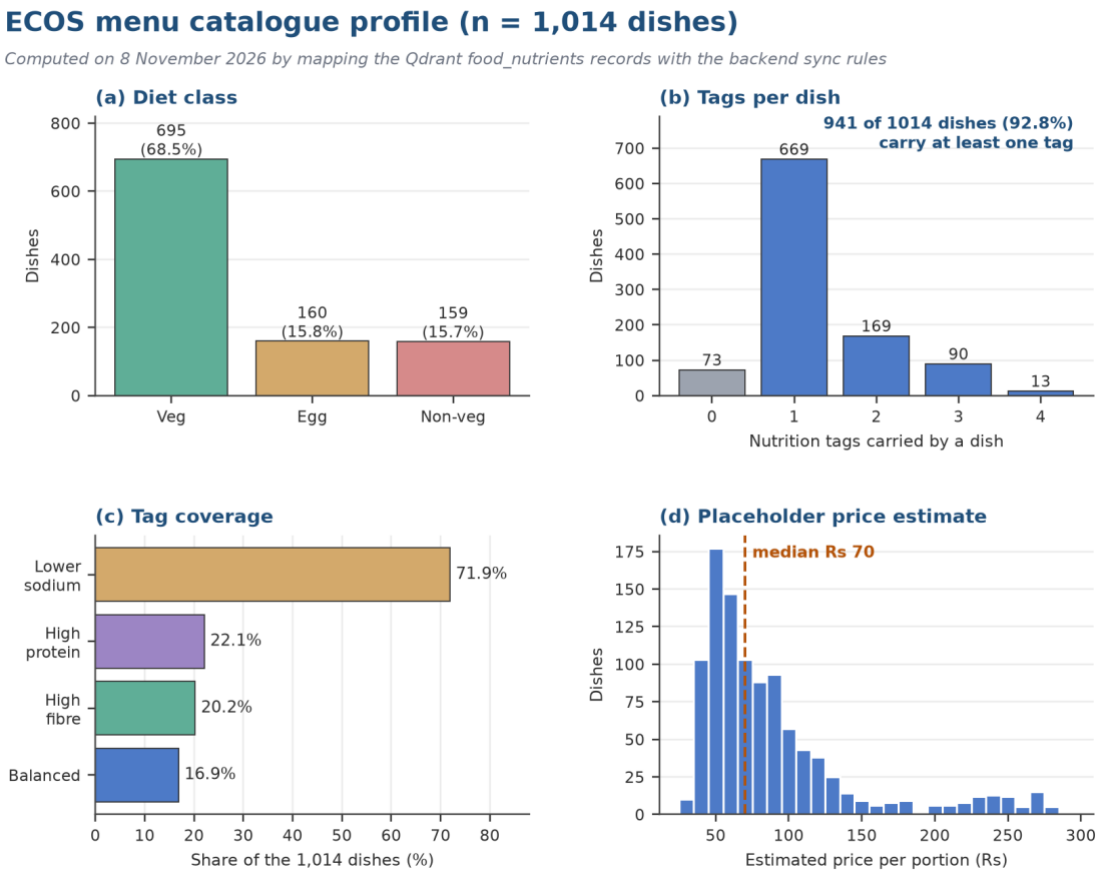


Figure 19. Catalogue profile: diet mix, declared allergens, nutrition tags and estimated prices.

Three management implications follow. First, **a third of the diet labels are inferred**, so diet filtering is only as reliable as the inference rule and should be stewarded (risk R1). Second, **41% of dishes declare no allergen**, which must be presented as unknown rather than safe. Third, the prices are modelled estimates, so any rupee figure derived from them is a demonstration value.

15.7 Security and privacy controls as implemented

Table 34. Security controls in the verified implementation

Control	Implementation	Evidence
Password storage	scrypt key derivation (n = 2^14, r = 8, p = 1); no plain-text passwords	Executed
Sessions	Random token; only its SHA-256 digest is stored; 30-minute sliding idle expiry; TTL index removes expired sessions	Executed
Brute-force protection	Five consecutive wrong passwords lock the account for five minutes (HTTP 429)	Executed
Authorisation	Server-side role checks: employees reach only their own orders and wellness; status changes, stock edits, uploads, forecasts and copilot are manager-only	Executed
Safety gates on the server	Diet and allergen eligibility are re-checked at checkout; per-dish and per-order limits are enforced	Executed

Control	Implementation	Evidence
Input validation	Pydantic request models; stock and recipe uploads are validated before any change is committed	Executed
Language-model minimisation	Only compact facts (dish and ingredient names, the rule answer and its evidence) are sent; the copilot is read-only	Code-inspected
Secrets	Supplied through environment variables; `env.template` lists them	Documented

Chapter 16: Role-Based Service Design and Operations

16.1 Employee experience

The employee workspace has five destinations: Home, Menu (For you, Full menu, Plan my day), My orders, My wellness and Profile. The journey runs from discovery to commitment to feedback: the employee sees dishes that are available and eligible for the declared diet and allergens, reads why a dish was recommended, places an order whose payment is the mock state `demo_paid` (the interface states that no money is taken), follows its status, rates it, and sees a monthly food-choice scorecard.

Three design properties matter for management. **Safety is enforced on the server**: eligibility is re-checked at checkout, so a manipulated client cannot order a dish that the profile excludes. **Recommendations are explainable**: each card carries up to five reasons (for example “Matches your high protein goal”, “One of your regulars (3 orders)”, “Popular this fortnight”) and a match percentage capped at 99 so that it is never shown as certainty. **Choice is preserved**: the full menu is always one step away.

The My wellness page shows a monthly score between 0 and 100: 20 points to start, 10 for every portion that carries at least one nutrition tag, 5 for every other portion, capped at 100 (Formula 1). The score is computed over every non-cancelled order of the month, including orders that are confirmed but not yet collected, whereas the tag shares on the same page use collected orders only; the page caption says a pending order counts once collected. This inconsistency is Finding B1 in Chapter 18. The page also says plainly that it summarises recorded choices only and makes no medical or clinical assessment.

Proposed principles for employee service: availability that is clearly shown; accessible wording and interaction; open price and dietary information; easy-to-reach non-personalised options; simple correction and support; and nobody penalised at work for taking part in the wellness feature.

16.2 Manager work and decision rights

A manager's day is full of disturbances: demand swings, substitutions, deliveries, thin staffing, bad data and escalations from diners. Seven workspaces are provided, namely the Today board, the Live orders queue, Inventory, Menu management, Forecasting, the Executive view and the Copilot, which divides the immediate operational picture from planning and review. What is established is that these views exist. Frequency of use, adequacy of refresh speed and any gain in efficiency are not.

Each view must come with decision rights. Someone has to be entitled to pause an uncertain item, change a menu description, look into a stock gap, alter a planning assumption, confirm a production quantity from a forecast, and approve a substitute ingredient once allergens are rechecked. Where nobody is named, the dashboard is merely a pleasant window onto unresolved problems.

Table 35. Who decides what in the manager workspace

Decision	Required check	Accountable owner	Responsible operator
Mark an item unavailable	Confirm physical stock; tell employees	Cafeteria manager	Shift lead
Change a published menu entry	Check price, ingredients, allergens and effective date	Operations owner	Menu steward
Commit a stock adjustment	Reconcile with source evidence	Inventory owner	Authorised stock editor
Use a forecast as a planning input	Look at coverage, the baseline, exceptions and capacity	Operations owner	Planner or manager
Raise a supplier request	Apply delegated authority and supplier terms	Procurement approver	Procurement
Resolve an employee complaint	Record the outcome; protect personal data	Employee-service owner	Service desk or manager

The inventory area has four tabs: Stock, Edit stock, Today's consumption and Risk & impact. Spreadsheet edits are validated before they are committed, and the screen shows movement history, affected dishes and replenishment context, so that changes are less likely to go unnoticed. Before real records rely on this, check who may edit, what happens when two people edit at once, how physical counts are done, whether an edit can be rolled back and how long the audit trail is kept.

In the verification run the inventory group (17 checks) and the upload group (6 checks) passed. Rollback of a committed stock edit, retention of the audit trail and behaviour under concurrent editors were not tested.

16.3 Blueprint of the service and its operating model

A service blueprint lines up what diners experience, what staff do out of sight, the systems underneath and the evidence left behind. For ECOS this matters because a successful order needs a current menu, believable stock, interpreted order events and a person ready to handle exceptions, none of which is visible in the click itself. The blueprint exposes the steps where information or authority is missing.

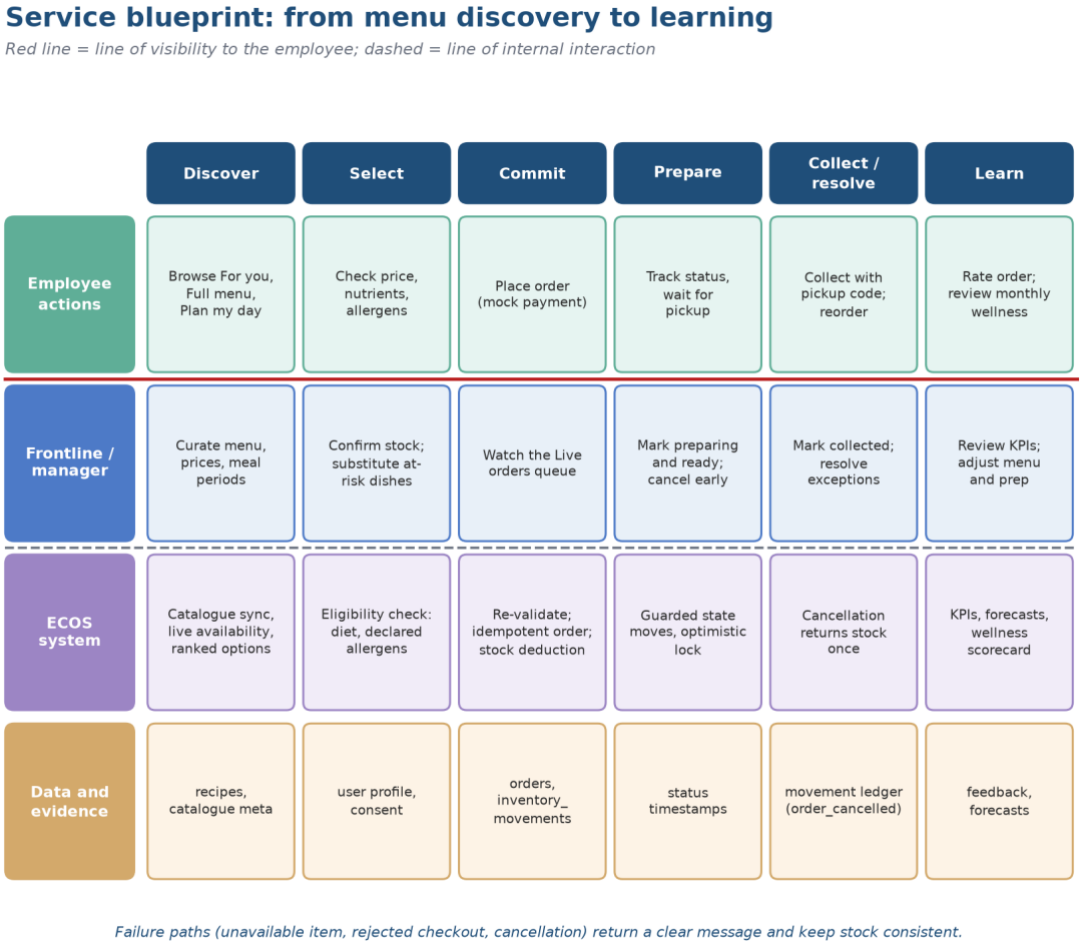


Figure 20. Service blueprint: employee touchpoints, kitchen activity, manager decisions and supporting data.

Table 36. Service blueprint by stage

Stage	Employee-facing action	Backstage work	Evidence or control
Discover	Looks through the menu and any suggestions	Staff keep dishes, prices, ingredients and availability up to date	Menu version history and a named attribute owner
Select	Reads an item and its declared details	Stock and substitutions are checked against the item	Timestamp on availability and a periodic review
Commit	Confirms a demonstration order	The order state is saved and shown to operations	Safe-to-repeat submission and a clear confirmation
Prepare	Waits for the order	The kitchen orders its work and deals with problems	History of status changes and a named escalation contact
Collect or resolve	Picks up the order or asks for help	A manager handles missing items, delays and swaps	Outcome code and a recovery log
Learn	Looks back at orders and may rate them	Behaviour is summarised and used for planning	Analytics with minimal personal detail

Every step should have a named service owner, frontline role, data steward, technical contact, escalation trigger and a fallback if the system is unavailable. Falling back to manual working is part of a resilient design rather than a defeat: an approved paper or shared menu and a written routine for updating stock keep service going during an outage.

16.4 Order lifecycle and service recovery

An order passes through states and cannot be reduced to one count. ECOS implements five: **confirmed, preparing, ready, collected** and **cancelled**, with cancellation reachable only from confirmed or preparing (Figure 21). Only managers change status. Each update is conditional on the previous status, so two managers acting at once cannot both succeed; the second receives a conflict response and must refresh.

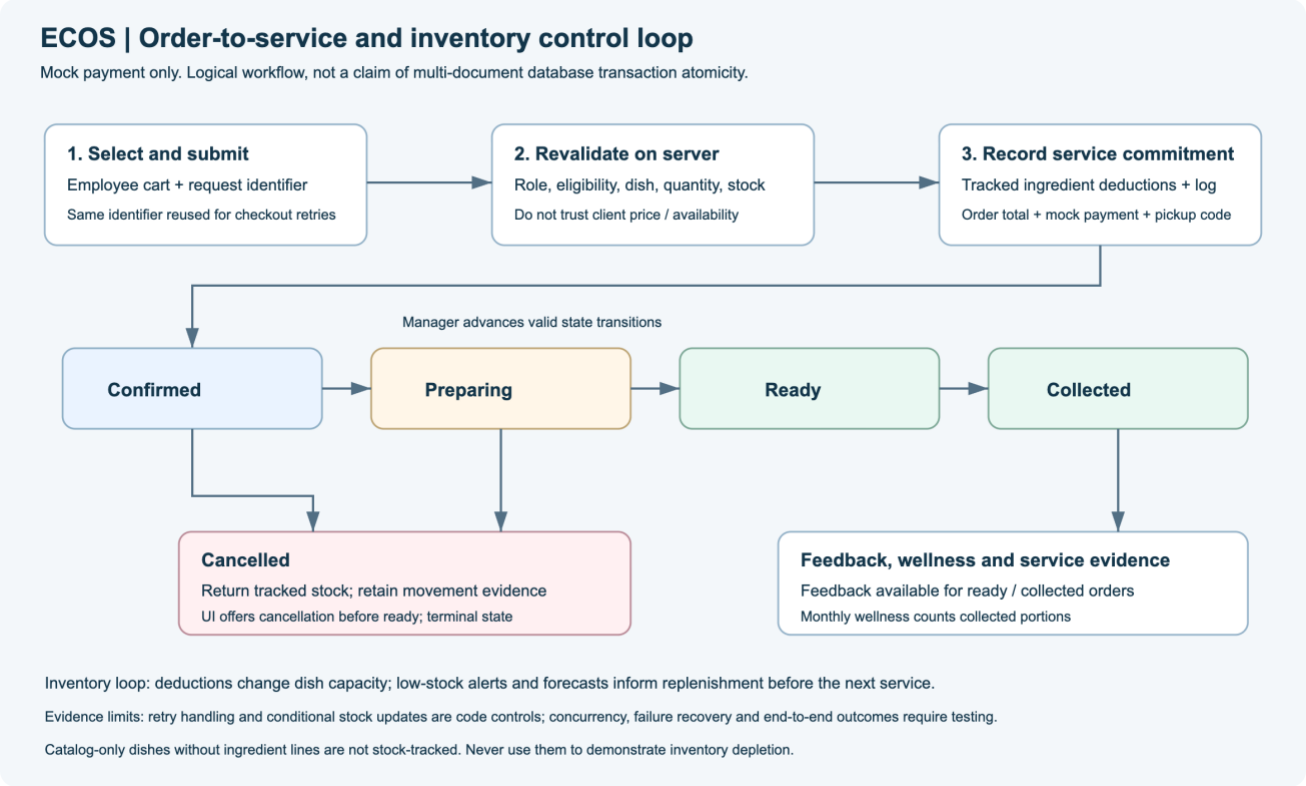


Figure 21. Order lifecycle (derived from the project asset docs/assets/order-lifecycle.svg).

Table 37. Order states and what they mean for management

State	What it means for management	Not to be mistaken for
confirmed	Order accepted; ingredients have been deducted from stock	Prepared, paid or collected
preparing	Kitchen has started the order	Ready for collection
ready	Waiting at the counter; cancellation is no longer possible	Collected

State	What it means for management	Not to be mistaken for
collected	Handed over; counts as completed for completion rate and tag shares	Paid or consumed
cancelled	Stock returned once; excluded from forecast training and from recommendation history	Zero demand or employee dissatisfaction

Table 38. Controls on the ordering and stock path

Control	How it works	Why it matters
Idempotent checkout	The client sends a `client_request_id`; a unique partial index on (`user_id`, `request_id`) makes a retry return the original receipt	Retries and double clicks cannot create duplicate orders
Atomic stock deduction	Each ingredient is decremented only if enough remains; on failure everything already taken is restored and a conflict is returned	No negative stock and no half-filled order
Compensation	If the order cannot be stored, the stock, the movements and the order are rolled back	A failed order never loses inventory
Cancellation	Allowed until ready; ingredients are returned once and recorded as `order_cancelled` movements	Stock stays reconcilable after recovery
Limits	At most 10 portions per dish and 15 lines per order	Guards against input errors and hoarding
Payment	Mock state `demo_paid`; no money is taken	Order totals are demonstration values, not revenue

Service recovery is a management capability. When a dish is unavailable the server names the dish and the reason in a clear conflict message. Because payment is a mock state there is no refund path; a paid production service would need an approved payment design and a ledger that reconciles exactly with the order states above.

16.5 Inventory control, waste and procurement

Stock policy has to tie together demand, replenishment lead time, shelf life, menu design, storage space and physical control. The manager's inventory area provides tabs for Stock, Edit stock, Today's consumption and Risk & impact. Checked spreadsheet uploads, movement history, affected-dish lists and replenishment hints give a base for disciplined review; they do not, however, make a complete ledger or an optimal replenishment policy.

First decide what the shown quantity means: physically on hand, promised to orders, reserved, on the way, or estimated use. Units need explicit conversion rules, because cases, kilograms, portions and recipe amounts cannot be compared without them. Every adjustment should capture the person, time, item, amount, unit, reason and supporting evidence. Compare periodic physical counts with system balances and investigate gaps instead of overwriting them.

Table 39. Inventory risks and controls

Risk condition	Decision owner	Preventive or detective practice
Out-of-date stock record	Inventory owner	Timestamp each balance; count on a fixed rhythm
Unit mismatch	Data steward	Controlled item master; review conversions
Unapproved adjustment	Operations manager	Role-based authorisation; reason code
Duplicate import	Authorised editor	File or version identifier; preview before commit
Hidden substitution risk	Menu steward	Show affected dishes and attributes
Waste not captured	Shift lead	Standard reason and quantity capture

A risk-and-impact view helps only when its score can be explained. A manager should be able to see what drives a high value: a thin balance, heavy forecast usage, an unreliable delivery window, or an ingredient shared across many dishes. Treat criticality as a plain operating rule. Calling it a stockout probability would need calibration against what actually happened.

Finding B5. ECOS records consumption caused by orders and manual stock movements, but it has no waste-logging transaction (spoilage, preparation waste, surplus) and no supplier, lead-time or purchase-order fields. Waste reduction and procurement savings, two of the three outcomes on the architecture poster, therefore cannot be evidenced from the system as it stands.

Purchasing converts a plan into a binding commercial act. The project papers call purchasing simplified: there is neither a full supplier master nor a purchase-order flow. Accordingly, a restock hint, an ingredient-risk display or a manager's choice to reorder should never be described as a supplier order, a commitment or an integration with purchasing.

Connecting purchasing later would require a governed item and supplier master, approved vendors, contract terms, price history, lead times, minimum order sizes, delivery calendars, substitutes, approval limits and invoice matching. The workflow must keep apart a forecast quantity, a suggested need, a requisition, an approved order, a supplier acknowledgement, a goods receipt and an invoice, since each carries different legal, operational and financial weight.

Table 40. Procurement maturity, from visibility to invoice matching

Stage	What the system can do	What must be shown before moving on
1. Visibility	Display stock levels and usage	Shared definitions of items and units
2. Suggestion	Offer a restocking scenario	Written assumptions and a human check
3. Requisition	Raise an internal request	An approval flow that stops duplicates

Stage	What the system can do	What must be shown before moving on
4. Purchase order	Commit through authorised buyers	Supplier records, contract terms, approvals and an audit trail
5. Receipt and matching	Log deliveries and invoices	Quantity and price agree; differences are resolved

16.6 Operating cadence and service management

Running a dependable service takes more than code: a daily routine, escalation paths, support, incident tracking and a way to recover. The root `app.py` waits for a local health check before it starts the dashboard. That suits demonstrations but says nothing about production monitoring, scaling, uptime, backups or incident handling.

Define service hours, a support channel, severity levels, target response times, who owns each shift and what the fallbacks are. If the dashboard fails mid-service, the cafeteria keeps serving from an approved offline menu with manual order and stock routines, and reconciles afterwards. Staff need to know what can safely be re-keyed, what must never be keyed twice, and who has the authority to declare normal service restored.

Table 41. Operational review rhythm

Cadence	Output	Review focus
Start of shift	Named owner; shift risks	Availability, open exceptions, data age
Before service	Availability list; substitution notice	Menu and stock confirmation
During service	Resolution; message to employees	Delays, stockouts, order problems
End of shift	Reconciliation; carried-over actions	Orders, stock movement, waste, incidents
Weekly	Approved process changes	Forecast error, menu mix, repeat issues
Monthly	Continue, change or stop decision	Scorecard, privacy and governance, benefits

16.7 Screen walkthrough

The plates that follow are screenshots of the running application captured with a headless browser on 8 November 2026. They show the synthetic demonstration identities and data described in Chapter 13; names, order values and stock levels are not real. They are included as evidence that the workflows described above exist and render, not as evidence of usability.

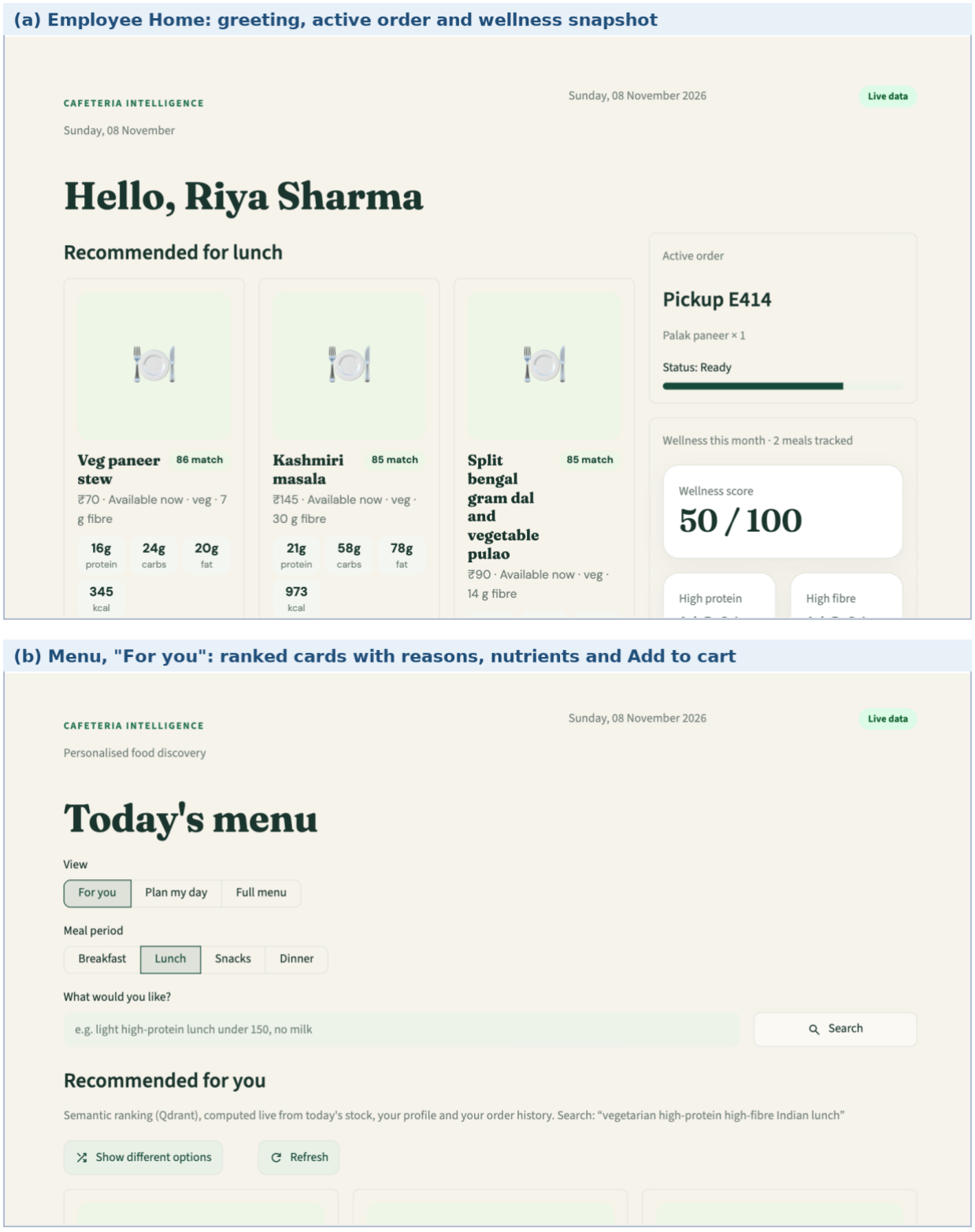
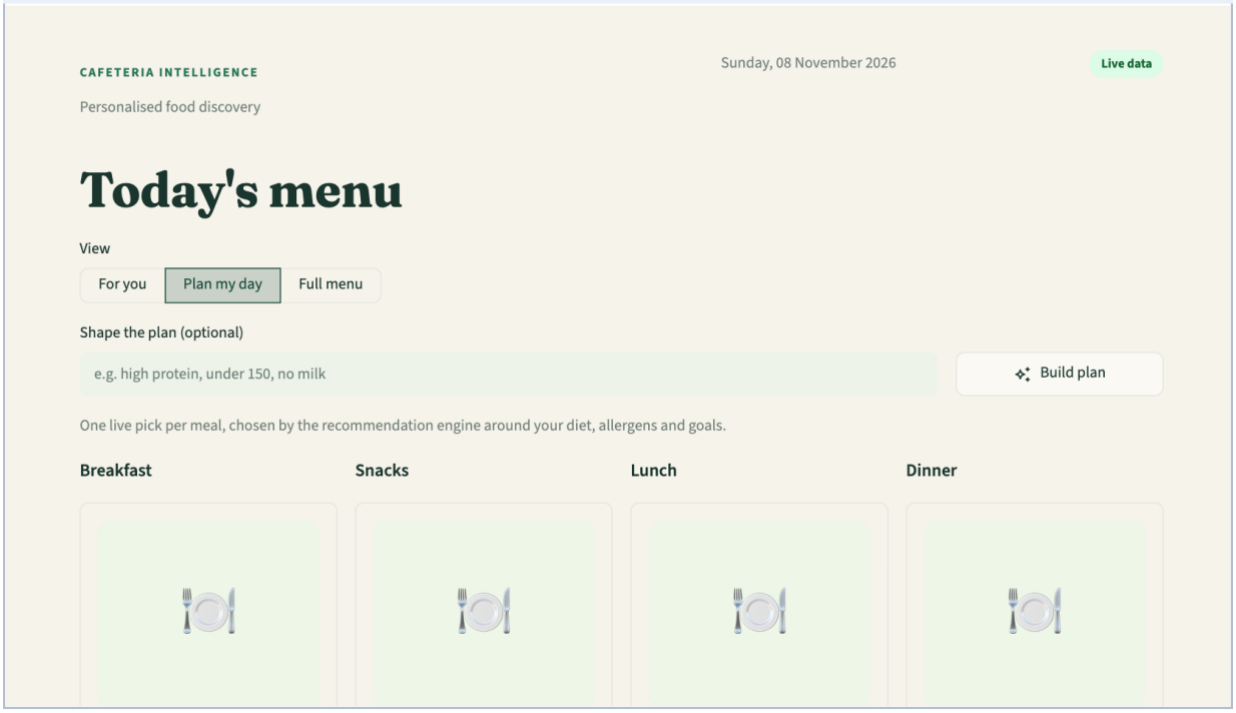


Figure 22. Employee workspace: (a) Home; (b) Menu, For you (synthetic demonstration data).

(a) Menu, "Plan my day": one live pick per meal period



(b) My orders: lifecycle tracker, reorder and feedback

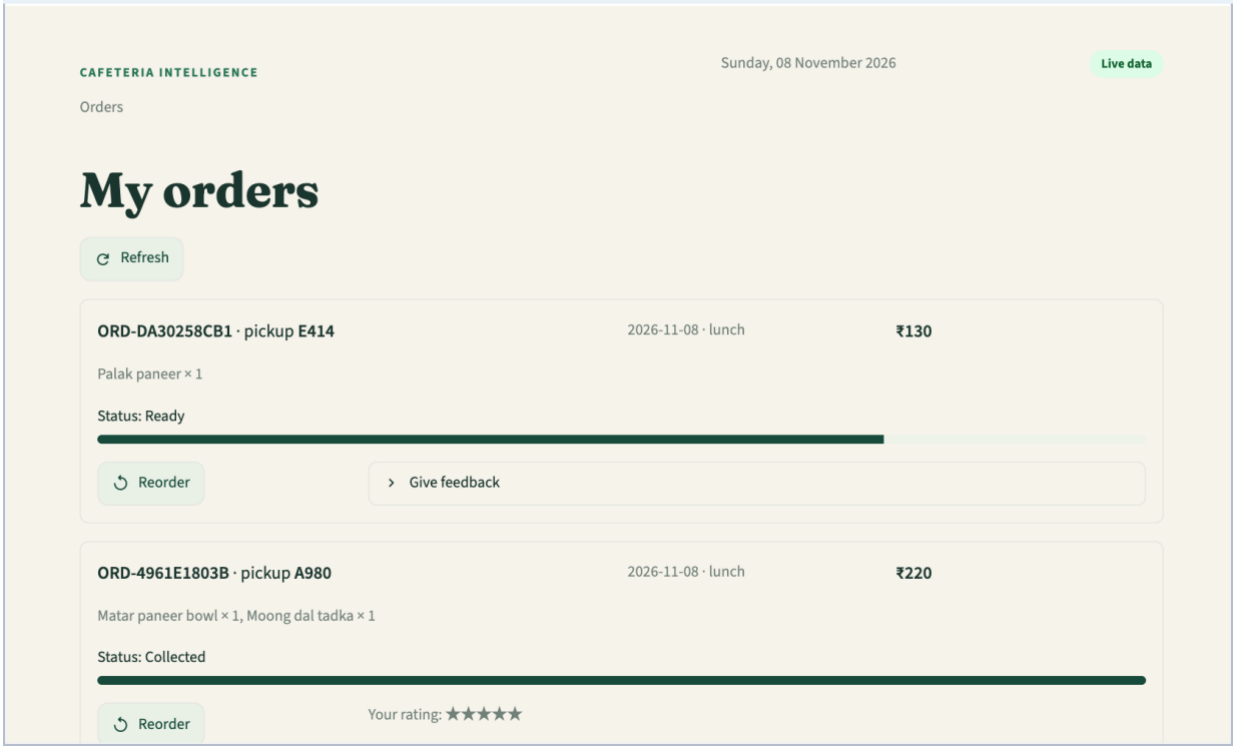


Figure 23. Employee workspace: (a) Plan my day; (b) My orders.

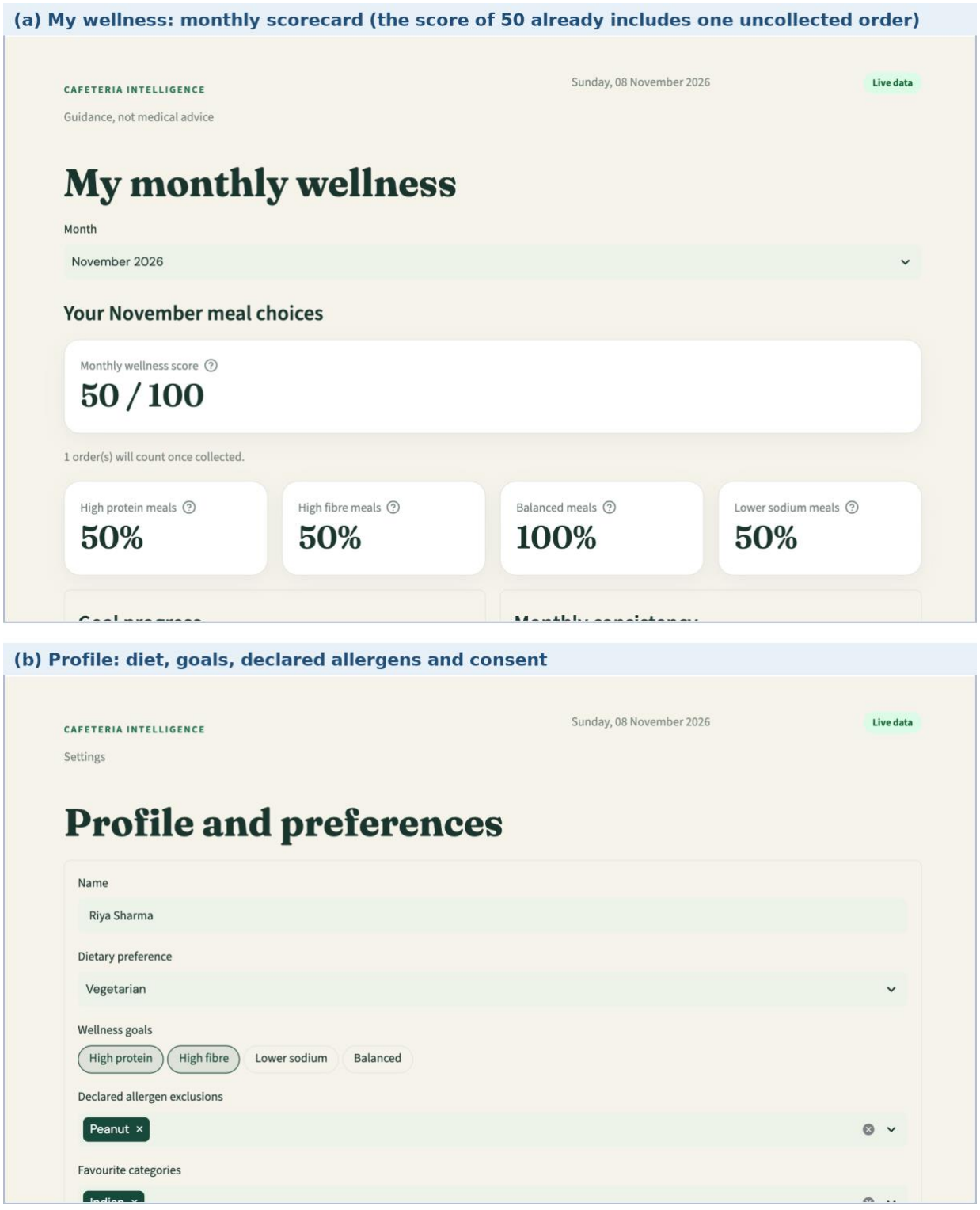


Figure 24. Employee workspace: (a) My wellness; (b) Profile and preferences.

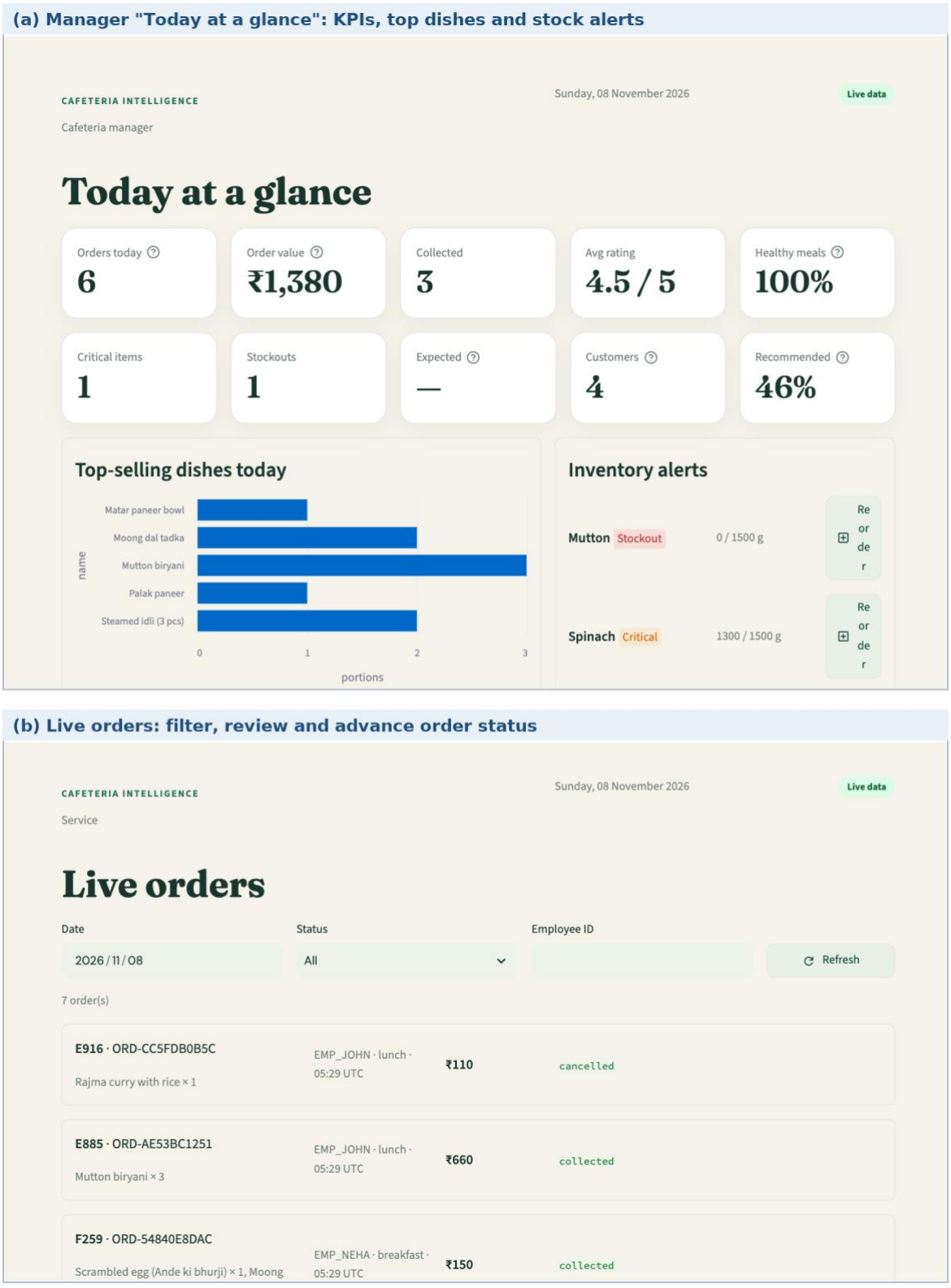
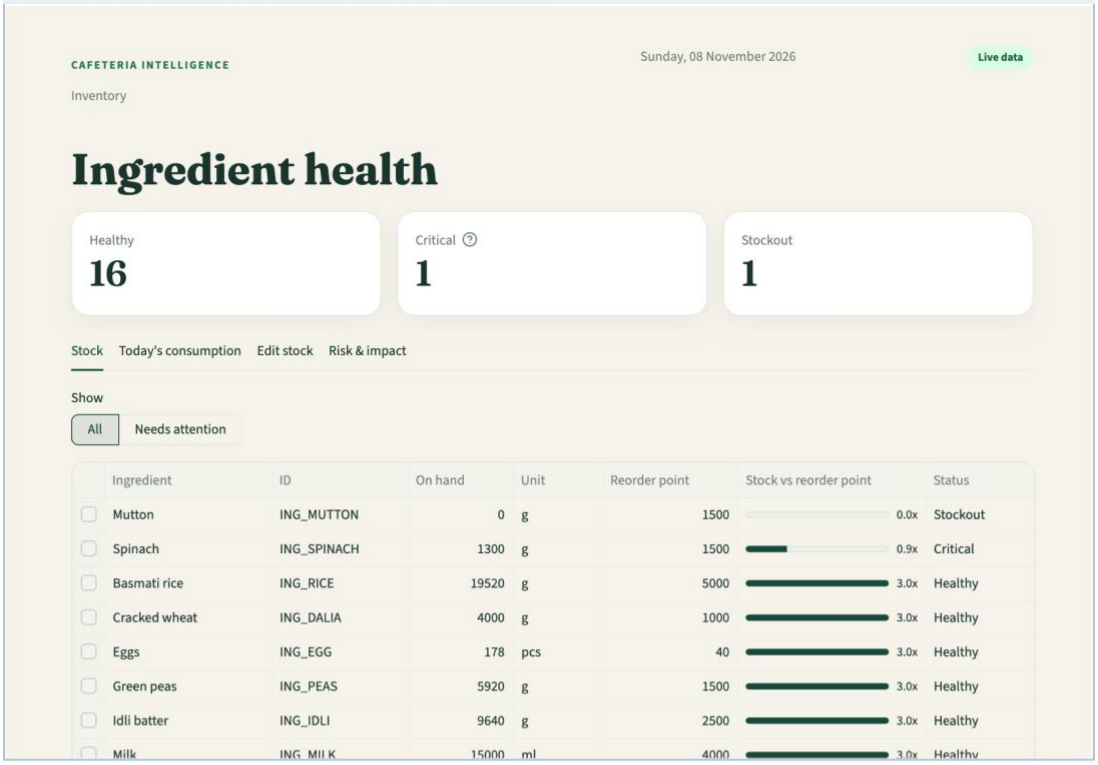


Figure 25. Manager workspace: (a) Today; (b) Live orders.

(a) Inventory, Stock: health status per ingredient



(b) Inventory, Risk & impact: dishes at risk, revenue at risk, swaps

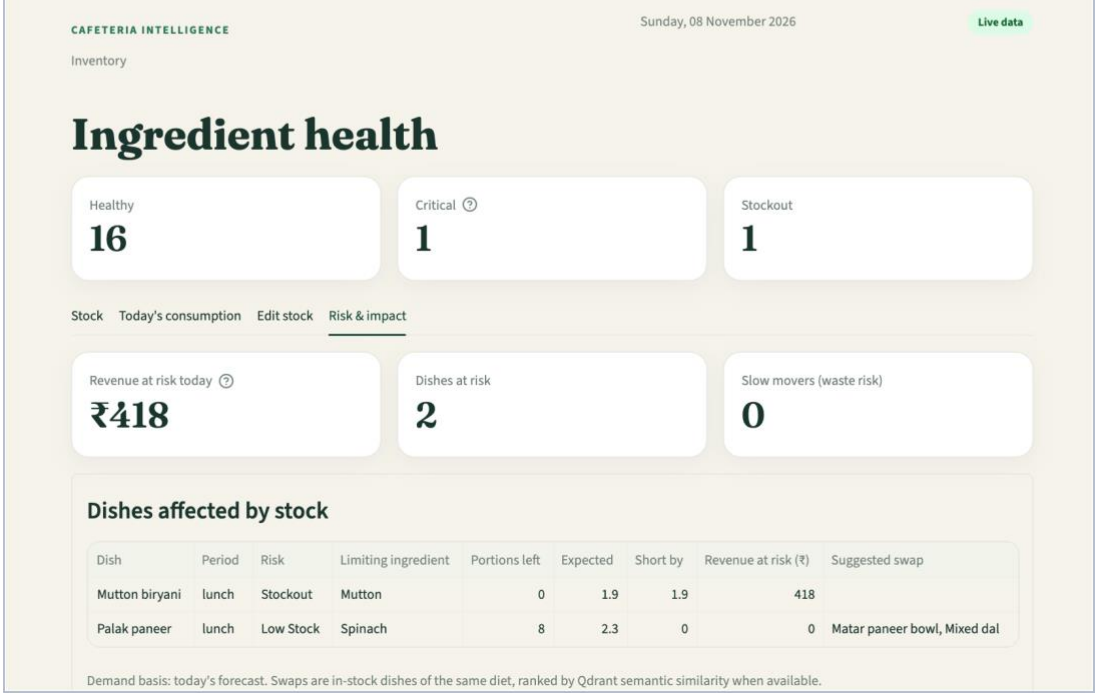


Figure 26. Manager workspace: (a) Inventory, Stock; (b) Today's consumption; (c) Risk and impact.

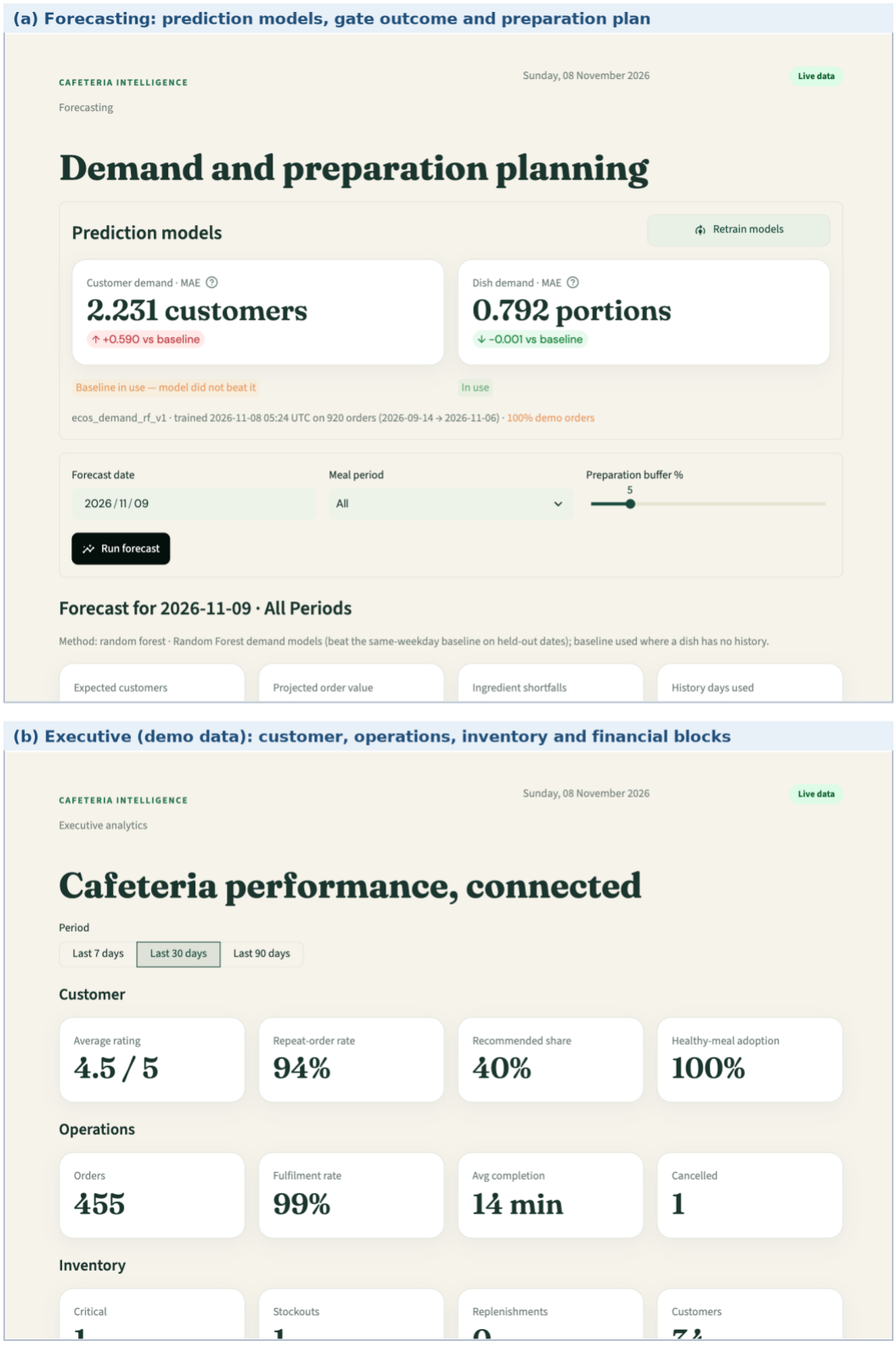


Figure 27. Manager workspace: (a) Forecasting; (b) Executive.

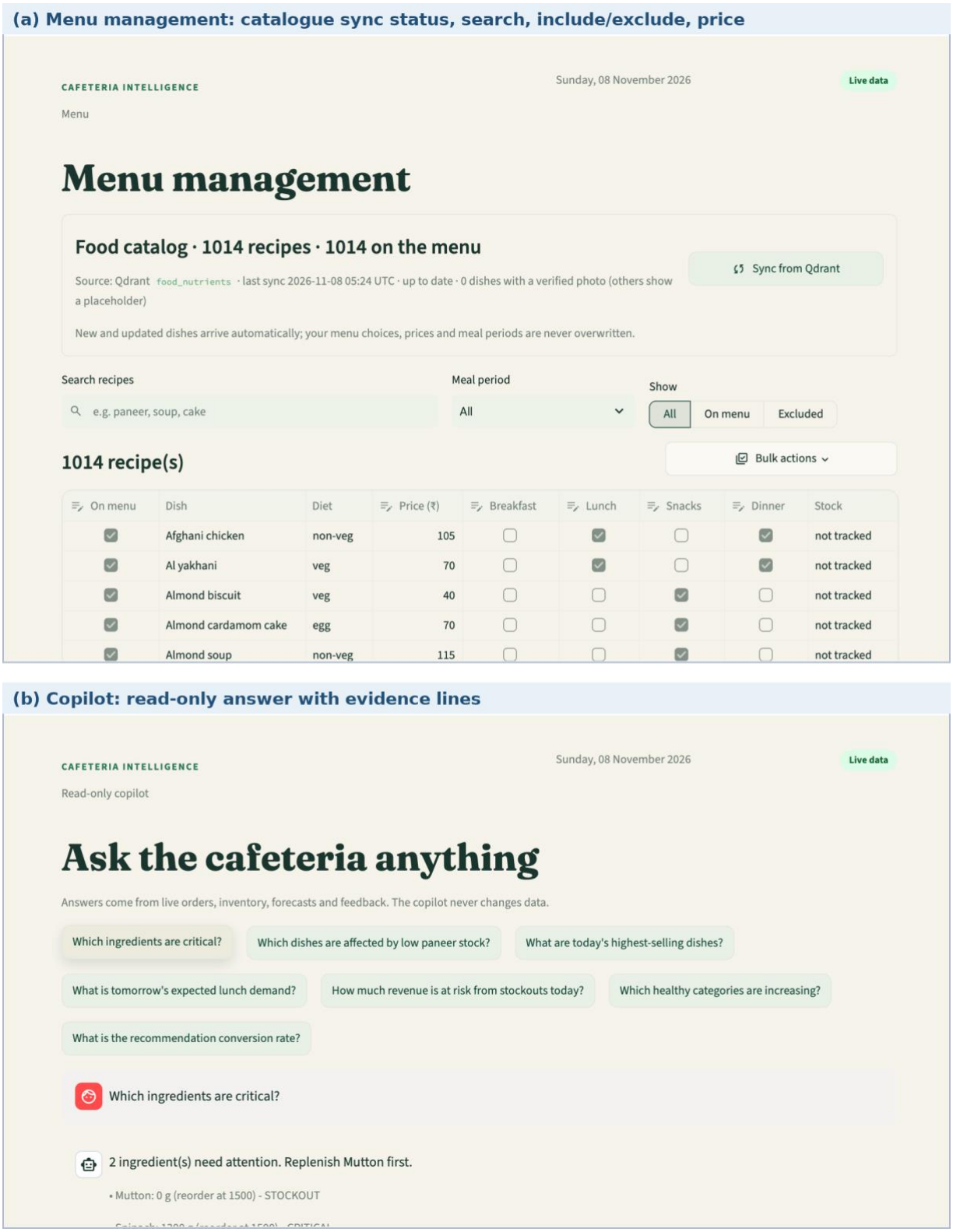


Figure 28. Manager workspace: (a) Menu management; (b) Copilot.

Chapter 17: Analytics and Decision Support

17.1 Recommendation engine

The recommender is a transparent two-stage pipeline (Figure 29). Stage one applies **hard filters** that are never traded off against ranking. Stage two scores the surviving dishes with an additive heuristic whose terms are shown to the user as reasons. The design choice is deliberate: safety attributes (allergens, diet, budget) are constraints, while taste and goals are preferences.

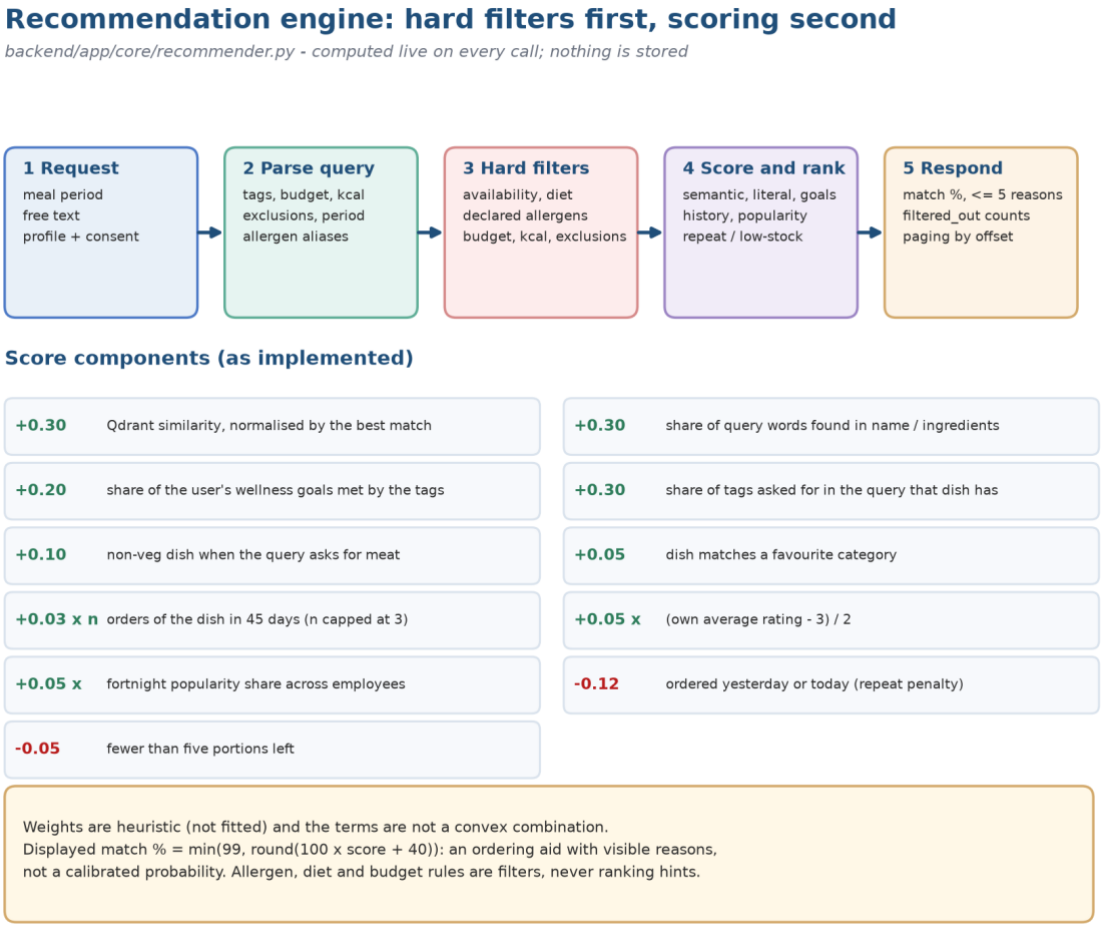


Figure 29. Recommendation pipeline: hard filters, scoring terms and the explanation shown to the employee.

Table 42. Hard filters applied before ranking (in order)

Filter	Rejects a dish when
Unavailable	It is off the menu for the period, or stock for any ingredient line is exhausted

Filter	Rejects a dish when
Seen	The caller has already used it, for example in the current day plan
Side item	It is a component (sauce, chutney, dough and similar) and the query names no search term
Diet or allergen	It conflicts with the profile; allergen names are canonicalised so that gluten also blocks wheat and dairy blocks milk
Allergen in the query	The text names an allergen the user excluded in words
Excluded term	The query says “without” a named ingredient that the dish contains
Diet intent	The query asks for vegetarian or vegan and the dish is not
Budget	Price exceeds the stated maximum
Calories	Energy is above the stated maximum or below the stated minimum

Table 43. Scoring terms of the recommendation heuristic

Term	Contribution	Reading
Semantic similarity	up to +0.30	Similarity to the query or profile text, relative to the best candidate; only when the embedding model is ready
Literal term match	up to +0.30	Share of query words found in the dish text
Goal match	up to +0.20	Share of the user's wellness goals met by the dish tags
Query tags	up to +0.30	Share of requested tags (for example high protein) met
Meat intent	+0.10	Query asks for meat and the dish is non-vegetarian
Favourite category	+0.05	Dish text matches a favourite category in the profile
Order history	+0.03 per order, at most 3	Dishes the user has ordered in the last 45 days
Rating	+0.05 x (rating - 3) / 2	Rating the user gave to the dish, default 3
Popularity	up to +0.05	Fortnight popularity relative to the most popular dish
Repeat penalty	-0.12	Ordered within the last day
Low stock	-0.05	Fewer than five portions left

The match percentage is `min(99, round(100 x score + 40))`. The constant 40 is a presentation offset, not a probability; the cap at 99 keeps the interface from implying certainty. No reviewer-labelled relevance test was run, so ranking quality (hypothesis H2 of Part I) remains untested; the executed checks show only that the filters and the ordering behave as specified.

17.2 Nutrition tags and the wellness score

A dish receives a nutrition tag when it meets a threshold: protein of at least 12 g (high protein), fibre of at least 5 g (high fibre), sodium above 0 and at most 300 mg (lower sodium), or protein of at least 7 g with fibre of at least 3 g and at most 550 kcal (balanced). The four tags together define a “healthy” portion. Because 92.8% of the catalogue qualifies, almost every portion earns the full 10 points, and the score saturates at 100 after eight portions in a month (Figure 30). The score therefore measures ordering volume more than ordering quality (Finding B4).

$$\text{Score} = \min(100,\ 20 + 10h + 5o)$$

Formula 1. Monthly wellness score (h = portions with a nutrition tag, o = other portions)

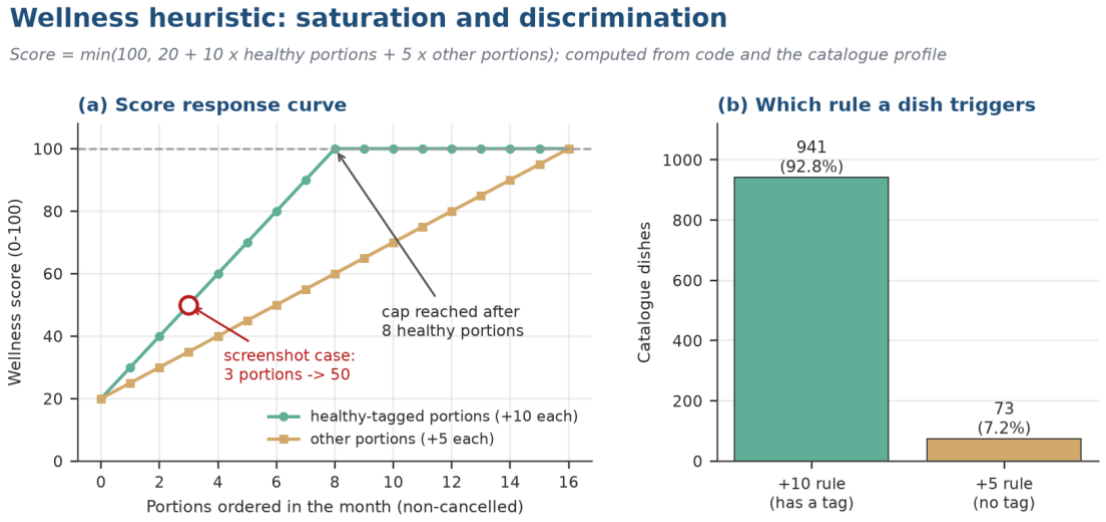


Figure 30. Wellness score against the number of tagged portions ordered in a month.

17.3 Demand forecasting

The forecasting module predicts customers per service date and meal period, and portions per dish. Each lag feature is computed only from earlier services of the same group, so no future information leaks into a prediction (Figure 31). The inputs are the meal period, the day of the week, the previous value, the four-service rolling mean and `same_weekday_4`, the mean of up to four preceding services on the same weekday. A Random Forest (200 trees, depth 8, minimum leaf 2, seed 42) is trained on the first 80% of service dates and tested on the last 20%. Training is refused below 12 customer rows, 20 dish rows or five service dates. The model is served only if its mean absolute error does not exceed that of the baseline (Formula 2); otherwise the baseline is used.

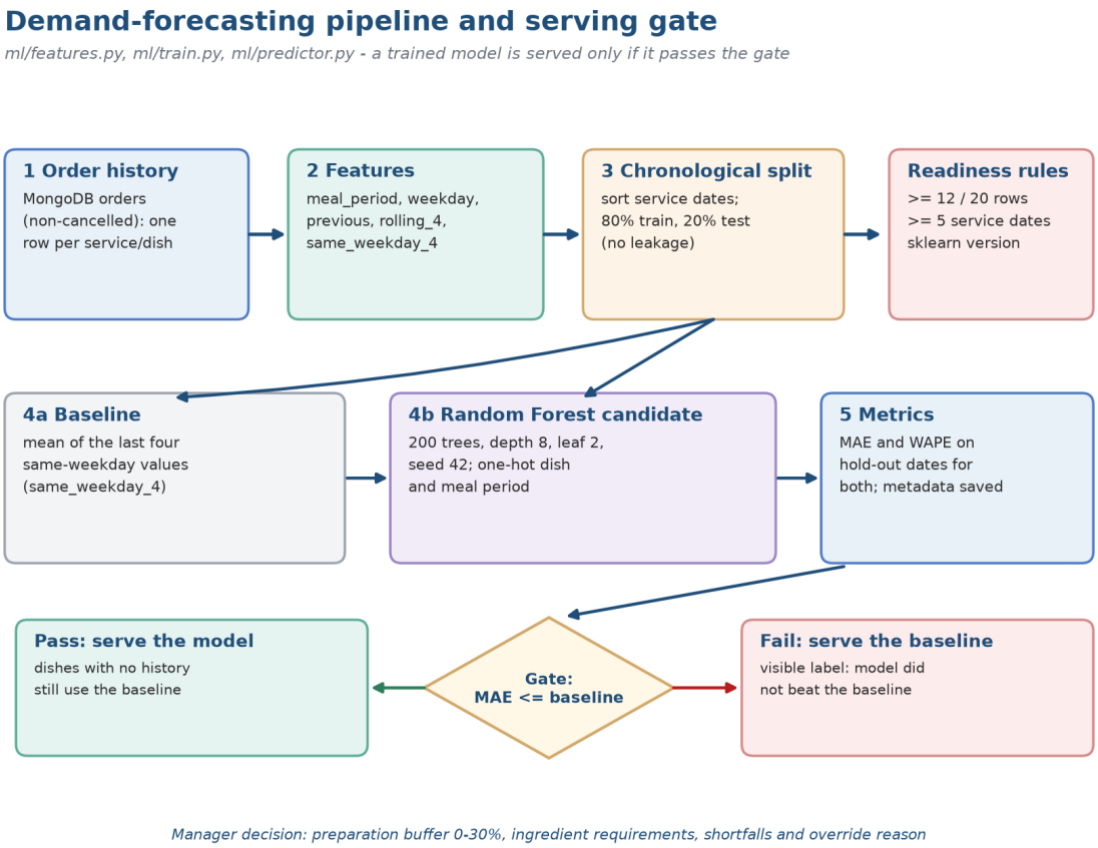


Figure 31. Forecast pipeline: features, chronological split, evaluation and serving gate.

$$\text{Serve model} \Leftrightarrow \text{MAE}_{\text{model}} \leq \text{MAE}_{\text{baseline}}$$

Formula 2. Serving gate for the demand model

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

Formula 3. Mean absolute error

$$\text{WAPE} = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{\sum_{i=1}^{n} y_i} \times 100\%$$

Formula 4. Weighted absolute percentage error

The baseline is itself one of the model's inputs, so on thin history the model can at best refine it. The manager chooses a safety buffer of 0 to 30% in the interface (the API accepts 0 to 50%), and the preparation quantity is the rounded-up product of the forecast and the buffer (Formula 5).

$$\text{Preparation} = \left\lceil \hat{d} \times \left(1 + \frac{b}{100}\right) \right\rceil$$

Formula 5. Preparation quantity (forecast d, buffer b in per cent)

A forecast matters only if someone uses it to decide something: it needs a defined horizon and a person answerable for the call. On the Forecasting page a manager picks dates, meal periods and a buffer between 0 and 30 per cent, and can retrain the models. These controls allow planning scenarios to be explored; they do not prove accuracy or show that any buffer is best.

Weigh the forecast against expected attendance and event calendars, and also against past demand, menu changes, availability limits, local conditions and kitchen capacity. State the horizon and the unit (orders or servings), and warn when history is thin or has changed shape. The buffer is a setting the manager chooses; it is not a statistical confidence band.

Table 44. Record to keep when a forecast drives a decision

Field	What it must contain
Forecast	Predicted demand for a stated dish, meal period, date and unit
Baseline	A simple comparator, for instance past demand on the same weekday
Buffer	The extra margin chosen, with who chose it and why
Limits	Stock on hand, staffing, preparation time, menu availability
Decision	Whether the manager accepted, changed or overrode it, and the reason
Result	What actually happened, measured with the same definition

What was run. The pipeline was trained on the synthetic history (920 orders on 39 service dates from 14 September to 6 November 2026, 100% demonstration orders), with the test period starting on 28 October. Table 45 gives the result and Figure 32 compares the errors.

Table 45. Forecast evaluation on synthetic demonstration history

Target	Rows train / test	MAE model	MAE baseline	WAPE model	WAPE baseline	Gate	Bootstrap MAE difference (95% CI)
Customers per meal period	124 / 32	2.231	1.641	39.22%	28.85%	Fails	+0.590 [0.050, 1.118]

Target	Rows train / test	MAE model	MAE baseline	WAPE model	WAPE baseline	Gate	Bootstrap MAE difference (95% CI)
Dish portions	589 / 152	0.792	0.793	66.12%	66.21%	Passes by 0.001	-0.0011 [-0.0868, 0.0869]

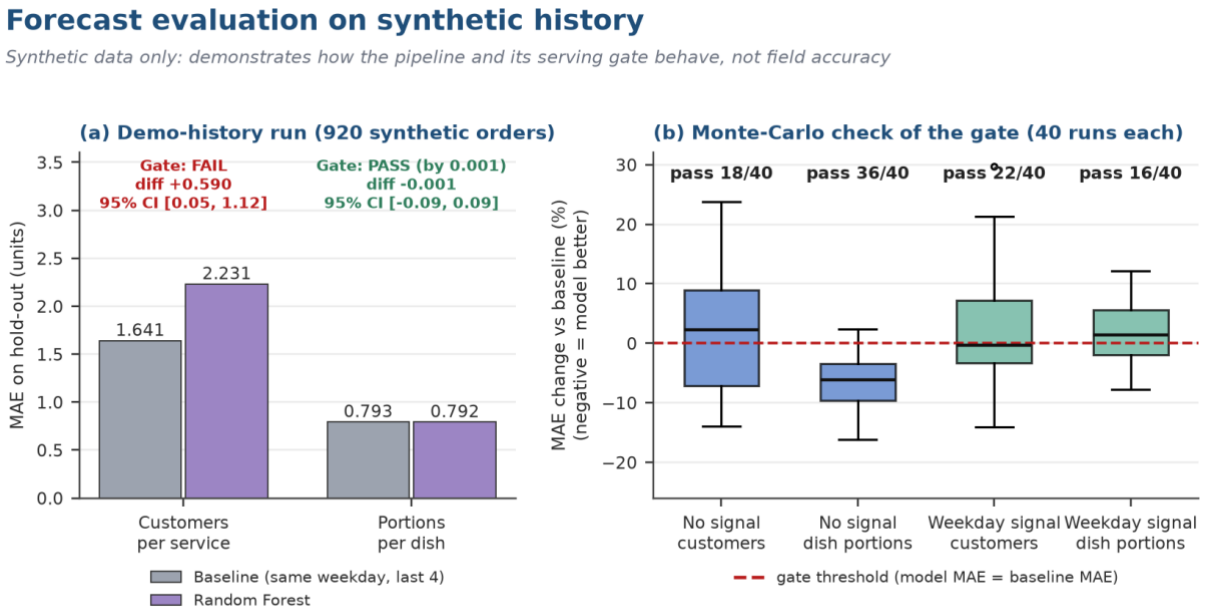


Figure 32. Forecast errors against the baseline and the stability of the serving gate.

The customer model is clearly worse than its baseline, and the dish model is statistically indistinguishable from it: its confidence interval spans zero and the gate is passed by one thousandth of an order. Because the data are random demonstration orders, a good model is not expected; the useful question is whether the **gate** can tell a useful model from a useless one.

Table 46 answers it with a Monte-Carlo study: 40 synthetic histories per scenario, each of about eight weeks. When demand has no signal at all, the dish model passes the gate in 90% of runs and the customer model in 45%. When demand has a genuine weekday pattern, the dish model passes in only 40% of runs and the customer model in 55%. A gate that approves noise and rejects signal at similar rates is not a control (Finding B3).

Table 46. Stability of the serving gate (40 simulated histories per scenario)

Scenario	Target	Gate passes (of 40)	Median MAE difference vs baseline	P10 / P90
No demand signal	Customers	18	+2.2%	-11.2% / +13.9%
No demand signal	Dishes	36	-6.1%	-11.2% / -0.3%

Scenario	Target	Gate passes (of 40)	Median MAE difference vs baseline	P10 / P90
Weekday signal	Customers	22	-0.3%	-7.3% / +11.6%
Weekday signal	Dishes	16	+1.3%	-5.9% / +8.3%

> **Recommended hardening of the gate**
> Require at least 12 to 16 weeks of history before a model can be served; demand a minimum improvement margin (for example 5% lower MAE); require the bootstrap interval of the difference to exclude zero or use rolling-origin evaluation; add a second naive benchmark (seasonal naive); report results by meal period; and show the source of every displayed number (model, baseline or fallback).

17.4 Inventory risk and replenishment

Each ingredient has a reorder point. The Stock tab compares on-hand quantity with it, Today's consumption derives usage from orders, and Risk and impact lists the ingredients under pressure together with the dishes they affect. The suggested replenishment is a transparent rule (Formula 6), not an optimiser: it restores stock to twice the reorder point. It is a prompt for a human decision, not a purchase order.

$$\text{Suggested replenishment} = \max(2 \times \text{reorder point} - \text{on-hand},\ 0)$$

Formula 6. Suggested replenishment quantity

17.5 Copilot

The copilot is available to managers only and is read-only. It recognises eight kinds of question (dishes hit by a shortage, stock risk, inventory alerts, forecasts, the trend in healthy choices, recommendation conversion, best-selling dishes and a sales summary) and otherwise says it cannot help. Every reply carries the intent, an evidence list, the source (`rules` or `llm`), a read-only flag and an as-of timestamp. When a language model is configured it rephrases the rule answer from a compact fact sheet; when it is not, the rule answer is returned unchanged. Table 47 states the interaction standard.

Table 47. Standard for copilot answers

When the manager asks	The answer should
about stock risk	give the rule used and the age of the balance
for a forecast	say whether it comes from the model, the baseline or a fallback, and give the horizon and unit
about a substitute	show the ingredient and declared-attribute checks
for wellness advice	refuse clinical advice and point to menu facts and support

When the manager asks	The answer should
to place a purchase order	explain that supplier ordering lies outside this simplified workflow
and data are missing	say so plainly instead of guessing

Chapter 18: Verification, Testing and Findings

18.1 Test strategy and results

Two executable suites ship with the repository. The backend smoke test drives the API against an empty and a seeded database; the user-interface test drives the Streamlit application with the framework's AppTest harness. Both were run on 8 November 2026 against throwaway databases (Table 24). **All 172 checks passed: 138 backend and 34 user-interface** (Table 48, Figure 33).

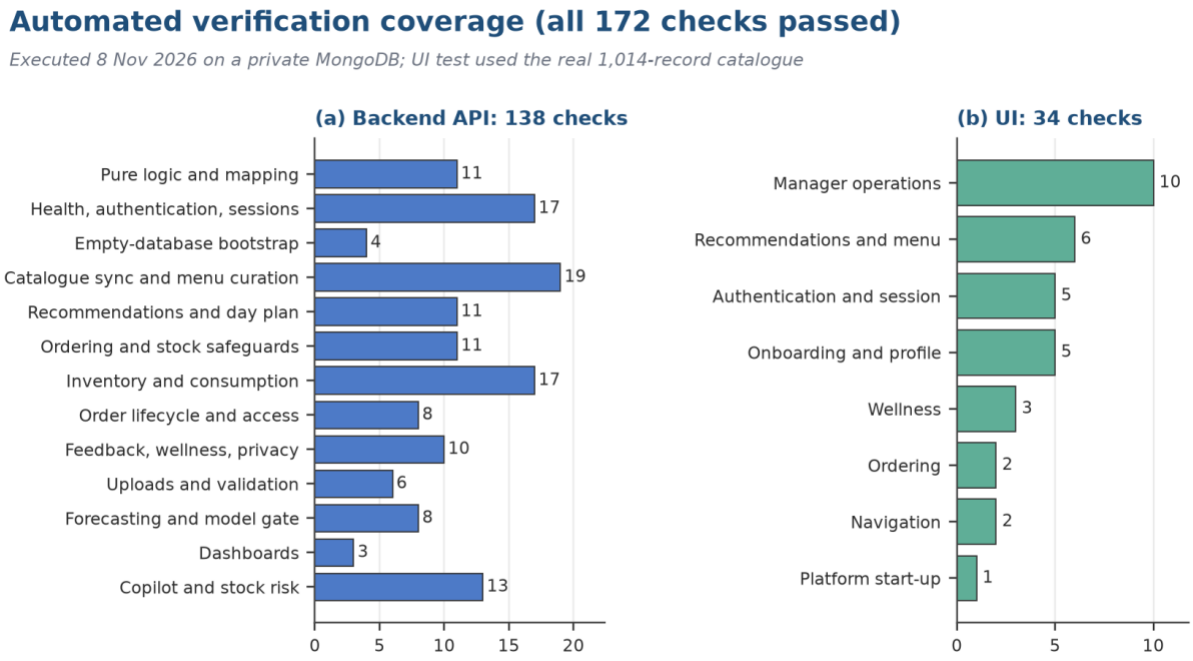


Figure 33. Executed checks by group (138 backend and 34 user-interface, all passed).

Table 48. Executed checks by suite and group

Suite	Group	Checks	Coverage
Backend	Pure logic	11	Scoring, tag rules, allergen canonicalisation and price estimate helpers
Backend	Authentication and sessions	17	Login, lockout, session refresh and expiry, logout, role gates
Backend	Empty database	4	Safe behaviour before any data exists
Backend	Catalogue sync and menu	19	Synchronisation of 1,014 recipes, menu flags, availability and periods
Backend	Recommendations and day plan	11	Hard filters, ranking, explanations, meal plan
Backend	Ordering	11	Checkout, idempotent retry, stock deduction, limits
Backend	Inventory	17	Stock edits, alerts, movements, affected dishes, replenishment
Backend	Order lifecycle	8	Allowed and refused transitions, cancellation and stock return
Backend	Feedback, wellness, privacy	10	One rating per order, scorecard, access to other users' data

Suite	Group	Checks	Coverage
Backend	Uploads	6	Validation before commit for stock and recipe files
Backend	Forecast and model	8	Run, latest, model status, refusal on thin history
Backend	Dashboards	3	Employee, manager and executive payloads
Backend	Copilot and risk	13	Intents, read-only contract, evidence, stock risk
User interface	Manager operations	10	Today, live orders, inventory, forecasting, executive, copilot pages
User interface	Recommendations and menu	6	For you, full menu, plan my day
User interface	Authentication and session	5	Login gate, session restore, logout
User interface	Onboarding and profile	5	Preferences, allergens, goals
User interface	Wellness, ordering, navigation, platform	8	Scorecard page, cart and checkout, role navigation, reconnect banner

18.2 What the tests do not prove

- **Performance and scale.** No load, soak or failure-injection test was run; the persistence layer is synchronous.
- **Usability.** The user-interface suite checks that pages render and behave, not that people can use them.
- **Forecast usefulness.** The model tests check mechanics on synthetic history, not accuracy on real demand.
- **Data truth.** Allergen and nutrition values are catalogue estimates; no laboratory or supplier verification was performed.
- **Security assurance.** Controls were exercised functionally; no penetration test or dependency audit was performed.

18.3 Findings register

The verification surfaced seven findings (Table 49). They are product-quality observations for the owner. None invalidates the purpose of the prototype, but B1 to B3 should be closed before any pilot report quotes the indicators they affect.

Table 49. Findings from the verification run

ID	Finding	Evidence	Severity	Recommended action
B1	The wellness score counts orders that are not yet collected, while the page caption says it counts collected orders	`wellness_score()` includes every non-cancelled order; `month_stats()` uses collected only; captured wellness screen	Medium	Count collected orders only or change the caption; add a regression check
B2	Evidence status is set to “confirmed” for every dish; the Part I three-state match status is not carried into the application	Qdrant payload has no `match_status` field	Medium	Carry the status into the payload and show it next to nutrition values

ID	Finding	Evidence	Severity	Recommended action
B3	The forecast serving gate is a bare comparison on one split and does not discriminate	`ml/train.py`; Table 46	High	History minimum, improvement margin, interval or rolling-origin test, second baseline
B4	The healthy-dish rule is permissive: 92.8% of dishes qualify and the score saturates after eight portions	Table 33; Figure 30	Medium	Tier the tags or weight by nutrient quality; re-baseline the KPI
B5	No waste log and no supplier or lead-time fields	Table 32	Medium	Add a waste transaction with reason codes and a supplier master before benefit claims
B6	Documentation is stale: API_ENDPOINTS.md states 33 endpoints, 60+ checks and 21 of 21 passing	Actual: 42 operations on 39 paths; 138 backend and 34 UI checks	Low	Generate the endpoint list from the OpenAPI export and quote counts from a run
B7	A manager can read any employee's monthly wellness and trend through the API	`require_self_or_manager` on both wellness endpoints	Medium	Limit managers to aggregates above a minimum group size and log privileged access

Reading the findings

The ordering and stock path, the part of ECOS that handles money-like and safety-critical state, passed every check and shows careful engineering (idempotency, conditional updates, compensation). The findings cluster in the **interpretive layer**: indicators whose meaning is more generous than their definition (B1, B3, B4) and evidence that is stated more strongly than it is carried (B2). That is the usual failure pattern of analytics products, and it is cheap to fix before a pilot and expensive to fix after.

Chapter 19: Consumer Behaviour, Ethics, Privacy and Governance

ECOS touches three sensitive areas: personal food choices, declared diet and allergens (which can reveal religion or health), and a wellness score. Governance is therefore treated as part of the product. The technical controls are in Chapter 15; this chapter sets out the management rules that must accompany them.

19.1 Consumer behaviour and choice architecture

The ordering screen doubles as a marketing channel. What people pick is shaped by availability, perceived price, wording, social cues, default placement, suggestions, convenience and timing. A “For you” shelf helps discovery but also directs attention, so ethical practice asks whether diners receive a fair and understandable choice, not just whether engagement goes up.

Counting recommended items is not the same as measuring conversion. An impression or a click does not show that the person chose, completed, collected or ate the dish, nor that they would have chosen differently without the suggestion. A trustworthy funnel tracks each stage on its own: exposure, then engagement, then selection, then the committed order, then fulfilment, with cancellations, substitutions and repeat use recorded alongside, each under a single event definition and privacy-aware aggregation.

Table 50. Ladder for measuring consumer behaviour

Step	Example measurement	What it does not tell you
Visibility	Count of times a suggestion was shown	Only that it could be seen
Interest	Opening a dish page or saving it	Attention rather than a wish to buy
Selection	Adding a dish to the cart	The cart may still be abandoned
Commitment	Placing the order	It may not be fulfilled or paid
Fulfilment	Order handed over	Whether the food was eaten
Repeat	Choosing the dish again later	Habit or preference, not proven loyalty

ECOS records whether each order line came from a recommendation (the `recommended` flag), so the share of ordered lines that were recommended can be measured. Impressions and detail views are not logged, so the earlier stages of the measurement ladder cannot yet be observed.

19.2 Wellness ethics and inclusion

Wellness analytics at work raise special sensitivity because power is unequal. A feature intended kindly can still feel mandatory if managers see named choices or staff fear an effect on appraisals, benefits or standing. ECOS should hence handle wellness information as an optional service signal with tight purpose limits, never as a score for managing people.

In ECOS the wellness score is a rule-based, voluntary indicator shown to the employee, with an explicit non-clinical disclaimer. Two design facts need policy attention: the score rewards volume more than quality (Finding B4), and the API allows a manager to read any individual's score (Finding B7). Purpose limitation therefore has to be enforced by policy and by access control, not left to good intentions.

The interface keeps estimated nutrition apart from measured values and warns that declared allergens do not guarantee freedom from cross-contact. That warning should remain visible where choices are made. In production, add source provenance, review dates, version history, a way to correct errors and an escalation route for people who need a reliable answer; a kitchen that cannot verify cross-contact controls must not imply a guarantee.

Table 51. Guardrails for wellness features

Principle	What it requires in practice
Free choice	Declining or ignoring the feature carries no penalty
Purpose limits	Data never reused for appraisal, promotion or discipline
Data minimisation	Gather as little as possible and aggregate where it can be done
Openness	Explain how scores work, what they leave out and where data come from
Inclusion	Every diet, culture and access need keeps its full range of choices
Right to challenge	People can fix wrong data and ask a human to look again
No clinical role	Nothing is diagnosed, treated or inferred about health

Management must also decide whether wellness features start switched off, sit in a store separate from operational data, or are reported only when groups are large enough. HR, privacy, legal, staff representatives and security should all review. Legal compliance is outside this report and needs qualified advice for the jurisdiction and the data flows concerned.

19.3 Data governance and privacy

Data quality, meaning and ownership set an upper limit on what the platform is worth to management. A dish code that shifts between menus, mismatched stock units, absent service dates or outdated allergen declarations can mislead even if the software runs as written. Governing data is thus part of operating the service, not a preliminary IT chore.

Begin with an inventory of datasets that records for each the owner, steward, source, refresh frequency, retention, sensitivity, definition and permitted purpose, together with a dictionary for each field that feeds a decision. Spell out what “order count” includes (cancelled? pending?), the unit and timestamp for “stock”, how “consumption” differs from sales, and who approved the rule behind “healthy portion”.

Table 52. Minimum data-quality dimensions

Dimension	Owner	Example test
Completeness	Menu steward	All required dish fields filled
Validity	Data steward	Units, dates and values obey defined rules
Consistency	Product or data owner	Same item, unit and status meaning in every view
Timeliness	Operations owner	Age of the data is visible and within threshold
Uniqueness	Application owner	No duplicated item or event identifiers
Accuracy	Operational controller	A sampled record matches the authoritative source
Lineage	Analytics owner	Each output traces to a source version and transformation

Privacy governance begins with the reason for each field and the decision it feeds. A field needed to fulfil an order does not thereby justify every other use. Wellness analytics, personalisation, recovery of failed service, forecasting and finance reconciliation each need a stated purpose and access rule; nothing should be retained or linked just because it might be useful someday.

Before a live pilot, carry out a privacy-by-design review: list categories, identifiers, sources, users, purposes, retention, storage, transfers, deletion and access; ask whether any field hints at diet-related restrictions, religious practice or health; and test whether individual-level analysis is truly needed or an aggregate suffices. Remember that small teams can be identified even in aggregates.

Table 53. Checklist before any data use

Question	Answer needed before the pilot starts
What is the reason for collecting it?	A specific purpose approved for the service or the evaluation
Who may see identifiable records?	Named roles holding only the access they need

Question	Answer needed before the pilot starts
How long will it be stored?	A retention schedule, with proof of deletion
Can an employee fix errors?	A clear contact point and correction route
Could it influence employment decisions?	It must be ruled out explicitly for wellness participation and scores
Does an outside provider touch it?	Review of data transfer and contract
What happens after misuse or a breach?	Defined steps for escalation, containment and notification

The NIST Privacy Framework offers risk-management guidance and is neither a certificate nor a legal safe harbour. The NIST AI Risk Management Framework could guide governance if more advanced AI arrives. Today the copilot is rule-based and read-only, with a language model only optionally rewording answers, and it should be described that way. Indian law and company policy call for review by qualified specialists.

19.4 Analytics and model governance

Analytics governance asks who owns the business question, who owns the method, where the data come from, who approves changes, who acts on the result and how mistakes are corrected. It covers plain rules as well as statistical models, since a rule-based score can mislead when its assumptions stay hidden.

Table 54. Controls across the analytics lifecycle

Step	Evidence of control
Define the use	The decision it supports, who uses it and what it must not be used for
Prepare data	Origin, lineage, quality checks and what was excluded
Build	Code under version control with reproducible dependencies
Test	Baseline comparison, time-ordered holdout, error and subgroup review
Approve	Named sign-off from business and analytics
Release	Artefact identity, start date and rollback plan
Watch	Coverage, drift, outputs, errors, overrides and incidents
Retire	Why it was withdrawn, where it is archived and what users are told

For the Random Forest, keep a governance file containing feature definitions, training cut-off, chronological split, baseline calculation, MAE and WAPE by segment, minimum-history rule, library version, artefact hash and activation decision. The serving gate is a helpful guardrail but should be read alongside segment results and the cost of errors, because a model that only just clears the aggregate bar may still be unfit for a high-risk dish or a quiet period.

Chapter 20: Financial Appraisal, Benefits and Balanced Scorecard

20.1 Cost model

Management accounting moves the discussion from features to the allocation of resources. Start by separating demonstration figures from accounting records: the Executive view labels demonstration order values as unaudited revenue and the demonstration checkout says nothing is charged. Keep those labels visible in screenshots and slides. An order value is useful for illustrating the interface; it is not booked sales, cash in hand, margin or audited revenue.

A future cost model should distinguish one-off from recurring cost. One-off items cover training, integration, discovery, build, data clean-up, change management and privacy assessment. Recurring items cover support, model monitoring, hosting, the database service, stewardship of menu and data, security review, supplier integration and the staff time absorbed by exceptions. Opportunity cost counts too: time a manager spends validating data is a real resource even though no bill arrives.

Table 55. Cost categories for a lifecycle view

Cost category	Evidence source	Example driver
Technology	Approved estimate or invoice	Hosting, storage, database, monitoring
Implementation	Time records and vendor statements	Engineering, integration, migration
Operations	Staff hours and service logs	Support, incidents, menu and data stewardship
Change	Attendance and effort records	Training, communication, backfill
Control	Approved engagement and time	Privacy, audit, testing, legal review
Transition	Project plan and process records	Parallel running, manual fallback, retirement

20.2 Investment appraisal with illustrative inputs

The appraisal below is entirely hypothetical. It uses round numbers that are **not** estimates for ECOS or for any cafeteria, and exists to show how the decision logic works and how sensitive it is. Assume an up-front investment of 12 lakh rupees, a constant annual net benefit for three years, and no tax, inflation or residual value. The net present value is given by Formula 7, and the annual benefit at which the project exactly breaks even by Formula 8.

$$NPV = -I_0 + \sum_{t=1}^{n} \frac{B_t - C_t}{(1+r)^t}$$

Formula 7. Net present value

$$B^{*} = \frac{I_0}{\sum_{t=1}^{n} (1+r)^{-t}}$$

Formula 8. Break-even annual net benefit (B) over n years at discount rate r*

Table 56. Illustrative appraisal under three scenarios (rupees in lakh; investment 12 lakh, three years)

Scenario	Annual net benefit	Discount rate	NPV	IRR	Simple payback	Discounted payback	Break-even benefit	Profitability index
Downside	3	12%	-4.79	-13.1%	4.0 years	Not within 3 years	5.00	0.60
Base illustration	8	10%	+7.89	44.6%	1.5 years	1.72 years	4.83	1.66
Upside	12	8%	+18.93	83.9%	1.0 year	1.09 years	4.66	2.58

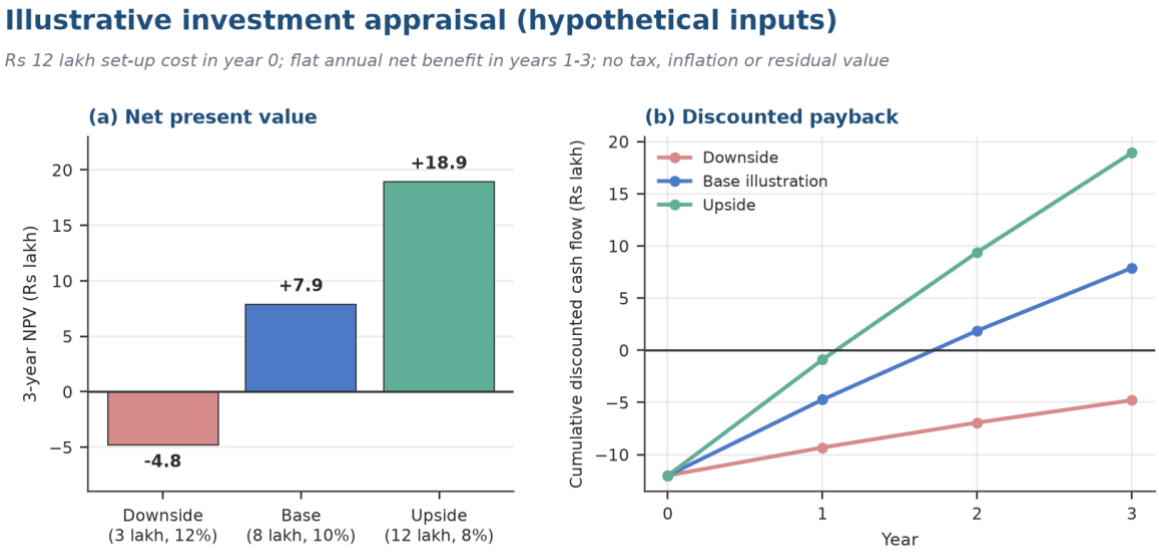


Figure 34. Cumulative discounted cash flow and NPV under the three illustrative scenarios.

Reading the result. The decision turns on one number: whether the true annual net benefit exceeds roughly 4.7 to 5.0 lakh. In the base illustration the NPV stays positive until the benefit falls about 40% below the assumed 8 lakh. For scale, 5 lakh a year is about 1,900 rupees per working day (260 days); for a cafeteria serving 300 meals a day, an illustrative figure, that is roughly 6 rupees of net benefit per meal. Whether waste reduction, shorter exception handling and fewer stock-outs can deliver that, net of the cost of stewardship, is precisely what the pilot must measure. Because B5 means waste is not yet measurable inside ECOS, the benefit side cannot currently be evidenced from the system.

A genuine appraisal swaps each assumption for approved local evidence, compares against leaving things as they are and against a simple shared spreadsheet, and sees a small pilot as an option purchase: a bounded cost that buys the information needed for a later scale decision. Benefits of employee experience should be reported on the scorecard, not converted into invented rupee values.

20.3 Benefits realisation

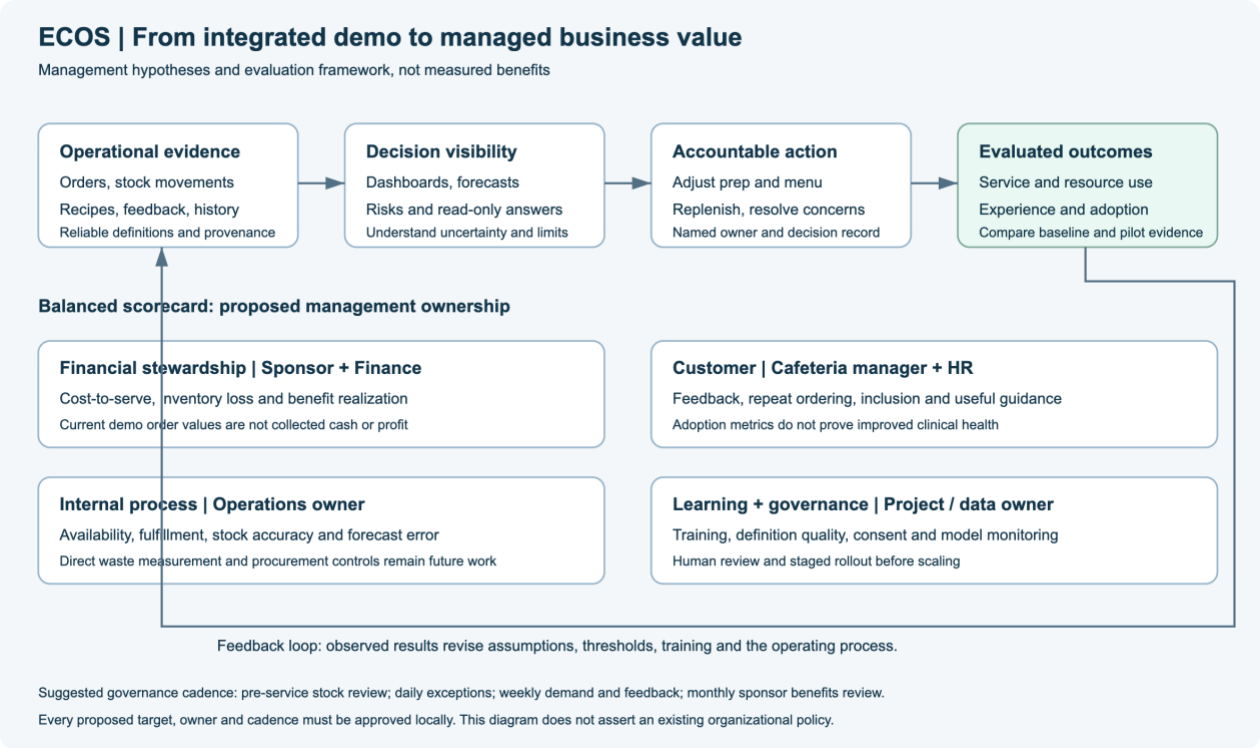


Figure 35. Management value chain (project asset docs/assets/management-value-chain.svg): where ECOS benefits are expected to arise.

Realising benefits has to be planned before the system is built. Each benefit needs an owner, baseline, method for setting targets, metric definition, evidence source, measurement schedule, dependencies and a due date. Software can be delivered without the benefit following if processes, training, decision rights and data quality stay the same; after handover the sponsor, not the developer, should hold the benefit case.

Table 57. Register of benefit hypotheses

Benefit hypothesis	Accountable owner	Outcome evidence	Leading indicator
Faster awareness of exceptions	Operations manager	Resolution time and service log	Time from event to visible queue
Better stock decisions	Inventory owner	Availability, variance and waste data	Completeness of counts and reconciliations
More informed menu choice	Menu or employee-service owner	Employee feedback; completed selections	Attribute completeness; views
Better demand planning	Planner	Actual outcomes; override reasons	Forecast coverage; baseline comparison
Stronger management review	Executive sponsor	Decisions recorded; actions closed	KPI definitions and freshness

20.4 Balanced scorecard

The Balanced Scorecard ties strategy to financial, customer, internal-process and learning measures. For ECOS its job is to stop leaders from improving one result while harming another, not to fill four quadrants with hopeful targets. Kaplan and Norton emphasise measures that turn strategy into action. The scorecard here should be built in stages: definitions and baselines first, targets second, review of outcomes and trade-offs third.

Table 58. Pilot scorecard by perspective

Perspective	Question to ask	Candidate measure	Safeguard
Financial	Is the money spent and saved responsibly?	Validated lifecycle cost and cashable benefit	Demonstration order value never counted as revenue
Employee	Is the service clear and trusted by diners?	Tasks completed, problems resolved, voluntary comments	The whole menu stays available; wellness can be declined
Internal process	Do service and stock decisions stay under control?	Availability, reconciliation and closing of exceptions	Accuracy and safety checks
Learning	Is the organisation getting better from evidence?	Completeness of data, review of model against baseline, staff readiness	No scaling up before a gate decision

Chapter 21: KPI Framework and Measurement Plan

Definitions have to come before dashboards. This chapter fixes the pilot's definitions, using the formulas executed in Chapter 17, and checks which of them ECOS can already compute from the data it stores (Table 61).

21.1 Service and adoption KPIs

A metric is worth only as much as its definition. The starter set below needs local validation. Every metric should have a stable name, a purpose, a formula with numerator and denominator, inclusion rules, a time grain, segments, an owner, a way of setting targets and a statement of its limits.

Table 59. Service and adoption KPIs

KPI	Interpretation limit	Definition
Order completion rate	Rules for cancellation, substitution and eligibility must be fixed first	Orders that reach the completed state as a share of eligible submitted orders
Availability rate	Requires a recorded history of availability	Share of item-periods on offer that were actually available
Stockout incidence	Depends on how an event is defined and reported	Confirmed stockouts relative to the volume served
Exception resolution time	Quote the median and the spread rather than the mean alone	Time from logging an exception to resolving it
Menu data completeness	A full field is not necessarily a correct one	Required fields that are filled, as a share of all required fields
Employee task success	The sample and study design must be reported	People who finish a set task unaided, as a share of those who try
Adoption by role	Logging in is not value; do not set coercive targets	Eligible users who perform a defined qualifying action, as a share of all eligible users
Repeat selection	Does not prove loyalty or satisfaction	People who pick the same dish again within a set window

$$\text{Completion rate} = \frac{\text{orders in the completed state}}{\text{eligible submitted orders}} \times 100\%$$

Formula 9. Completion rate

Exclude from the denominator only cases that are documented, such as duplicate test records, and report how many were excluded. Where demonstration orders are involved, call the result demonstration behaviour rather than sales conversion. If the completed state only means collected, avoid describing it as a paid conversion.

21.2 Analytics and control KPIs

Alongside outcomes, this scorecard tracks how good the decision process is. Forecast error, comparison with the baseline, data coverage, stock reconciliation and the reasons behind overrides each reveal a different way things go wrong, so a single blended score would hide them.

Table 60. Analytics and control KPIs

KPI	Control interpretation	Definition
MAE	In the units of the target; report by segment	Average of the absolute gap between actual and forecast on a defined holdout
WAPE	Becomes unstable when total demand is zero or very small	Total absolute gap divided by total actual demand
Forecast coverage	Count fallbacks and gaps separately	Share of eligible item-periods that received a valid forecast
Baseline improvement	Positive only if the candidate has the lower error	Reduction in MAE relative to the baseline MAE
Forecast bias	The sign matters for surplus versus shortage cost	Average of forecast minus actual
Reconciliation variance	Note unit and count date; investigate rather than smooth away	Absolute difference between recorded and counted quantity
Edit audit completeness	A complete log does not show that the change was correct	Share of stock changes that record who, why, when and on what source
Model override rate	Capture reasons; never use as a penalty	Share of reviewed outputs that were changed or rejected

$$\text{Baseline improvement} = \frac{\text{MAE}_{\text{baseline}} - \text{MAE}_{\text{candidate}}}{\text{MAE}_{\text{baseline}}}$$

Formula 10. Baseline improvement

When baseline MAE is zero the ratio has no meaning and must not be quoted as a percentage. Clearing the gate on overall MAE does not make a model safe: look for failures in individual segments, thin data and unequal costs of over- and under-forecasting, and report coverage so that low error on easy cases is not read as general usefulness.

In Chapter 17 the baseline comparison was reproduced on the synthetic history: the customer model had a baseline improvement of about -36% (it was worse), the dish model about 0%. These are the values this framework would report, and they illustrate why the interval and the segment view matter more than the sign.

Control KPIs need thresholds with matching actions, because a threshold nobody can act on is merely an alarm. If reconciliation variance goes beyond the agreed tolerance, the owner checks item mapping, unit conversion, timing, shrinkage and whether the process was followed. If forecast coverage drops, managers fall back to a labelled baseline or a manual plan.

21.3 Data contract: what ECOS records today

Table 61. Availability of KPI data in the current system

KPI	Source in ECOS	Available now	Gap to close
Order completion rate	`orders.status`, `created_at`, `collected_at`	Yes	Define eligible orders and separate demonstration data
Service time	`created_at` to `ready_at` and `collected_at`	Yes	Define the outlier rule
Availability rate	Computed from stock when a menu is requested	No history	Persist availability snapshots
Stockout incidence	Not recorded as events	No	Log stockout events with reason
Menu data completeness	Fields on `recipes`	Computable	Approve the required-field list
Recommendation conversion	`recommended` flag on order lines	Partial	Impressions are not logged
Forecast MAE and WAPE	Orders, `forecasts`, model metadata	Training run only	Rolling evaluation in operation
Reconciliation variance	`inventory_movements`	Partial	Physical counts are not modelled
Waste	None	No	Finding B5
Adoption by role	`sessions`, `orders`	Partial	Define the qualifying action

Chapter 22: Pilot Design, Change Management and Roadmap

22.1 Pilot design and decision gates

The recommended path is a four-phase, stage-gated programme (Figure 36): charter and baseline, controlled demonstration, single-site pilot and a scale decision. Each gate needs evidence, a named approver and an explicit stop condition, and the default when evidence is missing is to adapt or stop, never to proceed. Table 62 sets the pilot parameters.

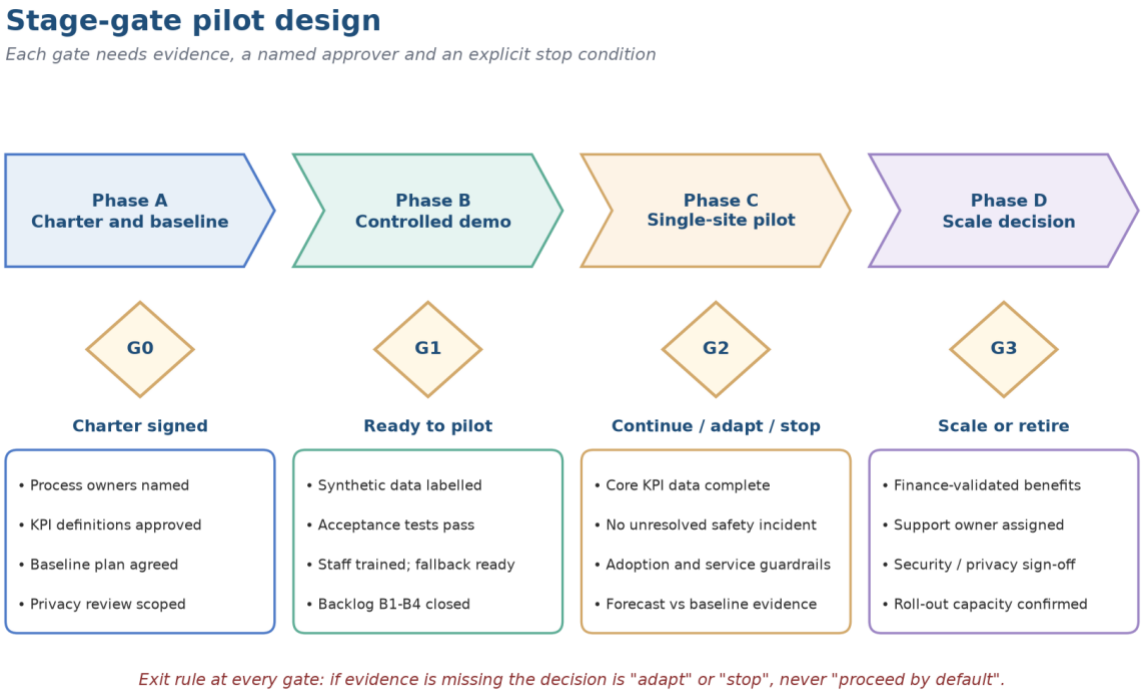


Figure 36. Stage-gate pilot design with gate criteria.

Table 62. Proposed pilot parameters

Parameter	Proposal
Site and owners	One cafeteria with a named menu steward, inventory owner and shift lead; operations sponsor as approver
Timeline	Baseline measurement from week 0 to week 8; pilot operation from week 8; interim review at week 12 (G2*); evaluation to month 6; scale-or-retire decision at month 11 to 12 (G3)
Population	Employees who opt in; the full menu stays available to everyone; no employment use of wellness data
Data	Real orders only, flagged by source; demonstration data excluded from every KPI
Primary KPIs	Order completion rate, availability rate, exception resolution time, menu data completeness
Guardrails	Cancellation and stockout rates, opt-out rate, privacy incidents, forecast override rate
Comparison	Pre-pilot baseline and, if a second site exists, a staggered start; otherwise report association, not causation

Parameter	Proposal
Stop conditions	Unresolved allergen or safety incident; privacy breach; stock data integrity failure; breach of an adoption guardrail

22.2 Adoption and change management

Technology adoption depends on perceived usefulness and ease of use (Davis, 1989), and change efforts usually fail for organisational rather than technical reasons (Kotter, 1995). Table 63 translates the eight steps of Kotter's model into pilot actions.

Table 63. Kotter's eight steps applied to the ECOS pilot

Step	Action in the pilot	Evidence of progress
1. Create urgency	Share the baseline on exception handling, stock accuracy and menu data gaps	Baseline report accepted by the sponsor
2. Guiding coalition	Operations, finance, IT and privacy, and an employee representative	Charter signed (G0)
3. Vision	A cafeteria that decides with timely, explainable information while humans stay accountable	One-page vision approved
4. Volunteer army	Shift leads and menu stewards as champions; employees opt in	Named champions per shift
5. Remove barriers	Close B1 to B4 and B7; agree fallback procedures; train on the stock edit flow	Gate G1 passed
6. Short-term wins	Visible exception queue and cleaner menu data within the first weeks	Interim review (G2*)
7. Sustain	Weekly review of forecast, stock and incidents; adapt the KPI dictionary under version control	Review minutes and closed actions
8. Anchor	Move ownership of definitions and benefits to the business sponsor	Handover accepted (G3)

22.3 Roadmap

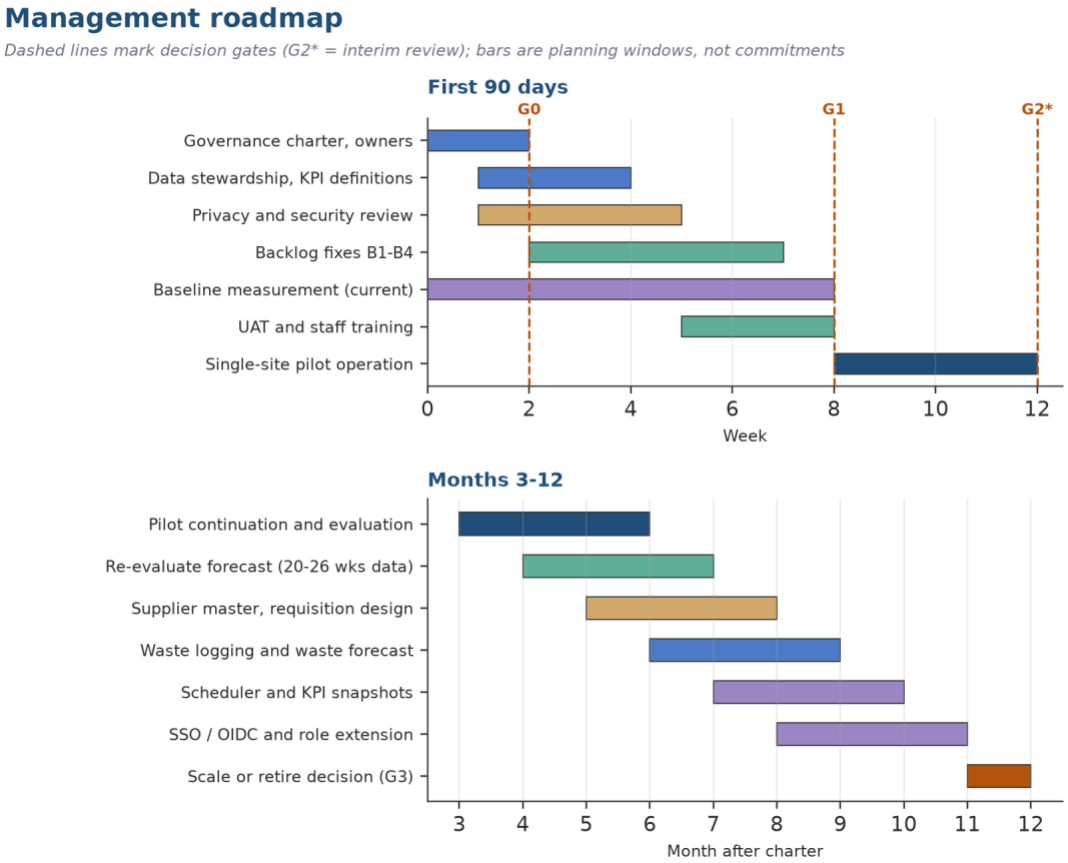


Figure 37. Management roadmap: first 90 days and months 3 to 12 (planning windows, not commitments).

A good lifecycle provides governance from first discovery to handover. The report favours stage gates over building until finished: each gate asks for business evidence, technical evidence, a control review, a named approver and written exit criteria. Merged code alone is not a reason for a feature to move on.

Table 64. Lifecycle stages and gate evidence

Stage	Gate evidence	Core work
Discover	Approved problem and process owner	Interview stakeholders; map current state; plan baseline
Define	Signed pilot charter with acceptance criteria	Scope, requirements, data contracts, risk assessment
Design	Review by operations, IT, finance, privacy	Service blueprint; role, privacy and control design
Build	Versioned release; test evidence	Implement priority workflows with traceability
Verify	Defects resolved; readiness decision	Functional, usability, data, performance, recovery tests
Pilot	Baseline; monitoring; incident and feedback logs	Controlled operation and measurement
Evaluate	Finance and business benefit review	Compare outcomes, costs and limits

Stage	Gate evidence	Core work
Hand over or retire	Operational acceptance or safe shutdown	Ownership, support, documentation, decommissioning

22.4 Ownership and accountability

Table 65. Responsibility assignment for the pilot (R responsible, A accountable, C consulted, I informed)

Activity	Operations sponsor	Cafeteria manager	Menu and inventory owner	Data and IT owner	Finance	Privacy and HR
Menu and allergen data accuracy	A	C	R	C	I	I
Stock adjustments and counts	I	A	R	C	I	-
Forecast activation and override	A	R	C	C	I	-
KPI definitions and dashboard	A	C	C	R	C	I
Privacy and wellness rules	C	I	-	R	-	A
Incident response	A	R	C	R	-	C
Benefits validation	A	C	C	C	R	I

Chapter 23: Risks and Limitations

23.1 Risk assessment

Fourteen risks were scored for likelihood and impact on a five-point scale before pilot controls (Figure 38, Table 66). The scores are prioritisation aids, not probabilities. The highest-scoring risk is stale or wrong menu attributes (R1); the most frequent are the weak forecast gate, the evidence status that is not carried, and the dependency on one maintainer (R4, R12, R14).

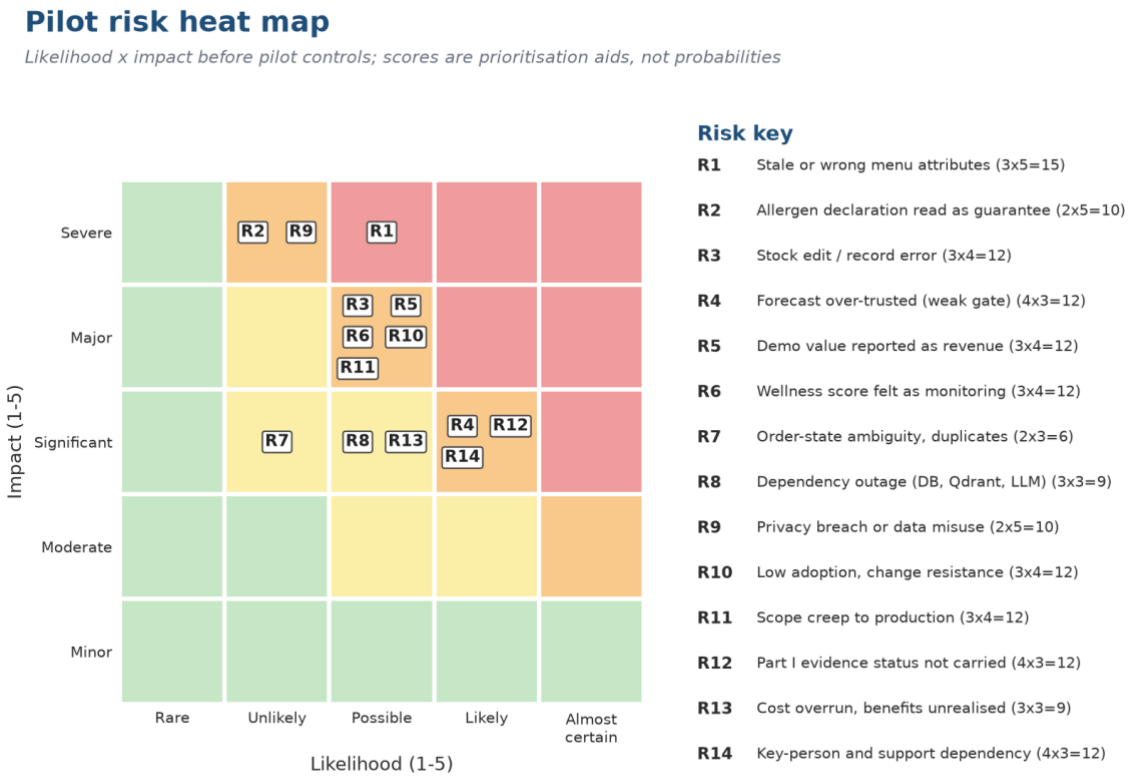


Figure 38. Pilot risk heat map (likelihood by impact, before controls).

Table 66. Risk register for the pilot

ID	Risk	L	I	Score	Principal control	Owner
R1	Stale or wrong menu attributes	3	5	15	Stewarded approval, effective dates, completeness review	Menu steward
R2	Allergen declaration read as a guarantee	2	5	10	Prominent “declared, not guaranteed” label; unknown shown as unknown; staff route	Food-safety owner
R3	Stock edit or record error	3	4	12	Role access, validation preview, reason codes, reconciliation	Inventory owner
R4	Forecast over-trusted (weak gate)	4	3	12	Harden the gate (B3); label the source; log overrides	Analytics owner

ID	Risk	L	I	Score	Principal control	Owner
R5	Demonstration value reported as revenue	3	4	12	Persistent demo label; metric dictionary; finance review	Finance owner
R6	Wellness score felt as monitoring	3	4	12	Voluntary use; aggregate-only manager access (B7); no employment use	Privacy and HR
R7	Order-state ambiguity, duplicates	2	3	6	Idempotency and conditional transitions (implemented); exception report	Service owner
R8	Dependency outage (database, Qdrant, language model)	3	3	9	Manual fallback menu; health checks; labelled degradation	IT owner
R9	Privacy breach or data misuse	2	5	10	Minimisation, retention schedule, access logging, breach process	Privacy owner
R10	Low adoption, change resistance	3	4	12	Co-design, training, observed task checks	Change lead
R11	Scope creep to production	3	4	12	Stage gates, change log, adapt-or-stop default	Sponsor
R12	Part I evidence status not carried	4	3	12	Carry the three-state status (B2) and show it beside nutrition values	Product owner
R13	Cost overrun, benefits unrealised	3	3	9	Lifecycle cost model, benefits register, interim review	Finance and sponsor
R14	Key-person and support dependency	4	3	12	Runbook, documentation, second maintainer	IT owner

Risk management connects uncertainty to an owner, prevention, detection, response and what remains afterwards. A risk with nobody in charge is not controlled, and a control with no evidence cannot be tested. The register above is a design for the pilot and should be tuned in a workshop with operations, finance, purchasing, IT, privacy and staff representatives.

23.2 Limitations and threats to validity

Table 67. Threats to validity and how they are handled

Type	Threat	Handling
Construct	The wellness score and the match percentage are proxies, not measures of wellbeing or relevance	Defined, caveated and listed as findings (B1, B4)
Internal	Demonstration orders are random, so forecast errors reflect the generator, not behaviour	The study tests the gate with simulation rather than claiming model accuracy
External	One catalogue, one machine, no real users or site	Pilot design; replication at a second site
Statistical	Small test sets (32 and 152 rows) and a single run	Bootstrap intervals; Monte-Carlo repetition for the gate
Reliability	One verification day	Versions and artefacts recorded in Annexure F; repeat before the pilot

Chapter 24: Conclusions and Recommendations

24.1 Answers to the research questions

Part I posed four research questions with testable hypotheses and stated that none was proven at mid-semester. Table 68 records what the final evidence supports.

Table 68. Outcome of the research questions and hypotheses

RQ / H	Statement (abridged)	Evidence at final stage	Verdict
RQ1 / H1	Heterogeneous data consolidated into an evidence-aware catalogue with unique keys and traceable evidence status	1,014 records synchronised under a unique `dish_id` index; evidence status not carried into the application (B2)	Partly supported
RQ2 / H2	Embeddings find semantically comparable foods beyond keyword search	Embeddings (256 dimensions) feed the ranking; no reviewer-labelled relevance test was run	Implemented, not tested
RQ3 / H3	A property graph makes dish, ingredient and allergen relationships queryable	No property graph; relationship queries (affected dishes, allergen exclusion) answered through document links	Not tested as stated
RQ4 / H4	A dashboard exposing evidence and filters is preferred to one that only ranks	Reasons, hard filters and disclaimers are shown; no preference study; evidence status is not shown (B2)	Implemented, not tested

24.2 Closure of the approved objective themes

Table 69. Status of the approved objective themes

Objective theme	Final status	Basis
Operational analytics	Partly delivered	Demand forecasting instead of preparation-time modelling; gate needs hardening (B3)
Menu intelligence	Delivered	Semantic ranking, component handling, explainable reasons
Food relationships	Partly delivered	Document references for ingredients, allergens and tags; no graph database
Explainable decision support	Partly delivered	Reason strings and evidence lists; no Graph RAG
Supply-chain dependency	Delivered (simplified)	Affected-dish and stock-risk views; no supplier master
Nutrition-aware choice	Delivered with caveats	Tag rules are permissive (B4); score definition inconsistent (B1)
Dashboard for management	Delivered	5 employee and 7 manager views; 172 executed checks

24.3 Contributions

- **Managerial.** A decision lens that separates safety constraints from preferences, indicators from outcomes, and demonstration values from revenue, together with a gated pilot design, a KPI dictionary and a risk register.
- **Methodological.** An evidence-class discipline and a Monte-Carlo test of a serving gate, which shows that a bare comparison with a baseline cannot separate signal from noise on eight weeks of history.

- **Practical.** A verified, integrated prototype with an auditable ordering and stock path, 172 executed checks, a profiled catalogue and a findings backlog ready for the owner to act on.

24.4 Recommendations

Table 70. Prioritised recommendations

#	Recommendation	Owner	By
1	Close findings B1 to B4 and B7 before the pilot (wellness score, evidence status, forecast gate, healthy-dish rule, manager access to wellness)	Product owner	G1
2	Approve the KPI dictionary and the data contract; start baseline measurement	Operations sponsor	G0
3	Name stewards for menu attributes, stock definitions and KPI logic	Operations sponsor	G0
4	Pilot with real orders flagged by source and exclude demonstration data from every KPI	Cafeteria manager	G2
5	Treat forecasts as advisory: show the source label and keep the baseline until the hardened gate passes on 12 to 16 weeks of history	Analytics owner	Months 3 to 7
6	Add waste logging and supplier fields before any benefit claim (B5)	Product owner	Months 6 to 9
7	Keep the wellness score voluntary and aggregate-only for managers	Privacy and HR	G1
8	Replace stale documentation with artefacts generated from the running system (B6)	Product owner	G1

24.5 Future research

- Evaluate a property graph or Graph RAG against the relational approach used here, with explicit relevance and latency criteria.
- Estimate the causal effect of recommendations on choice with a stepped rollout rather than a before-and-after comparison.
- Test forecasting methods designed for sparse, intermittent demand and for hierarchical series (meal period, dish).
- Replace the permissive tag rule with a nutrient-quality score validated by dietitians, and study its fairness across dietary and cultural patterns.
- Model waste explicitly and test menu optimisation that trades waste, cost and nutrition.

> **Final managerial position**
>
> ECOS is a credible, well-engineered decision-support prototype whose safety-critical path works as designed and whose analytical claims are not yet strong enough to carry decisions. The responsible next step is a governed single-site pilot with the findings closed, real data flagged by source, and stage gates that default to adapt or stop.
>
> The project's contribution is as much in what it refuses to claim as in what it built: no accuracy, savings or health effect is asserted, and each is linked to the evidence that would justify it.

Annexure F: Evidence and Reproduction Register

Table 71 lists the artefacts behind the executed and code-inspected claims of Part II. All executed results date from 8 November 2026.

Table 71. Evidence register

Artefact	What it evidences	Class
`backend/smoke_test.py`	138 backend checks across 13 groups	Executed
`dashboard/ui_smoke_test.py`	34 user-interface checks across 8 groups	Executed
`ml/train.py`, `ml/features.py`, `ml/predictor.py`	Forecast features, split, baseline, serving gate and fallback	Executed, code-inspected
`backend/app/routers/*.py` **(14 routers)**	42 operations on 39 paths; order, stock, wellness and copilot logic	Executed, code-inspected
`backend/app/core/recommender.py`	Hard filters, score terms and match percentage	Executed, code-inspected
`backend/app/core/catalog.py`, `vectors.py`	Catalogue synchronisation and embeddings	Code-inspected
Qdrant `food_nutrients` **collection**	1,014 vectors, 256 dimensions, cosine, 91 payload fields	Executed
`API_ENDPOINTS.md`, `api.json`	Published contract; counts found to be stale (B6)	Documented
`Architecture.png`, `docs/assets/*.svg`	Target and conceptual diagrams	Documented
`docs/FINAL_PROJECT_REPORT.md`	Management frameworks reused in Part II	Documented

Reproduction. Start a throwaway MongoDB instance, point the backend at it with `ECOS_MONGODB_URI`, seed demonstration history with `python -m backend.seed --history`, and run the two smoke tests. Use separate databases for the backend and the interface tests, and never point the application at a database that holds real data.

Annexure G: API Operation Catalogue

The service exposes 42 operations on 39 distinct paths under `/api/v1`, grouped here by router. The committed documentation lists 33 endpoints (Finding B6).

Table 72. Operations of the ECOS service

Method	Path	Router	Operation
GET	/api/v1/health	health	Health
POST	/api/v1/auth/login	users	Login
GET	/api/v1/auth/session	users	Refresh Session
POST	/api/v1/auth/logout	users	Logout
GET	/api/v1/users/{user_id}	users	Get User
PUT	/api/v1/users/{user_id}/preferences	users	Save Preferences
GET	/api/v1/users	users	List Users
GET	/api/v1/recipes	recipes	List Recipes
PATCH	/api/v1/recipes	recipes	Update Recipes
POST	/api/v1/recipes/bulk-menu	recipes	Bulk Menu
GET	/api/v1/catalog/status	recipes	Catalog Status
POST	/api/v1/catalog/sync	recipes	Sync Catalog
GET	/api/v1/menu	menus	Get Menu
POST	/api/v1/recommendations	recommendations	Recommendations
POST	/api/v1/recommendations/meal-plan	recommendations	Day Plan
POST	/api/v1/orders	orders	Create Order
GET	/api/v1/orders	orders	List Orders
GET	/api/v1/orders/{order_id}	orders	Get Order
PATCH	/api/v1/orders/{order_id}/status	orders	Update Status
POST	/api/v1/feedback	feedback	Submit Feedback
GET	/api/v1/inventory	inventory	List Inventory
POST	/api/v1/inventory	inventory	Create Ingredient
PATCH	/api/v1/inventory/{ingredient_id}	inventory	Update Ingredient
GET	/api/v1/inventory/alerts	inventory	Inventory Alerts
POST	/api/v1/inventory/{ingredient_id}/replenish	inventory	Replenish
GET	/api/v1/inventory/{ingredient_id}/affected-dishes	inventory	Affected Dishes
GET	/api/v1/inventory/{ingredient_id}/movements	inventory	Inventory Movements
GET	/api/v1/inventory/risk	inventory	Stock Risk
GET	/api/v1/inventory/consumption	inventory_consumption	Consumption
POST	/api/v1/upload/inventory	uploads	Upload Inventory
POST	/api/v1/upload/recipes	uploads	Upload Recipes
POST	/api/v1/forecast/run	forecasts	Run Forecast
GET	/api/v1/forecast/latest	forecasts	Latest Forecast
GET	/api/v1/forecast/model	forecasts	Model Status
POST	/api/v1/forecast/model/train	forecasts	Retrain Models
GET	/api/v1/wellness/{user_id}/monthly	wellness	Monthly
GET	/api/v1/wellness/{user_id}/trend	wellness	Trend
GET	/api/v1/dashboard/employee/{user_id}	dashboard	Employee Dashboard

Method	Path	Router	Operation
GET	`/api/v1/dashboard/manager`	dashboard	Manager Dashboard
GET	`/api/v1/dashboard/executive`	dashboard	Executive Dashboard
POST	`/api/v1/copilot`	copilot	Ask
GET	`/api/v1/copilot/suggestions`	copilot	Suggestions

Annexure H: Selected Logic Excerpts (Part II)

The three excerpts restate, as pseudocode, the ordering safeguard, the wellness score and the forecast serving gate; the source files are listed in Table 71.

H.1 Atomic stock deduction with rollback (ordering logic)

```
for each ingredient required by the order:
    run one conditional update: subtract the amount only if stock >= amount
    if nothing was updated:
        add back every amount already subtracted for this order
        reject the order (conflict) and name the ingredient that ran short
    remember (ingredient, amount) so that a cancellation can return it once
```

H.2 Monthly wellness score (wellness logic)

```
orders = the employee's orders in the month whose status is not cancelled
points = 0
for each order line in orders:
    points = points + quantity x (10 if the dish has any nutrition tag, else 5)
score = min(100, 20 + points)
```

H.3 Chronological split and serving gate (training logic)

```
cut-off = service date at the 80 per cent position of the sorted dates
train   = rows dated before the cut-off
test    = rows dated on or after the cut-off
fit the Random Forest on train and predict test
beats_baseline = MAE(model, test) <= MAE(same-weekday-4 baseline, test)
serve the model only when beats_baseline is true
```

Annexure I: Demonstration Script and Viva Preparation

Table 73. Eight-step demonstration script

Step	Action	Point to make
1	Log in as an employee	Role-based navigation; session handling; synthetic identity
2	Menu, For you	Reasons on each card; match percentage capped at 99; diet and allergen filters are hard
3	Plan my day	Day plan built from the same recommender
4	Place a demonstration order	“No money is taken”; pickup reference; idempotent retry
5	Manager: Live orders	Advance the status; cancel before ready and show that stock returns
6	Inventory: Stock, Risk and impact, Edit stock	Validation before commit; affected dishes; movement history
7	Forecasting	Source label, buffer, preparation quantity; model against baseline
8	Executive and Copilot	Demonstration-value caption; read-only answers with evidence

Table 74. Likely questions and defensible answers

Question	Defensible answer
What is new since mid-semester?	Integration into ECOS, 172 executed checks, a catalogue profile, a forecasting study, a financial and governance appraisal and a pilot roadmap
Why no knowledge graph?	It was re-scoped: relationships are held as document references and answer the needed queries; a graph would add cost without a demonstrated benefit yet
Is the forecast accurate?	No accuracy is claimed. Data are synthetic, the baseline is in use, and the study shows the serving gate is weak
Is the wellness score medical?	No. It is a voluntary, rule-based summary of recorded choices with a disclaimer; its definition has known weaknesses (B1, B4)
What is the business case?	Illustrative only: break-even needs about 4.7 to 5.0 lakh a year of net benefit; the pilot exists to measure it
How is allergen safety handled?	Diet and allergen rules are hard filters enforced on the server; declarations are described as declared, not guaranteed
What would you do first in a pilot?	Close B1 to B4 and B7, approve the KPI dictionary, start baseline measurement
What are the limits?	Synthetic operations data, no users, one run on one machine, hypothetical finance

Annexure J: References Added in the Final Report

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Annexure K: Glossary Additions

Table 75. Terms introduced in Part II

Term	Meaning in this report
ECOS	Enterprise Cafeteria Operating System: the integrated employee and manager service built on the Part I catalogue
Hard filter	A rule that removes a dish from consideration before ranking (availability, diet, allergen, budget, energy)
Idempotent checkout	Submitting the same request twice yields one order, because the request identifier is unique per user
Serving gate	The rule that decides whether the trained model or the baseline is used: model MAE not above baseline MAE
Baseline (same-weekday-4)	Mean of up to four preceding services on the same weekday, falling back to a four-service rolling mean
MAE, WAPE	Mean absolute error; weighted absolute percentage error (total absolute error over total actual demand)
Bootstrap interval	A confidence interval obtained by resampling the test rows; used here for the difference in MAE
Evidence class	The basis of a claim: executed, code-inspected, documented, user-reported, proposed or illustrative
NPV, IRR	Net present value and internal rate of return of a cash-flow series at a given discount rate
Profitability index	Present value of benefits divided by the initial investment
Break-even benefit	The constant annual net benefit at which NPV equals zero
Stage gate	A decision point with evidence, a named approver and an explicit stop condition
Synthetic demonstration data	Generated users, orders and stock flagged `demo_seed`; they show mechanics, not behaviour